

LABORATORY BEAMS CARRYING ORBITAL ANGULAR MOMENTUM

Allen *et al.* (3.1, **Paper 2.1**) in 1992 were the first to recognise that a beam with helical wavefronts has an orbital angular momentum in the direction of propagation. They showed that all beams with helical phasefronts characterised by an $\exp(il\phi)$ azimuthal phase have an orbital angular momentum equivalent to $l\hbar$ per photon.

Laser beams are usually described by use of a Hermite–Gaussian basis set, but a beam with helical wavefronts is best described in terms of the Laguerre–Gaussian basis set which has an explicit $\exp(il\phi)$ phase term. Hermite–Gaussian ($\text{HG}_{m,n}$) modes are characterised in terms of the mode indices m and n which denote the order of the Hermite polynomials in the x and y directions. Laguerre–Gaussian (LG_p^l) modes are described by the indices l and p where the l -index relates to the azimuthal phase and the p -index to the number of additional concentric rings around the central zone. The form for the LG mode field amplitude is given by

$$u_p^l(r, \phi, z) = C_p e^{-ikr^2/2R} e^{-r^2/w^2} e^{-i(2p+|l|+1)\tan^{-1}(z/z_r)} e^{-il\phi} (-1)^p (r\sqrt{2}/w)^{|l|} L_p^{|l|}(2r^2/w^2)$$

When $l \neq 0$, the helical phasefronts result in a phase discontinuity on the beam axis and a corresponding zero in the beam intensity. Consequently, the $l \neq 0$ modes have the appearance of an annular ring for $p = 0$, or rings when $p > 0$. Note also the Gouy phase term $\exp(-i(2p+|l|+1)\tan^{-1}(z/z_r))$ which describes the phase change as a beam moves through the beam waist. Padgett *et al.* (3.2, **Paper 3.1**) illustrate a number of examples of HG and LG modes and show interferograms formed between the LG modes and a plane wave where the helical phasefronts of the modes are revealed in the form of spiral fringes.

As both Hermite–Gaussian and Laguerre–Gaussian are complete orthonormal sets, any arbitrary beam may be described by a linear superposition of either set. As long ago as 1963 it was recognised that an LG_0^1 mode could be expressed as the phase-quadrature superposition of $\text{HG}_{1,0}$ and $\text{HG}_{0,1}$ modes (3.3). However for most actual laser cavities, astigmatism breaks the frequency degeneracy between the $\text{HG}_{1,0}$ and $\text{HG}_{0,1}$ modes and, in practice, an LG output is never observed. Deliberate removal of the cavity astigmatism allows the two modes to become degenerate and frequency lock together. If an on-axis intensity block is introduced, the modes become locked in phase-quadrature to give an LG output. Just as the handedness of circular polarisation depends upon the relative phase of the two composite linear polarisations, so the handedness of the LG mode depends upon the relative phase of the composite HG modes.

Vaughan and Willets (3.4, **Paper 3.2**) produced a laser mode with an annular intensity profile in 1983, but made no observation concerning the potential of the beam to carry

orbital angular momentum. In 1990 Tamm and Weiss (3.5, **Paper 3.3**) designed an astigmatism free laser cavity to produce an LG_0^1 mode and used selective feedback to define its handedness. However, again the orbital angular momentum properties of the resulting beam were not recognised. Slightly more recently, Harris *et al.* (3.6, **Paper 3.4**) succeeded in interfering an LG output from a laser with its own mirror image to produce spiral interference fringes, which showed the helical phase structure of the mode.

Laguerre–Gaussian modes were first produced in order to demonstrate their orbital angular momentum content, by Beijersbergen *et al.* in 1993 (3.7, **Paper 3.5**). This work used the results of Abramochkin and Volostnikov (3.8) in their investigation of the near field for astigmatic optical elements, to establish a general algebraic relationship with which any LG mode can be expressed as a linear superposition of HG modes and vice versa. Beijersbergen *et al.* demonstrated a lossless mode converter, consisting of a pair of cylindrical lenses, which transformed any HG mode into an LG mode. It relies on the fact that when a beam is focused between two cylindrical lenses a Gouy phase shift is introduced that depends upon the indices of the mode. This generalised the approach of Tamm and Weiss (3.5) but now, just as appropriate focusing of the incident beam and the canonical separation of the lenses introduces a $\pi/2$ phase shift between $\text{HG}_{1,0}$ and $\text{HG}_{0,1}$ modes to produce an LG_0^1 mode, so the same lens configuration converts every HG mode into an LG mode with $l = m - n$ and $p = \min(m, n)$. The action of the cylindrical lens $\pi/2$ -mode converter for orbital angular momentum is, therefore, analogous to that of a quarter wave plate for spin. Beijersbergen *et al.* also explained how the angular momentum transfer between the lens and the light could be explained in terms of the asymmetry of the incident HG mode. This can be shown to produce a gradient force which exerts a torque on the cylindrical lens and a subsequent transfer of angular momentum.

An increase in the separation between the lenses increases the Gouy phase shift between the modes. When the spacing is twice the focal length, the phase shift becomes very close to π and any incident LG mode is converted to the same mode with the opposite sign of l . This transformation is effectively a mirror inversion of the beam and has associated with it a reverse of the orbital angular momentum content. It should be noted here that a Dove prism achieves the same inversion without any assumption of diffraction free propagation (3.9). Recently, Molina-Terriza *et al.* (3.10, **Paper 3.6**) pointed out that on either side of a cylindrically focused LG mode, the sign of the vortex is reversed. As no angular momentum transfer can occur in free space, this emphasises that the angular momentum content is associated with the inclined phasefronts and not with the vortex itself. The phase distribution and transfer mechanism has been further analysed by Padgett and Allen (3.11, **Paper 3.7**).

Cylindrical lenses have also been used to impart orbital angular momentum to elliptical Gaussian beams. This produces an intensity profile which rotates on propagation (3.12). In such beams, the orbital angular momentum density is non-uniform and increases with increasing distance from the beam axis. Despite their very high total angular momentum, any resulting forces or torques are difficult to observe because of the presence of the gradient force, see section 4, which results from the beam profile.

The idea of mode transformation arising from the introduction of a phase delay between orthogonal components has also been exploited in a stressed fibre, where the resulting change in the guide index can introduce a phase delay between $\text{HG}_{1,0}$ and $\text{HG}_{0,1}$ modes, to again produce an LG mode (3.13).

At around the same time as the early work on the orbital angular momentum of LG modes, a number of groups were using computer-generated holograms for the production

of beams containing optical vortices. A hologram can be considered as a modified diffraction grating, where the first order diffracted beam has a specific phase and amplitude structure. The required form of the diffraction grating can be produced experimentally by interfering the light reflected from the desired object with a reference beam. Once photographically developed, the film is again placed in the reference beam and the resulting diffraction pattern gives a beam identical to that reflected from the original object. When the desired beam has a simple form, the first step in this process can be avoided as it is straightforward to calculate the interference pattern directly. If a calculated interference pattern is transferred to holographic film, or another recordable phase medium, a computer generated hologram or diffractive optic is produced. As with all holograms, and indeed diffraction gratings, the efficiency with which light can be diffracted into the chosen order is enhanced if the hologram is blazed. This modifies the phase of the transmitted light rather than just its amplitude. A perfect phase hologram is, in principle, 100% efficient and efficiencies in excess of 50% have been experimentally demonstrated.

The use of computer generated holograms for the generation of optical beams containing vortices was reported in 1990 by Bazhenov *et al.* (3.14, **Paper 3.8**) and in 1992 by Heckenberg *et al.* (3.15, **Paper 3.9**). As is now common practice, their designs were calculated for the two interfering beams meeting at an angle. This gives a diffraction grating appearance to the interference pattern, and results in the various diffraction orders appearing at differing angles, allowing easy selection of the desired beam. An alternative approach adopted by Heckenberg *et al.* (3.16) calculated the hologram for the case when the desired beam is focused. The result is that when the hologram is illuminated with a plane wave, the various diffraction orders focus to different axial positions, again allowing the desired diffraction order to be selected.

Holograms are now widely used to produce beams containing any manner of phase singularities for the studies of orbital angular momentum and related effects. Programmable spatial light modulators are now commercially available and can be configured to act as holograms that can be changed under software control. This allows virtually every aspect of the beam, including the orbital angular momentum, to be modified in real-time. In 1996 Soskin *et al.* (3.17, **Paper 3.10**) used computer generated holograms to create beams containing optical vortices and studied both vortex interaction and the effects of interference on the beam profile.

The spiral phase plate gives results similar to the computer generated hologram. It is an optical element with an optical thickness that increases with azimuthal angle such that, upon transmission, an incident quasi-plane wave emerges with a helical phasefront. It does in fact perform an identical operation to the ideal phase hologram although it requires sub-wavelength mechanical tolerance to ensure that the step height corresponds exactly to a whole number of optical wavelength. The first use of a spiral phase plate was by Beijersbergen *et al.* (3.18, **Paper 3.11**), but the extreme tolerances required in the optical regime meant that the quality of the resulting mode was below that which could be obtained using the cylindrical lens mode converter. In the mm-wave region of the spectrum, however, the required accuracy is obtainable and Turnbull *et al.* (3.19, **Paper 3.12**) successfully used this approach to generate LG modes using a waveplate fabricated from PTFE with a step height of several mm. This paper also includes a simple derivation based on refraction at the inclined surface of the waveplate to show that even a ray optic model predicts that the orbital angular momentum is indeed $l\hbar$ per photon. The same technique was later used by Courtial *et al.* (3.20, **Paper 6.6**) in their

experiments to demonstrate and measure precisely the rotational Doppler effect (see Section 6). A spiral phaseplate has also been used recently as one of the mirrors in a laser cavity, resulting in the formation of a helical mode within the laser cavity itself (3.21).

Although both the holograms and spiral phaseplates produce LG modes with a single l value, it should be appreciated that even when using an HG ($m = 0, n = 0$) mode as the input light, the resulting radial intensity distribution does not correspond to that of a single LG mode. The resulting beam is a superposition of LG modes with the same l -index but with a range of p -indices. Under optimum conditions approximately 90% of the energy can be contained in the $p = 0$ mode (3.22).

High-order Bessel beams also have helical wavefronts characterised by an $\exp(il\phi)$ phase term and so carry an orbital angular momentum of $l\hbar$ per photon. Such beams can be produced using computer generated holograms, or by passing an LG mode through a glass cone known as an axicon (3.23). Recently, in an experiment similar to those performed with LG modes (3.24, Paper 2.7), high-order Bessel beams have been used to set particles into orbital motion around the beam axis (3.25).

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An experiment to observe the intensity and phase structure of Laguerre–Gaussian laser modes

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We outline an easily reproduced experiment that allows the student to investigate the intensity and phase structure of transverse laser modes. In addition to discussing the usual Hermite–Gaussian laser modes we detail how Laguerre–Gaussian laser modes can be obtained by the direct conversion of the Hermite–Gaussian output. A Mach–Zehnder interferometer allows the phase structure of the Laguerre–Gaussian modes to be compared with the phase structure of a plane wave with the same frequency. The resulting interference patterns clearly illustrate the azimuthal phase dependence of the Laguerre–Gaussian modes, which is the origin of the orbital angular momentum associated with each of them. © 1996 American Association of Physics Teachers.

I. INTRODUCTION

Lasers form a key topic within an increasing number of undergraduate physics and optoelectronic degrees. This paper presents an easily reproduced experiment that allows the student to investigate the amplitude and phase structure of various transverse laser modes. In general, transverse laser modes are best described by a product of a Hermite polynomial and a Gaussian and are known as Hermite–Gaussian (HG) modes. However, in this paper we show how other modes can be generated and their phase distribution analyzed. Specifically, we present results relating to the Laguerre–Gaussian (LG) transverse laser modes. Laguerre polynomials are more frequently encountered within quantum mechanics as the radial term in the solution to Schrödinger's time-independent wave equation for a harmonic oscillator potential (e.g., the hydrogen atom problem). As in the quantum mechanical example the presence of an azimuthal phase term may be interpreted as indicating the presence of orbital angular momentum, hence the current interest in these unusual laser modes^{1–4}

II. LONGITUDINAL AND TRANSVERSE MODES WITHIN LASER OSCILLATORS

All lasers essentially consist of a gain material within which stimulated emission amplifies the light and an optical

resonator to provide the feedback needed to form an oscillator. The electromagnetic field within the laser resonator must satisfy a number of boundary conditions. In the steady state, there is the requirement that it reproduce itself in phase after one round trip of the laser cavity. This gives rise to the longitudinal mode structure of the laser. For a two-mirror linear cavity of optical length L , the operating wavelength of the laser, λ , must satisfy the equation

$$m(\lambda/2) = L, \quad (1)$$

where m is an integer.

The second boundary condition constrains the transverse nature of the electromagnetic field. In free space, there is a requirement that the electromagnetic field falls to zero away from the axis of the laser cavity. This gives rise to the transverse mode structure within the laser cavity and the observed intensity profile of the laser output. The analytical form of the allowed laser modes must satisfy these conditions and be a scalar solution to the paraxial limit of Maxwell's wave equation. (The paraxial limit assumes that the light diverges at only a small angle with respect to the optical axis of the system.⁵) This requires both the field amplitude and its derivative to fall to zero at a large distance from the axis of the cavity.

The rectangularly symmetric HG modes are described in part by the product of two-independent Hermite polynomi-

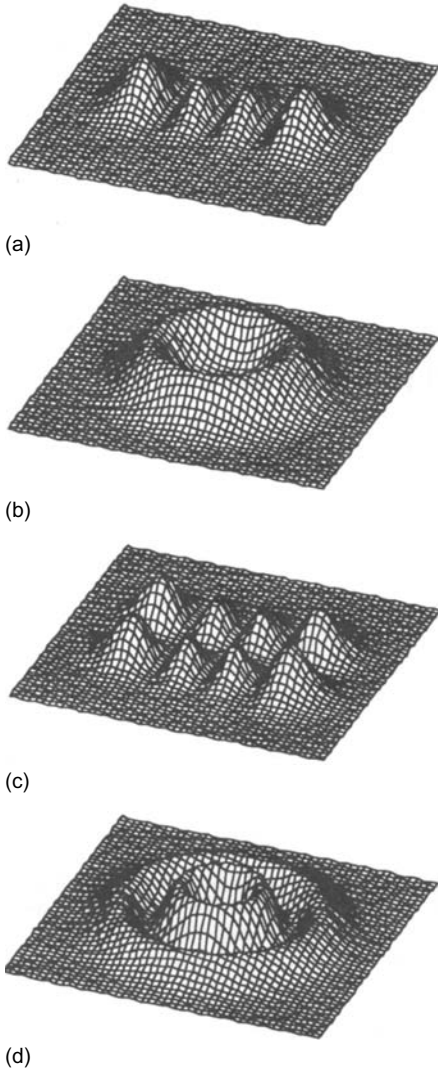


Fig. 1. Intensity ($|E|^2$) distribution of (a) a Hermite-Gaussian mode $m=3$, $n=0$, (b) a Laguerre-Gaussian mode $l=3$, $p=0$, (c) a Hermite-Gaussian mode $m=3$, $n=1$, and (d) a Laguerre-Gaussian mode $l=2$, $p=1$.

als, for the field distribution in the x and y directions, respectively. They are characterized by the integer subscripts m and n which give the order of the two polynomials. The values of m and n correspond to the number of nodes in the electromagnetic field. For example, the $HG_{3,1}$ mode has three field nodes in the x direction and one in the y direction [See Figs. 1(a) and 1(c)].

The circularly symmetric LG modes are similarly described by a single Laguerre polynomial with the superscript l and the subscript p . The index l gives the number of 2π cycles of phase in the azimuthal direction around the circumference of the mode, while $p+1$ gives the number of nodes across the radial field distribution. The LG_1^2 mode has a 4π

phase variation around the circumference of the mode and two radial nodes one of which is on axis. [See Figs. 1(b) and 1(d)].

As frequently encountered within quantum mechanics, it is the symmetry of the potential well and associated boundary conditions that determines the set of functions that most conveniently describes the eigenfunctions of the system. In the case of the simple harmonic oscillator, rectangular symmetry gives rise to HG eigenfunctions and circular symmetry results in LG eigenfunctions. In the case of a laser system, it would seem probable that given circular mirrors, the LG functions would provide the most accurate description of the transverse modes in a real laser, but this is not the case. Slight asymmetries in the laser cavity such as the inclusion of a Brewster window give rise to rectangular symmetry which results in a product of Hermite polynomials providing a more appropriate description. A full description with many examples of transverse laser modes can be found in Siegman's book.⁶ As an example of what the eigenfunctions look like, the field amplitudes, E , transverse to the direction of propagation for the $HG_{3,1}$ and LG_1^2 modes, by way of example, are given by

$$\begin{aligned} E(HG_{3,1}) &\propto \exp\left(-ik \frac{(x^2+y^2)}{2R}\right) \\ &\times \exp\left(\frac{-(x^2+y^2)}{\omega^2}\right) H_3\left(\frac{\sqrt{2}x}{\omega}\right) H_1\left(\frac{\sqrt{2}y}{\omega}\right) \psi(z), \\ E(LG_1^2) &\propto \exp\left(-ik \frac{r^2}{2R}\right) \exp\left(\frac{-r^2}{\omega^2}\right) \left(\sqrt{2} \frac{r}{\omega}\right)^2 \\ &\times L_1^2\left(\frac{2r^2}{\omega^2}\right) \exp(-2i\phi) \psi(z), \end{aligned} \quad (2)$$

where R is the radius of curvature of the near spherical wave front, k is the wave number of the electromagnetic wave, while x , y , and r are the transverse positions and radii within the beam, ϕ is the azimuthal angle within the beam, and ω is the beam radius at which the Gaussian term falls to $1/e$ of its on axis value. In both of the above expressions, the first factor relates to the phase change that results from the curvature of the wave front, and the factor $\psi(z)$ is the Gouy phase which we shall consider in detail later. The polynomials in these cases are

$$\begin{aligned} H_3\left(\frac{\sqrt{2}x}{\omega}\right) &= 8\left(\frac{\sqrt{2}x}{\omega}\right)^3 - 12\left(\frac{\sqrt{2}x}{\omega}\right), \\ H_1\left(\frac{\sqrt{2}y}{\omega}\right) &= 2\frac{\sqrt{2}y}{\omega} \end{aligned}$$

and

$$L_1^2\left(\frac{2r^2}{\omega^2}\right) = 3 - \frac{2r^2}{\omega^2}. \quad (3)$$

III. THE GENERATION OF HIGH-ORDER TRANSVERSE LASER MODES

In most uses of a laser, the multilobed or multiringed nature of a high-order transverse mode is undesirable. Steps are taken to force the laser to oscillate in the fundamental mode $m=0$, $n=0$, the amplitude of which has the form of a simple Gaussian centered on axis. As a laser will tend to oscillate in the transverse mode for which the losses are lowest, the fun-

damental mode is often preferentially selected by the inclusion of an aperture into the cavity. Similarly, if one of the higher-order HG modes is in fact required, then this is most simply achieved by inserting a cross wire into the laser cavity with the wires aligned with the nodes of the desired mode. However, obtaining a pure LG is not so easy because the azimuthal phase term can be either clockwise or anti-clockwise. The two modes have identical intensity distributions and a simple mask or aperture is therefore not sufficient to select one mode in preference to the other. Although a LG laser output has been reported, see for example the recent work by Harris *et al.*,⁷ the experimental difficulties are beyond what can be expected of a teaching laboratory and we show an alternative way in which such modes may be investigated.

Recent work by Beijersbergen *et al.*⁸ has demonstrated, following the work of Tamm and Weiss⁹ on low-order modes, that there is an extra-cavity way to obtain pure LG modes of any order. A mode converter consisting of two cylindrical lenses can transform the HG output of a conventional laser into the corresponding LG mode. In the experiment described here, we convert the HG output of a He-Ne laser into the desired LG mode and investigate the relative phase structure by superposing the LG mode on the HG mode derived from the same laser source. The resulting interference pattern allows the direct observation of the azimuthal phase dependence of the LG modes.

IV. THE OPERATING PRINCIPLE OF THE MODE CONVERTER

The HG and LG modes both form complete sets of solutions to the paraxial wave equation. Any arbitrary paraxial distribution can be described as a superposition of HG or LG terms with the appropriate weighting and phase factors. It follows that an LG mode can be described as a superposition of various HG modes weighted and phased accordingly, and vice versa. It has been shown^{1,10} surprisingly recently, that the relationship between the HG and LG modes is comparatively simple, and examples detailing how various LG modes can be synthesized from combinations of HG modes are presented in Ref. 8. Figure 2 shows how a $HG_{1,0}$ mode rotated at 45° to the x - y axis is equivalent to the sum of a $HG_{1,0}$ and $HG_{0,1}$ modes and how these two modes are related to the LG_0^1 mode. Specifically, the LG_0^1 can be formed by the superposition of the $HG_{1,0}$ and the $HG_{0,1}$ modes with a phase difference of $\pi/2$.

The propagation of laser beams with a HG or LG structure can be described in the usual language of Gaussian beams (see Ref. 5). The radius of the beam, ω , at which the Gaussian factor in the expression for the electric field falls to $1/e$ of its on axis value and the radius of curvature, R , of the near spherical wave fronts are given by,

$$\omega^2(z) = \frac{2(z_r^2 + z^2)}{kz_r},$$

$$R(z) = \frac{(z_r^2 + z^2)}{z}, \quad (4)$$

where the quantity z_r is the Rayleigh range and k is the wave number of the electromagnetic wave. The position $z=0$ corresponds to the smallest beam radius, and this position is

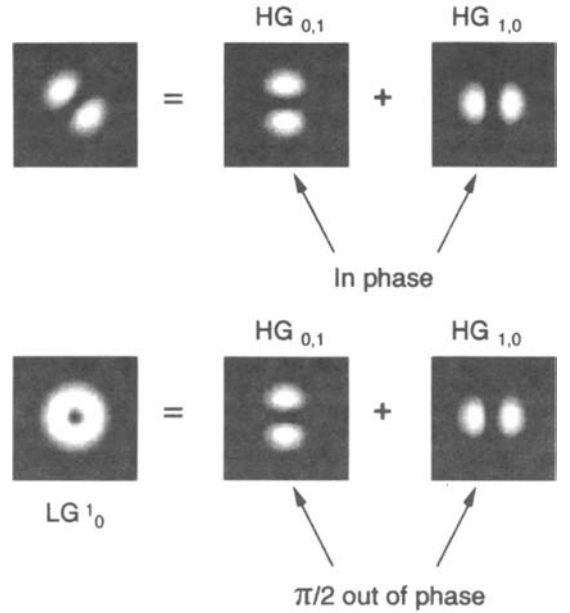


Fig. 2. The LG_0^1 mode can be formed by the superposition of the $HG_{1,0}$ and the $HG_{0,1}$ modes with a phase delay of $\pi/2$.

referred to as the beam waist. The radius of the beam waist, ω_0 , is related to the Rayleigh range by

$$\omega_0^2 = 2z_r/k. \quad (5)$$

In the vicinity of the beam waist, a Gaussian beam experiences a phase shift compared to that of a plane wave of the same frequency. This phase shift ψ is termed the Gouy phase.¹¹ For the simple case of a HG mode, described by mode indices m and n in the x and y directions respectively, and with identical Rayleigh ranges in the x - z and y - z planes, $z_{r(x-z)} = z_{r(y-z)}$, the Gouy phase is given by

$$\psi(z) = (n + m + 1) \arctan(z/z_r), \quad (6)$$

while for a LG mode the phase is written

$$\psi(z) = (2p + l + 1) \arctan\left(\frac{z}{z_r}\right), \quad (7)$$

where z is the distance along the axis from the beam waist in each case. If the Gaussian beam is focused by a cylindrical lens then the situation becomes more complex as the Rayleigh ranges in the x - z and y - z planes are not equal, $z_{r(x-z)} \neq z_{r(y-z)}$. Such a beam is termed an elliptical Gaussian beam and the corresponding Gouy phase shift for the $HG_{m,n}$ mode is given by

$$\psi(z) = \left(m + \frac{1}{2}\right) \arctan\left(\frac{z}{z_{r(x-z)}}\right) + \left(n + \frac{1}{2}\right) \arctan\left(\frac{z}{z_{r(y-z)}}\right). \quad (8)$$

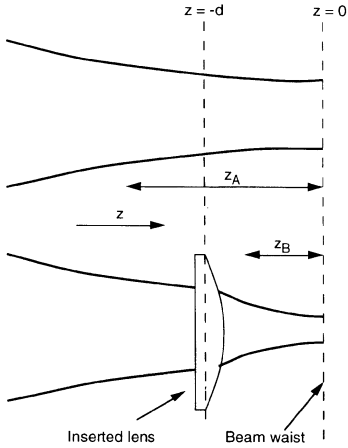


Fig. 3. A lens inserted into a Gaussian beam with a Rayleigh range z_A can be selected to change the Rayleigh range to z_B while leaving the position of the beam waist unaltered.

It is the Gouy phase shift in the presence of a cylindrical lens that forms the basis of the mode converter.

Consider a Gaussian beam traveling in the z direction with a Rayleigh range z_A and a beam waist at $z=0$. A lens inserted at $-d$ would change the radius of curvature of the beam and it proves possible to select a focal length which changes the Rayleigh range from z_A to z_B without affecting the position of the beam waist (see Fig. 3). A second, identical, lens positioned at $+d$ would change the Rayleigh range back to z_A . If the lenses are cylindrical, aligned with their axes parallel to the x direction, then only the Rayleigh range in the y - z plane is affected and consequently in the region between the lenses $z_{r(x-z)}$ is no longer equal to $z_{r(y-z)}$ and the beam waist is elliptical (see Fig. 4). The differing Rayleigh ranges in the x - z and y - z planes means that in transmission through the lens pair, the $HG_{m,n}$ and $HG_{n,m}$ ($m \neq n$) modes will undergo different Gouy phase shifts. For the generation of the LG_0^1 mode, we require that this phase difference be

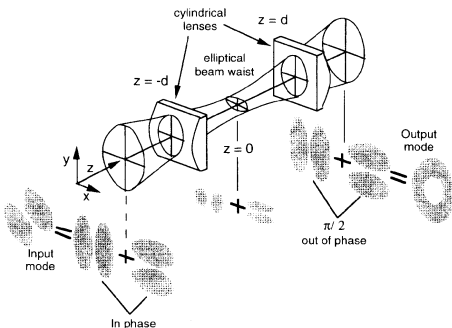


Fig. 4. The cylindrical lens mode converter. If the input $HG_{1,0}$ mode is oriented at 45° with respect to the cylinder axis of lens the mode is converted into the LG_0^1 mode.

$\pi/2$. Beijersbergen *et al.* show that this occurs when the separation, $2d$, of the cylindrical lenses of focal length f is given by

$$2d = f \sqrt{2}, \quad (9)$$

and the Rayleigh range, z_r , of the incident beam is given by

$$z_r = (1 + 1/\sqrt{2})f. \quad (10)$$

The correct Rayleigh range of the beam entering the mode converter is established using an additional spherical lens of an appropriate focal length, prior to the mode converter, positioned to form a beam waist at $z=0$. For a diverging input beam, with a wave-front radius of R_i and of radius ω , the required focal length, f_{in} , of the input lens is given by

$$\frac{1}{f_{in}} = \frac{\sqrt{2k\omega^2 z_r - 4z_r^2}}{k\omega^2 z_r} + \frac{1}{R_i}, \quad (11)$$

where k is the wave number of the light.

It can be shown that in addition to the transformation of the $HG_{1,0}$ mode, a mode-converter of the above design will transform any $HG_{m,n}$ mode, rotated at 45° to the axis of the lens, into the corresponding LG_p^1 mode with $l = |m - n|$ and $p = \min(m, n)$.

V. EXPERIMENTAL APPARATUS

The design of the mode converter used in this work is based on the simple equations given in the previous section and has the following specifications:

- 25.4 mm focal length cylindrical lenses (e.g., Newport CKX025¹²),
- 35.9 mm lens separation (between the principal planes)
- and an input beam with a Rayleigh range of 43.3 mm.

The lenses are mounted with their plane faces cemented to a 19 mm long tube to set the correct separation of their principal planes.

The optimum focal length for the input lens can be calculated from the diameter of the input beam and its wave-front curvature. However, in practice it is simpler to experiment with a number of different focal lengths and select the one that gives the best results. In our case, a single lens of 250 mm focal length is used to focus the output of the HeNe laser into the mode converter.

For the laser source, we use an unmounted He-Ne tube with a Brewster window and an external output coupler.¹³ By positioning the output coupler 100 mm from the Brewster window, an x - y translation stage carrying a crosswire (made from 10 μm diam tungsten wire) can be inserted into the laser cavity. With careful adjustment of the cross wire and alignment of the output coupler, the student can force the laser to oscillate in a variety of higher-order HG modes.

In order to observe the phase structure of the LG modes it was necessary to extend the experimental arrangement beyond that of the mode converter, as shown in Fig. 5. The experimental arrangement is based on a Mach-Zehnder interferometer, with the mode converter and input lens ($f=250$ mm) in one arm and a beam expansion telescope in the other. The purpose of the lens ($f=160$ mm) after the mode converter is to form a second beam waist outside the interferometer, so that the mode structure can be examined at various positions either side of a beam waist. The beam expansion telescope consists of two lenses ($f=40$ mm and $f=300$ mm) to give an expanded, collimated output beam. This beam is

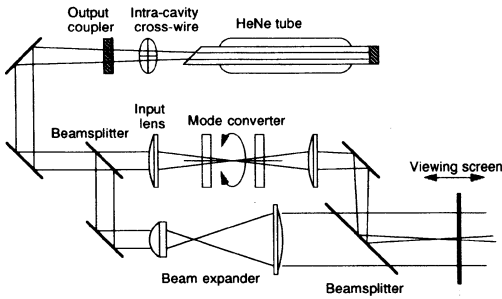


Fig. 5. The experimental arrangement for the observation of the intensity and phase structure of Laguerre-Gaussian transverse laser modes.

an expanded form of the input HG mode. Although having a Hermite Gaussian intensity profile, the large Rayleigh range now achieved for the mode results in a near-planar wave front that acts as a phase reference for the examination of the LG modes.

VI. EXPERIMENTAL OBSERVATION OF THE INTENSITY AND PHASE STRUCTURE OF LG MODES

Initial experiments can be completed with the beam expander arm blocked off. The results presented in Ref. 8 can be repeated; namely after selecting a high-order HG mode from the laser the student can examine the intensity distribution of the beam after it is transmitted by the mode converter. If the cylindrical lenses are rotated about the optical axis so that the axis of the cylinders are at 0° or 90° to the input HG mode then the transmitted light is also an HG mode. If, however, they are rotated to lie at 45° , the input $HG_{n,m}$ mode is transformed to the corresponding LG_p^l mode ($l = |m - n|$ and $p = \min(m, n)$).

After the various intensity patterns have been observed, the student can superimpose the expanded HG mode, aligned so that the LG mode falls within one of the greatly expanded lobes, and the resulting interference pattern can be observed. The expanded HG mode acts as a plane-wave reference and the observed fringes reveal the relative phase variations of the HG or LG mode in question.

Figure 6 shows the intensity distributions and interference patterns of a $HG_{3,0}$ mode and the corresponding LG_0^3 mode at various positions about the beam waist. These intensity distributions can be observed simply by placing a screen in the beam path. For the purpose of publication and student reports, permanent record of the intensity distributions can be obtained by directly exposing a CCD array interfaced to a framegrabbing card running on a desktop computer. Subsequently, the interferogram image can be printed or transferred into a variety of applications running on the same computer.

As can be seen from the figure, at the beam waist the 6π phase change around the circumference of the LG_0^3 mode gives rise to three dark radial fringes. Away from the beam waist, spiral fringes are observed. These arise from the combination of the radial phase variation due to wave-front curvature, and the tangential phase variation due to the azimuthal phase dependence. The sense of the spiral is reversed on either side of the waist in accordance with the curvature

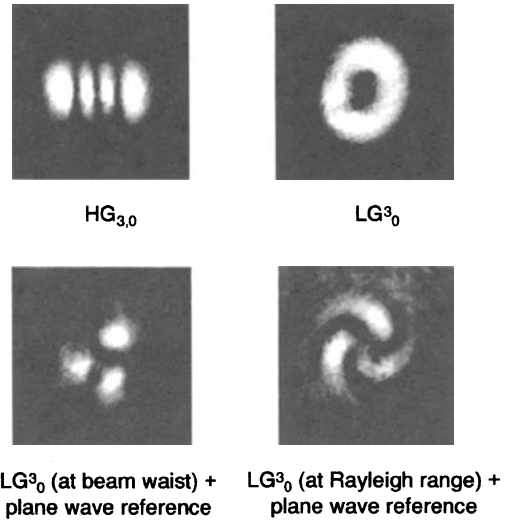


Fig. 6. Photographs of the Hermite-Gaussian $HG_{3,0}$ mode, the transformed Laguerre-Gaussian LG_0^3 mode and interferograms between the LG_0^3 mode and a plane wave.

of the wave fronts. In addition, the sense of the spirals can be reversed by changing the sense of the azimuthal phase variation. This is most readily achieved by rotating the mode converter about the optical axis by 90° (i.e., from $+45^\circ$ to -45° with respect to the input mode).

Figure 7 shows the intensity and interference patterns ob-

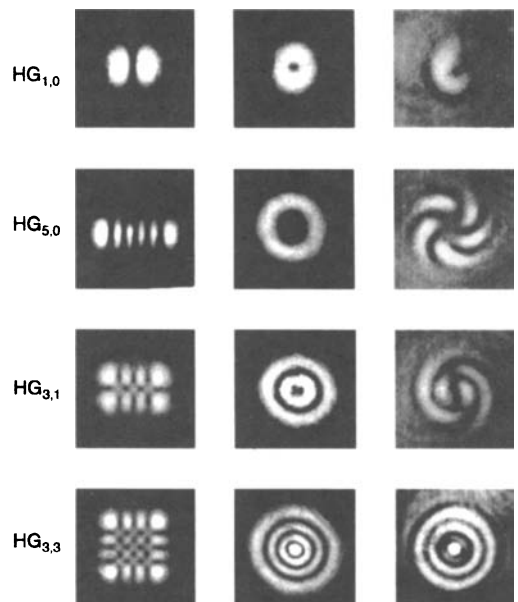


Fig. 7. Photographs of a number of Hermite-Gaussian laser modes, the corresponding Laguerre-Gaussian modes and interferograms between the Laguerre-Gaussian modes and a plane-wave reference.

tained for several higher-order HG and LG modes. In all cases the number of radial fringes is equal to l (i.e., $|m-n|$) and the number of radial nodes is equal to $p+1$ [i.e., $\min(m,n)+1$]. For the LG modes with multiple ring amplitudes (e.g., the LG_1^2 , which is obtained from the $HG_{3,1}$), the azimuthal position of the fringe maxima and minima are reversed between successive rings, corresponding to the change of phase of the Laguerre–Gaussian distribution. Note also that in the special case of $m=n$ there is an on axis intensity for the LG mode but no azimuthal phase dependence and hence the fringes are circular and not spiral. The radii of the fringes can be calculated from the wave-front curvature of the beam and are analogous to Newton's rings.

VII. CONCLUSIONS

This paper presents an easily reproduced experiment with which the student can investigate the intensity and phase structure of various transverse laser modes. In addition to the standard Hermite–Gaussian laser modes we detail how Laguerre–Gaussian laser modes can be obtained by the direct conversion of the Hermite–Gaussian laser output using a mode converter utilizing the Gouy phase shift in the region of an elliptical beam waist. Using a Mach–Zehnder interferometer, the phase structure of the Laguerre–Gaussian modes can be compared with that of a plane wave of the same frequency.

One of the current sources of interest in the LG laser modes is that they are predicted to carry orbital angular momentum in addition to the spin momentum associated with the polarization state of the photons.¹ By superimposing the LG mode with a plane-wave reference, the resulting interference patterns clearly illustrate the azimuthal phase dependence of the Laguerre–Gaussian modes which is the origin of the orbital angular momentum associated with these modes.

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Temporal and interference fringe analysis of TEM₀₁* laser modes

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The properties of TEM₀₁* doughnut modes have been examined by frequency analysis and by two-beam interference. Interpretation in terms of an evolving helix, made up of two orthogonal TEM₀₁ modes of different frequency, is supported by computer simulation of fringe patterns. These patterns are shown to correspond closely with photographic recordings; the implications of the phenomena for the crossed-beam technique in laser velocimetry are outlined. Finally, the possibility of developing a beam of pure helical cophasal surface is discussed and its interference patterns analyzed.

INTRODUCTION

In previous publications^{1,2} we have investigated two-beam interference patterns that are due to TEM₀₀ and TEM₀₁ laser modes of different relative power. It was suggested that the TEM₀₁* doughnut mode, composed of two TEM₀₁ modes in phase and space quadrature, may possess a cophasal surface of helical form. However, in order to explain the observed interference patterns derived from several high-gain Ar⁺- and Kr⁺-laser lines, it was necessary to suppose that the observed doughnut mode was an evolving helix, which switched rapidly back and forth between a left-hand and a right-hand character. In our earlier papers the interference theory of these modes was performed analytically in terms of phase contours; we now present new results on the temporal behavior of these modes that explain the previous assumptions of helical hand-switching. These results are incorporated into computer calculations and graphical constructions of the fringe profiles, which are found to be in good agreement with experimental photographic recordings. The calculations have been carried out both for evolving helices of the TEM₀₁* mode alone and for various admixtures of the TEM₀₀ mode. In the final section we propose a simple means of producing a stationary helical wave of either hand and draw attention to unusual interference properties that it should possess.

It may be remarked that interest in these modes is not confined solely to the unusual character of the wave surfaces and their interference patterns. These modes also have considerable practical importance in photon-correlation spectroscopy and in crossed-beam fringe velocimetry. In the first case considerable reduction in the strength of fluctuations may occur, leading to a reduced signal-to-noise ratio; this important problem is discussed in detail, both experimentally and theoretically, in Ref. 3. In velocimetry the reduced fringe visibility and changing phase may introduce considerable distortion of experimental data; this is outlined in the present paper. These effects are of course particularly troublesome if such modes are additionally present in a nominally TEM₀₀ laser. In the past such effects were widely found, particularly when high-gain ion lasers were used, but they have not hitherto been explained or evaluated in detail.

TEMPORAL MEASUREMENTS

The source employed in this work was a Kr⁺-ion laser oscillating at 476 nm, with an intracavity étalon for longitudinal-mode selection. It must be emphasized, however, that the phenomena are quite general for high-gain laser lines. Similar two-beam interference patterns have been seen in many lines from Ar⁺- and Kr⁺-ion lasers and have also been observed in the interference patterns that are due to a doughnut mode from a CO₂ laser operating at 10.6 μm, viewed with a thermal imager. For the present work the laser output frequency was analyzed in the 10–500-MHz range with a 20-cm scanning, confocal, Fabry–Perot interferometer. If the whole mode were allowed to enter the interferometer, it was found to consist of a single frequency when the laser was emitting a TEM₀₀ mode; by adjustment of étalon tilt and discharge current a pure TEM₀₁* mode, with characteristic dark center, could be obtained and was found to consist of two frequencies ω_2 and ω_3 separated by 128 MHz. The spatial distribution of these frequencies was investigated by isolating a small fraction of the mode in the (ρ, θ) plane perpendicular to the propagation direction. This was accomplished by scanning a small aperture in this plane and subjecting the light that passed through it to spectral analysis. Frequency ω_2 was found to be concentrated near the horizontal and ω_3 near the vertical axis, qualitatively consistent with spatial distributions of the kind $\rho^2 \exp(-\rho^2/\sigma^2) \cos^2 \theta$ and $\rho^2 \exp(-\rho^2/\sigma^2) \sin^2 \theta$ to be expected of TEM₀₁ modes.⁴

These observations were confirmed by beat-frequency analysis in the region between dc and 1.8 GHz by using a small silicon avalanche photodiode coupled to a Tektronix 7L12 spectrum analyzer. The magnitude of the beat-frequency signal at $|\omega_3 - \omega_2|$ (equal to 128 MHz) maximized along the $\theta = \pm 45^\circ$ directions and was zero along the vertical and horizontal axes, distributed qualitatively as $\rho^2 \exp(-\rho^2/\sigma^2) \sin 2\theta$. We conclude from the spectral data that the TEM₀₁* mode does indeed result from the superposition of two TEM₀₁ modes in space quadrature but that in our case they are non-degenerate in frequency, being adjacent longitudinal modes of the laser resonator whose length L is 117 cm ($c/2L = 128$ MHz). There appears to be no fundamental reason why both

modes should not be degenerate; in the present case the degeneracy is almost certainly lifted by cavity asymmetry introduced by the Brewster windows, dispersive prism, etc. The fact that the polarization direction, defined by the Brewster windows and the prism, is vertical and coincident with the axis of one of the TEM₀₁ modes lends support to this notion.

A consequence of the different frequencies of the two TEM₀₁ modes is that their resultant evolves from a left- to right-handed helix and back again via intermediate TEM₀₁ modes orientated along the bisectors of the ω_2 and ω_3 modes; the evolution repeats at 128 MHz. This fact explains the earlier assumption of an evolving helical mode of changing hand that was required to give the observed interference patterns. It is straightforward to show that a simple helix is equivalent to two TEM₀₁ modes of the same frequency in phase and space quadrature. More generally, two orthogonal TEM₀₁ modes of the same frequency form an equivalent basis set to two helical modes of opposite hand, also at the same frequency.

FRINGE-PATTERN FORMULATION

Calculations of fringe intensity are carried out with the coordinate basis shown in Fig. 1. The centers of the two interfering beams are at $(\pm R, 0)$, and the two beams are propagating into the plane of the paper to intersect at an angle 2α . We consider the general case in which the beams are composed of both a Gaussian (TEM₀₀) mode at frequency ω_0 and a doughnut (TEM₀₁*) mode oscillating initially at a single frequency ω_1 , i.e., operating as a pure helix. Then the expression for the electric field at any point in the plane is given by

$$E(\rho, \theta, t) = C[\exp(-\rho_1^2/2\sigma^2)\sin(\omega_0 t - \phi_1) + \exp(-\rho_2^2/2\sigma^2)\sin(\omega_0 t + \phi_2) + (\rho_1/\sigma)\exp(-\rho_1^2/2\sigma^2)\sin(\omega_1 t \pm \theta_1 - \phi_1) + (\rho_2/\sigma)\exp(-\rho_2^2/2\sigma^2)\sin(\omega_1 t + \theta_2 + \phi_2), \quad (1)$$

where the term within brackets represents the Gaussian contribution and the remainder the helical wave; we have used the fact that the phase of a helical wave is equal to the azimuthal angle θ . C is a constant representing the relative TEM₀₀ field strength, and σ describes the mode radius. In the third term, the plus applies when the helices are uninverted with respect to each other, the minus when they are inverted (Figs. 1b and 1c). The phase angles ϕ_i arise from the

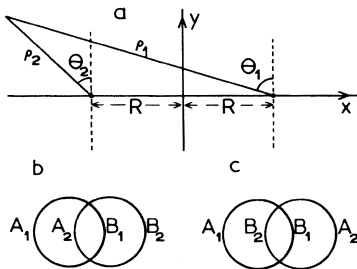


Fig. 1. (a) Coordinate system used for fringe intensity calculations. The angles θ_1 and θ_2 are constrained within the limits $-(\pi/2) < \theta_1 \leq (3\pi/2)$ and $-(3\pi/2) < \theta_2 \leq (\pi/2)$ chosen to simplify the analysis. (b) Uninverted beams, (c) inverted beams, both shown displaced sideways.

inclination α of the beams to the (ρ, θ) plane and are given by $(2\pi/\lambda)\sin \alpha \times \rho_i \sin \theta_i$ ($i = 1, 2$). Since the beams are inclined in opposite directions, the ϕ_i appear in Eq. (1) with opposite signs. The intensity arising from the field of Eq. (1) is obtained by taking the square modulus and time averaging in the usual way, with the result that

$$I \propto \frac{1}{2} \sum_{i=1}^2 (a_i^2 + b_i^2) + a_1 a_2 \cos(\phi_1 + \phi_2) + b_1 b_2 \cos(\phi_1 + \phi_2 \mp \theta_1 - \theta_2), \quad (2)$$

where

$$a_i = C \exp(-\rho_i^2/2\sigma^2) \quad i = 1, 2, \\ b_i = (\rho_i/\sigma) \exp(-\rho_i^2/2\sigma^2) \quad i = 1, 2.$$

This is the result for an unchanging helix. Our previous observations show that the helix switches hand at the intermode frequency $|\omega_2 - \omega_3|$ of 128 MHz. Writing the complete expression for this case shows that the third term in expression (2) should be replaced with

$$\frac{1}{2} b_1 b_2 [\cos(\phi_1 + \phi_2 \mp \theta_2 - \theta_1) + \cos(\phi_1 + \phi_2 \pm \theta_1 + \theta_2)].$$

Under these circumstances the intensity becomes

$$I \propto \frac{1}{2} \sum_{i=1}^2 (a_i^2 + b_i^2) + [a_1 a_2 + b_1 b_2 \cos(\theta_2 \pm \theta_1)] \cos(\phi_1 + \phi_2). \quad (3)$$

COMPUTATIONAL PROCEDURE

A Cartesian basis set was chosen that is related to the polar coordinates of Fig. 1 by

$$x = (-1)^{i+1} R - \rho_i \sin \theta_i, \quad i = 1, 2, \\ y = \rho_i \cos \theta_i, \quad i = 1, 2.$$

A program to display $I(x, y)$ was written in BASIC by using the Hewlett-Packard 9800 Series calculator and plotter. For fixed $y = y_0$, $I(x, y)$ was computed at a series of points in x differing by Δx , and these points were plotted to give a vertical section through the surface $I(x, y)$ at $y = y_0$; y_0 was then incremented to $y_0 + \Delta y$, and the computation was repeated. The results were plotted with a vertical shift on the chart of $K \times \Delta y$ ($K < 1$) and a horizontal stagger s . The process was repeated until the whole pattern was built up from a series of vertical sections to give the illusion of an isometric projection of the fringe pattern, as shown in Figs. 2a–2d. For small s , the apparent incidence angle of view is $\cos^{-1} K$. The input data to the program were the limits in x and y and the respective increments Δx and Δy , K and s , and the fringe data C , σ , R , and $(2\pi/\lambda)\sin \alpha$. In the examples shown, the centers of the beams are marked if they are separated; a scale of σ is also provided. $(2\pi/\lambda)\sin \alpha$ was chosen to give 20 fringes across the diagram, and K was 0.8, so that the apparent incidence angle was 37° .

The computations for various configurations may be compared with the experimentally obtained interferograms of Figs. 3a–3f. These photographs were obtained by two-beam interference, which provides a useful examination of spatial coherence.^{5,6} The inclusion of an intracavity étalon in the laser permitted longitudinal-mode selection and some degree of tuning of the relative intensity of the two TEM₀₁ modes.

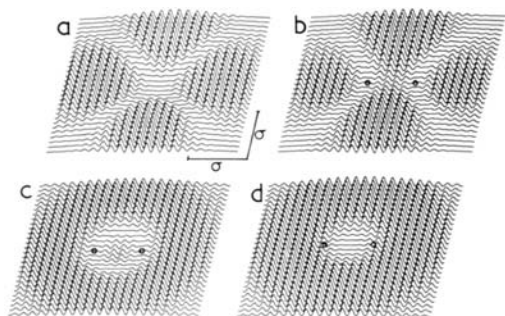


Fig. 2. Computer reconstruction of two-beam interferograms with a TEM_{01}^* laser mode: (a) beams inverted (i.e., Fig. 1c) but not displaced, (b) beams inverted and displaced horizontally, (c) beams uninverted and displaced horizontally (i.e., Fig. 1b); (d) like (c) but with the addition of 14% TEM_{00} mode. Note the half-cycle displacements in the fringes in different regions; see text for details of the perspective. The centers of the separated beams are marked by circles.

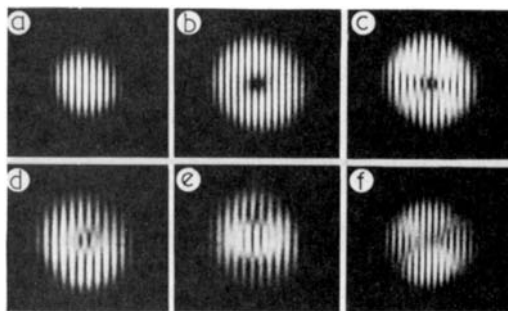


Fig. 3. Two-beam interference fringes. The top row shows the beams overlapped exactly without displacement: (a) Gaussian TEM_{00} mode, (b) doughnut TEM_{01}^* mode uninverted (i.e., Fig. 1b), (c) doughnut TEM_{01}^* mode inverted (i.e., Fig. 1c). The lower row shows the effect of lateral displacement of the beams: (d) doughnut TEM_{01}^* mode uninverted and displaced horizontally, (e) doughnut TEM_{01}^* mode uninverted and displaced vertically, (f) doughnut TEM_{01}^* mode inverted and displaced horizontally. Note in particular the half-cycle changes on the vertical fringes and compare them with Fig. 2.

Details of the procedure for photographing the fringe patterns have been described in Refs. 1 and 2. In essence, the original laser beam was divided at a prism beam splitter, and the two beams were enlarged and recombined on a translucent screen for ease of examination. The angle between the beams and the degree of overlap could be varied in both the inverted and the uninverted arrangements. The clearest fringe patterns, as might be expected, were obtained with the two TEM_{01} modes of nearly equal intensity. In these two figures the corresponding theoretical and experimental recordings are those of Figs. 2a and 3c, 2c and 3d, and 2b and 3f, respectively. It is clear that there is good correspondence between the appropriate pairs of diagrams; the fringe-smearing region where the phase difference between helices approaches $\pm\pi/2$ is clearly reproduced in Fig. 2 as an area of constant nonzero intensity separating regions in which the fringes are of opposite phase. Figure 2d illustrates that addition of TEM_{00} mode contracts the phase-changed circle of Fig. 2c; in contrast, for the inverted case it pushes the limbs of the phase-change cross

in Fig. 2a apart vertically. If the helices are separated to the extent shown in Fig. 2b, the addition of the same amount of TEM_{00} power ($C = 0.4$) used in these calculations gives rise to a pattern similar to that of Fig. 2a.

The precautions needed in using the crossed-beam technique in laser velocimetry have been outlined in Refs. 1 and 2 and are graphically illustrated in Figs. 2 and 3. The problem is readily appreciated by considering a small particle traversing the fringes shown in these figures. If the beams are recombined in an uninverted manner, then at the region of exact overlap (e.g., Fig. 3b), there is at least no change of phase, although the beam profile varies with the particular trajectory and is far from Gaussian. However, even in this uninverted case, there are changes of phase away from exact overlap (Fig. 3d); on the other hand, if the beams are inverted and recombined, phase changes occur throughout (Figs. 3c and 3f). In practice it is unlikely that such a pure doughnut mode would pass unnoticed. Much more insidious is the possibility of a residual TEM_{00} mode's filling in the hole of the TEM_{01}^* mode to give an apparently smooth Gaussian-like profile. This may be illustrated by the example with as low as 14% of the power in the TEM_{00} mode. At first sight it might be thought that, with 86% of the power in the TEM_{01}^* mode, a beam would appear clearly to be a doughnut; in fact, the central intensity is as much as 37% of the maximum, and, given the inherent nonlinearity of the eye (and photographic emulsion) particularly at high intensity, this central minimum might well pass unnoticed, with the beam being classed as Gaussian. Indeed, several workers have commented on the occasional sudden enlargement of a beam as it snaps into a larger, but apparently Gaussian, mode. It can now be recognized that this is almost certainly the growth, as the laser gain conditions become favorable, of a TEM_{01}^* mode around the lowest-order mode. Techniques of laser velocimetry employing the Doppler difference or fringes technique usually rely on accurate measurement of the time intervals between fringe peaks (for strong scattering) or on accumulating an intensity-correlation function (most particularly for weak scattering). Both methods rely on accurate knowledge of the fringe geometry; in the latter case, for work of the highest accuracy, reliable information is needed on the beam profile, usually considered to be Gaussian (indeed, this assumption, that all trajectories of particles through the beam give rise to Gaussian scattering profiles of equivalent form, is usually fundamental). Obviously the distortion of the fringe patterns introduced by higher-order laser modes must introduce considerable loss of precision and speed. The conclusion must be that stringent precautions should be taken when two-beam fringe velocimetry is used to ensure that the laser beam is in the lowest-order TEM_{00} mode. The precautions should include, at least for high-gain lasers, adjustment of an internal cavity aperture and, most importantly, detailed optical examination of the fringes under the actual conditions of use.

GENERATION OF HELICAL WAVE OF CONSTANT HAND

In an earlier section of this paper it was shown that the TEM_{01}^* mode described in our experiments evolved back and forth between left- and right-hand helices since the two constituent TEM_{01} modes were oscillating at different frequencies. In Fig. 4 we show an arrangement designed to produce

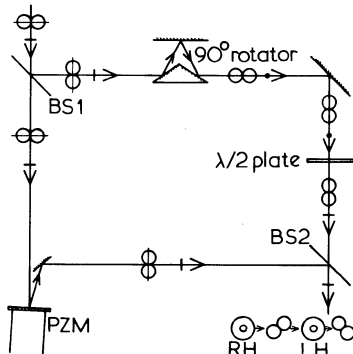


Fig. 4. Optical arrangement to produce a beam of pure helical cophasal form from a beam initially in a two-spot TEM_{01} mode of single frequency. The two paths between the beam splitters BS1 and BS2 should be arranged to be approximately equal. The axis of the two-spot mode lies in the plane of the paper when the modes are shown normal to the beam and is normal to the plane of the paper when the mode axis lies along the beam. The optical path and phase between the two beams may be varied by the piezoadjustable mirror at PZM. As the phase is changed the beam varies between a left-hand and a right-hand helix, as shown.

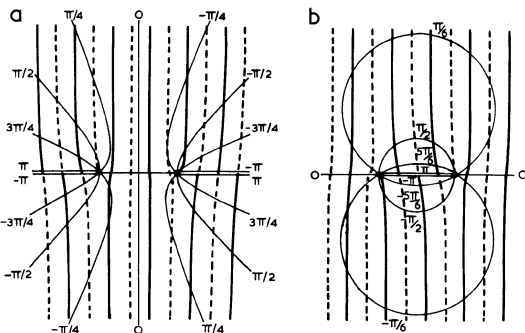


Fig. 5. Phase-difference contour maps (thin lines) and fringe patterns (thick lines, continuous and dashed) for two-beam interference of a true helical TEM_{01}^* laser mode: (a) inverted and separated beams, (b) uninverted and separated beams. The contours are labeled in phase difference.

a helical TEM_{01}^* mode of either hand. A single-mode linearly polarized laser is forced to oscillate on a TEM_{01} mode by inclusion of a pin within its resonator. Its output is fed into a Mach-Zehnder interferometer having in one arm an optical beam rotator, such as a Dove prism, so that the two beams are in space quadrature. Collinearity of output polarization is ensured by an appropriately orientated half-wave plate, and fine length tuning of one of the interferometer arms (e.g., by piezoelectric means) is provided so that the desired phase relation between the two TEM_{01} beams may be produced. As this phase relationship is varied, it is readily seen that the

resultant mode moves through a series of patterns. Starting in phase to produce a pure right-hand helix, a $\pi/2$ change produces a two-lobe distribution (with its axis lying between the axes of the two primary beams). A further $\pi/2$ change gives a left-hand helix, and another $\pi/2$ change produces a two-lobe distribution with its axis at right angles to the previous one. A final $\pi/2$ displacement, making 2π , brings the mode back to the original right-hand helix.

The fringe patterns to be expected from such a single-hand helical mode may be analyzed by recourse to the phase-difference diagram method developed in Ref. 1. In previous work calculations were performed modulo π since hand-switching phenomena were in evidence; for a single-hand helix the effects of the multivalued nature of the phase-difference surface must be included. Singularities called branch points⁷ arise at the centers of the helical modes, joined by a discontinuity in the surface, which is a branch line. Figure 5 shows the phase-difference contour maps for both uninverted and inverted displaced TEM_{01}^* modes with predicted fringe patterns overlaid. Branch lines are shown as double (where there is a cliff of altitude 2π), arranged to contract when a TEM_{00} mode is mixed in. The fringes are drawn continuous and dashed to distinguish between adjacent order numbers; the order number changes by one as a branch line is crossed, corresponding to the 2π phase change. A fringe terminates on each of the branch points, but, since these points are at zero intensity, the fringe will not terminate abruptly. Investigations of single-hand interference patterns will be the subject of a future publication.

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Bistability and optical switching of spatial patterns in a laser

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Bistability of the transverse configuration of the optical field is found for a helium–neon laser oscillating in the TEM₀₁* hybrid optical-resonator mode. Optically induced switching between the mirror-symmetric field patterns occurs when a fraction of the spatially rotated output field is reinjected into the laser resonator. These phenomena are described theoretically with laser equations modeling the mutual interaction of the two first-order transverse modes of the optical resonator. The experimentally observed bistability and switching of the field patterns is in full qualitative agreement with the theoretical results.

1. INTRODUCTION

The formation and competition of radiation patterns in nonlinear-optical systems is one manifestation of self-organization in nonequilibrium systems with spatial degrees of freedom.¹ In the case of transverse field patterns in nonlinear-optical resonators, the transverse variation of amplitude and phase of an outcoupled light field can yield complete information on the nature of the radiation patterns formed inside the cavity. Conversely, the conditions of pattern formation in nonlinear-optical resonators may be manipulated by injecting field patterns into the resonator.

For the interaction of nearly degenerate transverse modes of the optical field in nonlinear resonators, the occurrence of symmetry breaking and Turing instabilities and the formation of stationary transverse radiation patterns have been analyzed theoretically.^{2,3} It can be concluded from these studies, and also from one early publication,⁴ that lasers operating in a state of cooperative frequency locking³ of transverse resonator modes may show bistability or multistability, in the sense that it is possible to have stable laser output of any pattern from among a discrete set of different transverse field patterns for the same operating conditions.

This paper, for the first time to our knowledge, gives experimental evidence for the bistability of a transverse field pattern of a laser. Moreover, optically induced switching between the two stable field configurations is demonstrated. The experiment uses a laser that emits in the TEM₀₁* hybrid mode, i.e., in a superposition of TEM₁₀ and TEM₀₁ Gauss–Hermite optical-resonator modes⁵ of equal optical frequency that maintain a constant optical phase difference of either $+\pi/2$ or $-\pi/2$.⁶ It has been noted that these two values of optical phase difference lead to two physically different transverse field patterns that are mirror images of each other.⁷ It is shown here that laser oscillation in the TEM₀₁* hybrid mode is in fact bistable in these two field patterns. Hysteresis and switching between the stable states is observed when the transversally rotated output field pattern of the laser is reinjected into the laser resonator with a changing optical phase.

The experimental scheme is shown in Fig. 1. A single-

longitudinal-mode helium–neon laser operates in a stationary TEM₀₁* hybrid-mode configuration.⁷ The gain medium of this laser is inhomogeneously (Doppler) broadened; the optical resonator is of the Fabry–Perot type. The laser output is split into three beams. One beam is used for viewing the TEM₀₁* emission with its annular, rotationally symmetric intensity pattern. Another beam passes through two cylindrical lenses; this arrangement allows discrimination between the two possible spatial field configurations of the TEM₀₁* laser that have identical transverse intensity distributions (see below). A third beam passes through a combination of reflectors that has the property of rotating a given field or intensity pattern by 90° in the transverse plane. This beam is matched to the laser resonator mode with respect to polarization and beam parameters and is attenuated and fed back into the laser with a phase difference determined by the optical path length.

2. THEORETICAL MODEL

In order to describe the main features of the system, a simplified theoretical model is used that assumes a traveling-wave laser with a homogeneously broadened gain medium and, corresponding to the experimental situation, with a resonator geometry that supports only equally polarized TEM₁₀ and TEM₀₁ Gauss–Hermite transverse modes. When beam divergence and the curvature of wave fronts are neglected, the resulting optical field $\mathbf{E}(\mathbf{r}, t)$ inside the cavity is written as

$$\mathbf{E}(\mathbf{r}, t) = \hat{e}_p N \exp[-(x^2 + y^2)/w^2](A + A^*) \\ A = [F_{10}(x/w) + F_{01}(y/w)] \exp[i(\omega t - kz + \phi_0)]. \quad (1)$$

The transverse distribution of intensity $I(x, y)$ is then given by

$$I(x, y) = K \exp[-2(x^2 + y^2)/w^2] \\ \times [|F_{10}|^2(x^2/w^2) + |F_{01}|^2(y^2/w^2) + (F_{10}F_{01}^* + F_{10}^*F_{01})(xy/w^2)]. \quad (2)$$

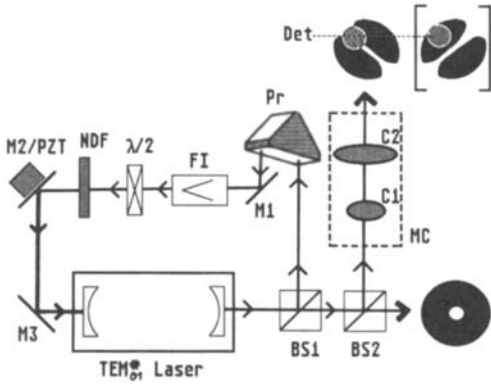


Fig. 1. Experimental setup: BS1, BS2, beam splitters; C1, C2, cylindrical lenses, forming the transverse-mode converter MC; Det, detector monitoring the local intensity in the bistable transverse intensity pattern—the detector position coincides with the nodal line of one pattern; Pr, rectangular reflecting prism—the plane of reflection of Pr is rotated out of the drawing plane by 45°; FI, Faraday isolator (optical diode); $\lambda/2$, $\lambda/2$ retardation plate; NDF, neutral density filter; PZT, piezoceramic transducer; M1, M2, M3, mirrors. For simplicity, mode matching and beam expanding lenses are not shown.

Here the unit vector $\hat{e}_p$ denotes the polarization direction, N and K are real constants, ϕ_0 is an arbitrary phase, z is the longitudinal coordinate, ω is the optical frequency, k denotes the modulus of the wave vector, w scales the transverse dimensions of the beam, and F_{10} and F_{01} are normalized complex field amplitudes of the TEM_{10} and TEM_{01} modes, respectively.

With the modal decomposition performed in Eqs. (1), one may now derive from the Maxwell–Bloch laser equations a system of coupled-mode equations³ for the field amplitudes F_{10} and F_{01} and for the atomic variables of a homogeneously broadened two-level medium. For the case when the feedback of a fraction of the rotated output field pattern is taken into account in the evolution equations for F_{10} and F_{01} , and assuming a cylindrically symmetric pumping profile, resonant tuning of the two first-order transverse modes, and negligible excitation of other transverse modes of field and polarization, the laser equations are of the form

$$\frac{\partial F_{10}}{\partial t} = -\kappa[F_{10} + \beta \exp(i\delta')F_{01} - 2CP_{10}], \quad (3a)$$

$$\frac{\partial F_{01}}{\partial t} = -\kappa[F_{01} - \beta \exp(i\delta')F_{10} - 2CP_{01}], \quad (3b)$$

$$\frac{\partial P_{10}}{\partial t} = -\gamma_{\perp}(P_{10} - \frac{1}{2}F_{10}D_0 - \frac{1}{4}F_{10}D_1 - \frac{1}{4}F_{01}D_2), \quad (3c)$$

$$\frac{\partial P_{01}}{\partial t} = -\gamma_{\perp}(P_{01} - \frac{1}{2}F_{01}D_0 - \frac{1}{4}F_{01}D_1 - \frac{1}{4}F_{10}D_2), \quad (3d)$$

$$\frac{\partial D_0}{\partial t} = -\gamma_{\parallel}[D_0 - 2\sqrt{\pi} + \frac{1}{4}(F_{10}P_{10}^* + F_{10}^*P_{10} + F_{01}P_{01}^* + F_{01}^*P_{01})], \quad (3e)$$

$$\frac{\partial D_1}{\partial t} = -\gamma_{\parallel}[D_1 + \frac{1}{4}(F_{10}P_{10}^* + F_{10}^*P_{10} - F_{01}P_{01}^* - F_{01}^*P_{01})], \quad (3f)$$

$$\frac{\partial D_2}{\partial t} = -\gamma_{\parallel}[D_2 + \frac{1}{4}(F_{10}P_{01}^* + F_{10}^*P_{01} + F_{01}P_{10}^* - F_{01}^*P_{10})]. \quad (3g)$$

Here $1/\kappa$ denotes the time constant of the laser resonator, β^2 is the ratio of feedback intensity to output intensity, δ' is the phase difference between the feedback field and cavity field, and C is the pumping parameter. P_{10} and P_{01} and $\gamma_{\perp}$ are the two normalized mode amplitudes and the decay rate of the atomic polarization, respectively. D_0 , D_1 , and D_2 denote spatial modes of the normalized atomic inversion $D(x, y)$ (decay rate $\gamma_{\parallel}$) and its possible azimuthal variation that is due to the coupling to the field and polarization modes:

$$\begin{bmatrix} D_0 \\ D_1 \\ D_2 \end{bmatrix} = \frac{2}{\pi w^2} \int dx dy \times D(x, y) \exp[-2(x^2 + y^2)/w^2] \begin{bmatrix} \sqrt{\pi}(x^2 + y^2) \\ \sqrt{2}(x^2 - y^2) \\ 2\sqrt{2}xy \end{bmatrix}. \quad (3h)$$

An analytical examination of Eqs. (3a)–(3g) in the limit of adiabatic elimination of the atomic variables shows that above laser threshold and for $\beta = 0$ (i.e., without feedback) the solution of the complete system of laser equations (3a)–(3g) is always stationary and bistable, with relations between the field amplitudes given by $F_{01} = +iF_{10}$ or $F_{01} = -iF_{10}$. This implies that for stable laser oscillation in frequency-degenerate TEM_{10} and TEM_{01} modes, both mode fields are locked in phase quadrature, a configuration that is known as the TEM_{01}^* hybrid mode.⁶ Numerical solutions of the laser equations (3a)–(3g) outside the adiabatic limit show that these characteristics of the bistable solutions persist for a broad range of typical operation parameters and that they do not change substantially for small perturbations of the cylindrical symmetry of the laser resonator.⁷

The two field configurations that are found as stationary solutions of Eqs. (3a)–(3g) are identical in their intensity patterns, which are rotationally symmetric in traveling and standing waves. With the use of Eqs. (1) it can be shown, however, that the equiphase surfaces of the traveling-wave fields are helices that are mirror images of each other. This corresponds to a breaking of space-inversion symmetry that makes the two field patterns physically distinguishable (see below). It may be noted that in an earlier work the presence of helical components in the output field of a laser was inferred from an analysis of interference patterns.⁸ Helical equiphase surfaces were also found in a recent study of optical vortices.⁹

The effect of the perturbation of the laser caused by feedback of the rotated output field pattern (see Fig. 1) is best shown in a new mode system that is adapted to the two steady-state field patterns of the TEM_{01}^* laser. With the mode amplitudes $F_+ = 2^{-(1/2)}(F_{10} + iF_{01})$ and $F_- = 2^{-(1/2)}(F_{10} - iF_{01})$ and a corresponding definition of P_+ and P_- , and with $d = D_1 + iD_2$, one obtains from Eqs. (3a)–(3g) ($\delta = \delta' + \pi/2$):

$$\frac{\partial F_+}{\partial t} = -\kappa[F_+ + \beta \exp(i\delta)F_- - 2CP_+], \quad (4a)$$

$$\frac{\partial F_-}{\partial t} = -\kappa[F_- - \beta \exp(i\delta)F_+ - 2CP_-], \quad (4b)$$

$$\frac{\partial P_+}{\partial t} = -\gamma_\perp(P_+ - \frac{1}{2}F_+D_0 - \frac{1}{4}F_-d), \quad (4c)$$

$$\frac{\partial P_-}{\partial t} = -\gamma_\perp(P_- - \frac{1}{2}F_-D_0 - \frac{1}{4}F_+d^*), \quad (4d)$$

$$\frac{\partial D_0}{\partial t} = -\gamma_\parallel[D_0 - 2\sqrt{\pi} + \frac{1}{4}(F_+P_+^* + F_-^*P_+ + F_-P_-^* + F_+^*P_-)], \quad (4e)$$

$$\frac{\partial d}{\partial t} = -\gamma_\parallel[d + (1/2)(F_+P_-^* + F_-^*P_+)]. \quad (4f)$$

Equations (4a) and (4b) show that for the two steady-state modes F_+ and F_- the optical feedback leads to decoupled interference terms of opposite sign in the evolution equations. It is interesting to note that without these interference terms the system of Eqs. (4a)–(4f) shows a strong similarity to the theoretical model of a bidirectional ring laser.¹⁰

The optical switching of transverse patterns in our experiment relies on selecting the transverse field configuration given by F_+ (F_-) by creating conditions of destructive interference for the competing mode F_- (F_+) through the choice of δ . To illustrate this point, Fig. 2 shows the result of a numerical experiment based on Eqs. (4a)–(4f) that demonstrates the hysteretic switching of the mode intensities $|F_+|^2$ and $|F_-|^2$ that is due to the variation of δ ; here the atomic relaxation parameters are chosen to be near the limiting case, where the atomic variables may be adiabatically eliminated. In the adiabatic limit, an (analytical) linear-stability analysis of Eqs. (4a)–(4f) yields the result that switching occurs under the conditions

$$|\beta \cos \delta| \geq \frac{1}{2} \left(\frac{C}{C_{th}} - 1 \right),$$

$$\cos \delta > 0 \quad \text{for } F_+ \neq 0, F_- = 0 \rightarrow F_+ = 0, F_- \neq 0,$$

$$\cos \delta < 0 \quad \text{for } F_+ = 0, F_- \neq 0 \rightarrow F_+ \neq 0, F_- = 0, \quad (5)$$

where C_{th} denotes the pump parameter at laser threshold.

The above suggests that we should also expect an analogous selective excitation of spatial field patterns for the case when the self-injection scheme considered here is replaced by the injection of field patterns from a separate coherent source.

In order to discriminate between the two stable spatial field configurations of the TEM_{01}^* laser, a mode converter relying on the astigmatic imaging of higher-order transverse modes¹¹ is used in the experimental setup shown in Fig. 1: A cylindrical lens, oriented with its cylinder axis parallel to the x axis, creates a beam waist for the TEM_{01}^* field in the y - z plane, leaving propagation in the x - z plane unaffected. As a result the TEM_{10} and TEM_{01} components of the TEM_{01}^* field propagate here with different Guoy phase shifts¹² $\phi_{10}(z)$ and $\phi_{01}(z)$. A second cylindrical lens, oriented with its axis parallel to and placed at a distance l from the first lens, is used to limit the region of astigmatic propagation. It is easy to show¹³ that the case $\phi_{10}(d) - \phi_{01}(l) = \pi/2$ corre-

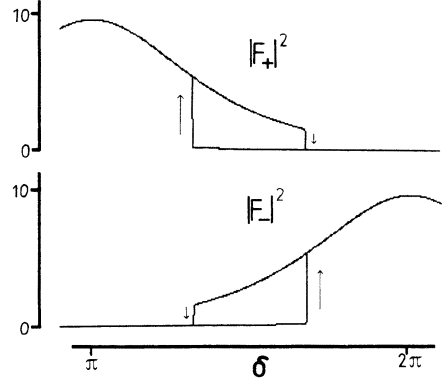


Fig. 2. Calculated normalized intensities $|F_+|^2$ and $|F_-|^2$ as a function of slowly varying feedback phase angle δ . Parameters: $\beta = 0.5$, $\gamma_\parallel = 10\kappa$, $\gamma_\perp = 20\kappa$, $C = 2C_{th}$, sweep rate $d\delta/dt \approx \kappa \times 10^{-5}$ rad; for definition of symbols see text. Small random fluctuations were introduced into the numerical calculation to stimulate the decay of unstable states. The arrows indicate the path followed in the hysteresis cycle.

sponds to the conversion of input fields described by the steady-state mode amplitudes F_+ and F_- into output fields given by $F_+' = 2^{-(1/2)}(F_{10} + F_{01})$ and $F_- = 2^{-(1/2)}(F_{10} + F_{01})$, respectively. (Common phase factors related to the optical path length l have been omitted.) With the use of Eqs. (1) and (2), it can be shown that the field amplitudes F_+' and F_- correspond to TEM_{10} and TEM_{01} fields, respectively, with nodal planes that are rotated by 45° with respect to the orientation of the cylindrical lenses. Thus, the two field patterns of a TEM_{01}^* laser appear as mirror-symmetric intensity patterns at the output of the mode converter.

3. EXPERIMENT

The laser used in the experimental setup shown in Fig. 1 consists of a 0.22-m-long, $\lambda = 633$ -nm, helium–neon laser tube (Melles–Griot type 05-LHB-290) terminated by Brewster windows. The Fabry–Perot laser resonator has a length of 0.25 m and uses mirrors with a reflectivity of 99.2% and a radius of curvature of 0.6 m. The eigenfrequency difference between the TEM_{10} and TEM_{01} modes of the optical resonator resulting from the astigmatism of the Brewster windows of the laser tube is compensated for by diffraction at the laser tube bore aperture, which is placed slightly noncentrally with respect to the optical axis of the resonator; TEM_{01}^* oscillation is obtained by placing a small absorbing dot (approximate size, $50 \mu\text{m}$) close to the optical axis of the resonator.⁷ Under these conditions the typical output power from each of the outcoupling mirrors is in the range of 30–100 μW , with the actual value depending on alignment, tuning, and the discharge current of the laser tube.

The effect of a mode converter of the type described above and the bistability of a TEM_{01}^* laser is illustrated by the photographs in Fig. 3; the practical realization of the mode converter is described in detail in Ref. 13. The change of the transverse field configuration of the laser that shows up in Figs. 3(b) and 3(c) is induced without external feedback by moving a beam stop through the laser resonator perpendicu-

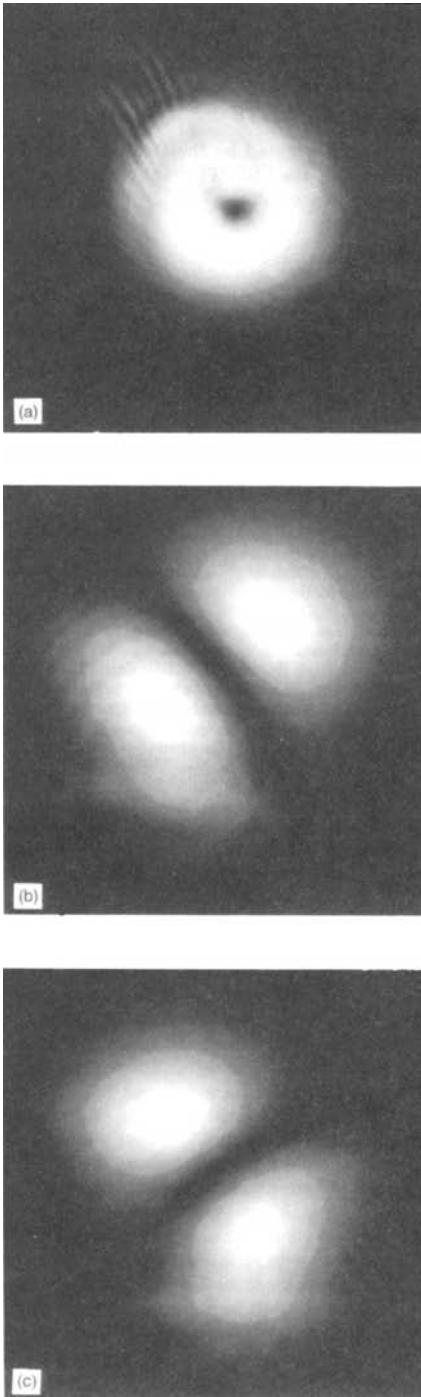


Fig. 3. Photographs of (a) transverse intensity distribution of the laser output and (b), (c) bistable intensity patterns at output of mode converter. The cylindrical lenses of the mode converter (see text) are oriented horizontally with respect to patterns (b) and (c).

lar to the optical axis: It is observed that the field pattern emitted by the laser can be selected by the direction of movement of the edge of the obstacle when the beam path becomes unblocked. This effect probably is caused by diffraction of the intracavity field at the edge of the beam stop: The corresponding perturbation of the cylindrical symmetry of the laser resonator may favor the onset of lasing in one of the two field patterns.

Hysteresis and switching between the two different field patterns is observed by monitoring the local (see Fig. 1) optical power density at the output of the mode converter as the optical feedback phase δ is swept back and forth by translation of a mirror in the feedback path. One experimental result that corresponds to a recording of a single hysteresis cycle in ≈ 0.1 sec is shown in Fig. 4(a). For unchanged experimental conditions, Figs. 4(b) and 4(c) show the switching transients with higher temporal resolution. Here the feedback coupling strength corresponds to $\beta \approx 0.12$, and the laser tube discharge current is approximately two times the laser threshold value.

It is found that the width of the observed hysteresis loop strongly depends on the discharge current; the width appears to be approximately proportional to the laser output power. No switching is observed for $\beta \lesssim 0.04$; above the switching threshold, the variation of the feedback coupling strength leads only to small changes of the shape of the hysteresis loop. The switching transients are always found, with typical rise and fall times of 2–5 μsec . This value is

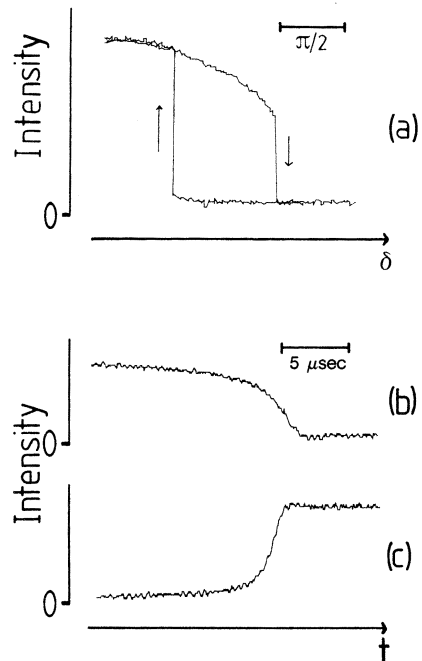


Fig. 4. (a) Experimentally determined local intensity at output of the transverse-mode converter (see Fig. 1) as a function of feedback phase δ . (b), (c) The same signal as a function of time t in the transient regimes. In all cases the sweep rate is $\delta/\text{dt} \approx 0.1$ rad/msec. For other experimental parameters, see text.

probably associated with the buildup time of the optical field in the laser resonator: with $1/\kappa \approx 0.1 \mu\text{sec}$ as the time constant of the laser resonator, derived from the resonator data given above, estimates of the field buildup time¹⁴ in our case yield values in the range of 1–10 μsec , which covers the range of experimentally observed switching times.

4. CONCLUSIONS AND OUTLOOK

The experimental findings are in full qualitative agreement with the results of the theoretical model described in Section 2. A significant quantitative difference is that in the experiment the fraction of the laser output field that is necessary to switch the system is much smaller than predicted. This seems to indicate a decreased stability margin of the experimental steady-state field patterns with respect to the perturbations induced by the injected field. It may be noted here that in the theoretical model atomic diffusion in directions perpendicular to the optical axis has not been taken into account. This process may play a crucial role for the stability of transverse optical patterns in gas lasers, since it washes out the differences in the transverse variation of inversion associated with different transverse intensity distributions.

The experimental techniques described in this paper may also be useful for more general studies on radiation patterns in nonlinear-optical systems. Complex stationary transverse patterns and multistability are to be expected when frequency-locked transverse modes of higher order are excited in a laser; until now, multiple-transverse-mode lasers have been studied mainly with respect to their dynamical behavior, i.e., under nonstationary conditions.¹⁵ Optical feedback techniques similar to the one described here may serve as a tool to probe the stability and to manipulate the generated transverse optical patterns.

Finally, one might consider a nonlinear-optical resonator whose transverse field structure is determined by an externally injected field pattern. Systems of this type could be

used in optical data processing and might even be capable of performing pattern recognition and associative memory operations.

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Optical helices and spiral interference fringes

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Very pure optical helices have been generated in an argon ion laser of low Fresnel number. The beam character, with continuous cophasal surface of helical form, is clearly demonstrated by spiral interference fringes produced in a novel interferometric arrangement. In addition to single-start helices the multistart fringe patterns establish both two-start and three-start helices (of pitch two and three wavelengths, respectively), and also the state of helicity (i.e. rotational hand) of the beams.

In a collimated monochromatic beam of light the surfaces of constant phase (modulo 2π) are usually considered to be a series of discs, normal to the beam axis, spaced one wavelength λ apart. However it has been known for some time [1–4], although not widely appreciated even among optical scientists, that monochromatic beams can possess a continuous cophasal surface of helical form extending through space. Such helical waves must of course satisfy Maxwell's equations; not surprisingly the phase discontinuity on axis necessitates a null in the intensity distribution, as readily seen in so-called laser "doughnut" modes. In the present work, techniques have been applied for generating very pure helices in an argon ion laser of low Fresnel number. Single-start helices (of pitch λ) and both two-start and three-start helices (pitch 2λ and 3λ , respectively) have been created. Novel interference methods demonstrate this helical character in a remarkable range of multistart spiral interference fringes.

The concept of screw and edge dislocations in wave fronts was first introduced by Nye and Berry [1] in 1974 originating in attempts to understand radio echoes from the bottom of Antarctic ice sheets; analogous experiments were carried out in the laboratory using ultrasound. In the first demonstration at optical frequencies [2,3], comparable helices in the lowest order doughnut mode of an argon ion laser were shown to be evolving between left- and right-hand helicity, at the frequency difference of the two constituent TEM_{01} modes. The impact of such spa-

tial coherence properties on various laser applications, including photon correlation spectroscopy, fringe velocimetry, speckle and coherent processing was also studied [4]. Subsequently frequency locking of two transverse modes in a He-Ne laser was achieved [5], and complex spatial and temporal instabilities in a CO_2 laser were analysed [6] in terms of nonlinear interaction of transverse cavity modes with the active medium. At the same time, in a theoretical study [7] based on the Maxwell–Bloch model, laser cavities of large Fresnel number were shown to exhibit optical vortices. In the most recent developments, studies of these optical helix/dislocation/phase-singularity/bifurcation/vortex-phenomena include bistability and optical switching of spatial patterns [8], waves of considerable spatiotemporal complexity in a CO_2 laser [9], the analysis of regular patterns of singularity arrays [10], the dynamics of optical vortices [11–14] and the creation of topologically stable patterns [15], and the theory of propagation in nonlinear media [16] and of arrays in free space [17]. Experimental studies have been made on a single longitudinal mode Na_2 ring laser [10,18–20] that generates families of transverse modes. In addition the design of optical and holographic phase converters for the generation of phase singularities has been developed [20–23]. Much of this work has concentrated on the properties of large arrays; the recent experimental evidence [14,17,19,20,21] for embedded vortices of first order consists of forked fringes employing plane-wave

and displaced-beam interference techniques (of intrinsically poor visibility). In addition single spiral fringes have been shown in an optical fibre [17] and by mixing with a spherical reference wave [24]. One of the first suggestions of multistart helices, and the only experimental evidence of a two-start, or charge 2, helix (derived from plane-wave fringes in a holographically generated doughnut), is contained in ref. [20]. The present work establishes that pure, higher order, helical modes may be generated in an active laser cavity.

The experimental arrangement is shown in fig. 1. The argon ion laser was a modified Lexcel Model 95 operating double-ended at 488 nm wavelength, with maximum Fresnel number ~ 10 , and a longitudinal mode spacing Δ of 156.02 MHz. Tuning to the required mode follows a technique related to that of Tamm [5]. Suppression of the on-axis modes is achieved by insertion of a small circular absorber into the cavity; a set of these with diameters ranging from 50 to 500 μm is deposited on a glass plate mounted on an X - Y translation stage. Similarly a set of clear apertures is inserted into the optical cavity. A range of transverse laser modes may be excited simultaneously; these are evident both from strong inter-mode beats and from the direct spectral display of the Fabry-Pérot interferometer. With suitable alignment of absorber and aperture these may be reduced to just two modes. For example these might be the Laguerre-gaussian 01 (two-lobe) modes in phase and

space quadrature which form the circular-symmetric first-order 01* doughnut mode. Characteristically these give a strong intermode beat signal within a few MHz of the longitudinal mode spacing Δ . The frequency of this beat signal may be tuned quite close to Δ by small lateral adjustments of the intracavity absorber and/or aperture, thus modifying the astigmatism of the optical cavity. The two modes thus approach degeneracy and at a certain point, usually within ~ 0.5 MHz of Δ , the beat disappears and the modes are locked to a single frequency which is typically stable over a wide range of laser power (up to 80 mW at full current of the laser discharge). A similar procedure, with suitably selected absorbers and apertures, permits selection of the higher-order, circular-symmetric, 02* and 03* doughnut modes.

The interferometric arrangement that unequivocally manifests the pure helical character of these beams by producing spiral interference patterns is shown in the lower half of fig. 1. The aim of this arrangement is three fold: after splitting at beam splitter BS1 the two equal beams are recombined at BS2 with (a) different curvature, (b) reversed handedness (i.e. helicity) and (c) equal size at the screen. Variations of curvature are achieved with a selection of lenses L_1 of different focal length whose precise position is adjusted to achieve condition (c) above. The overall change of handedness arising from the different reflections at beamsplitters and mirrors is indicated in fig. 1. It must be emphasised that this refers to the helicity of the beams; they remain linearly polarised throughout. As shown the two beams arrive with opposite helicity at the screen; common helicity is easily achieved with an additional mirror in the lower leg of the interferometer. The fringes are displayed on the screen, where they are recorded photographically. With beams of equal size and intensity the fringe visibility is 100%; for optimum fringe stability the optical paths between the beamsplitters are made as equal as possible and shielded from air movements.

Sets of photographs are shown in figs. 2 and 3. Of these fig. 2a shows the simple circular fringes expected for a lowest-order 00 gaussian mode. Arising as it does from beams of different curvature, the form of the fringes is essentially that of Newton's rings. Figure 2b similarly shows circular fringes for a 01* doughnut mode in which the two beams have been

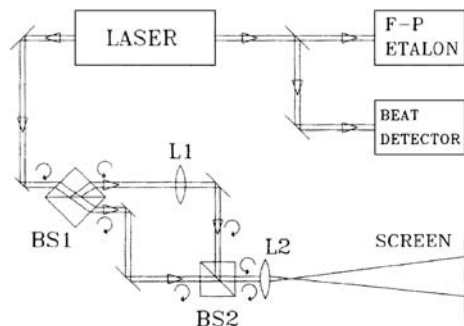


Fig. 1. Experimental arrangement. A circular absorber and aperture are inserted into the laser cavity and the laser output is monitored on the Fabry-Pérot interferometer and beat frequency detector. In the lower interferometric arrangement lens L_1 sets up a difference of curvature in the two beams which are recombined at beam splitter BS2. The helicity of each beam at different points is indicated.

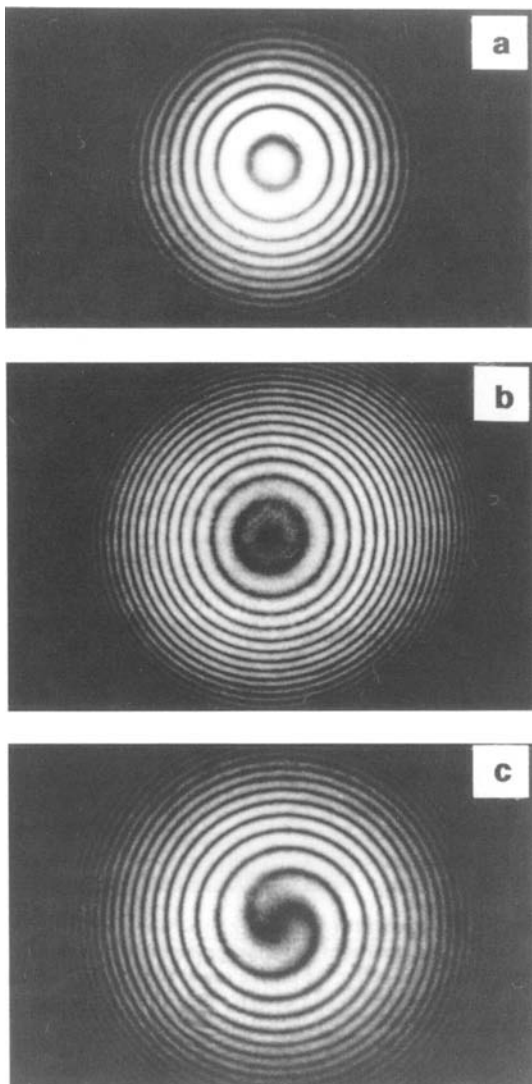


Fig. 2. Photographs of interference fringes: (a) lowest order gaussian mode, with circular fringes similar to Newton's rings; (b) circular fringes with a 01* doughnut mode in which the beams have been recombined with the same helicity; (c) the same as (b) but with the beams brought together with opposite helicity (as in fig. 1). Note the two-start spiral fringes.

brought together with the same handedness (i.e. an extra mirror has been introduced into the interferometer). In contrast fig. 2c shows the remarkable effect of bringing the beams together with opposite

helicity; the two-start spiral interference fringes directly manifest the helical character of the component beams. The sets of fringes in fig. 3 have been selected to demonstrate the effect of varying curvature difference for 01* and the higher order 02* and 03* modes. In the upper row the curvatures of the two beams are well matched. In consequence the fringes closely approximate the intensity patterns of individual TEM_{01,02,03} spot modes; their orientation is determined by the relative phase difference of the interfering beams. By contrast, in the middle row the small curvature difference amounts to less than one wavelength across the beams; only a fraction of a spiral extends across the beam front. In the lower row the curvature difference is a few wavelengths, leading to several complete spirals. The two-, four- and six-start spirals for the higher order modes are strikingly evident in each set of photographs.

The electric field at the screen that describes the interference of two helical beams of opposite hand is given for the 01* mode by

$$E = E_0 r \exp(-r^2/2a^2) \{ \exp \{ i[\theta + \phi(r)] \} + \exp \{ -i[\theta + \phi(r)] \} \} \quad (1)$$

$$= 2E_0 r \exp(-r^2/2a^2) \cos[\theta + \phi(r)], \quad (2)$$

where $2\phi(r)$ is the phase term shared equally between the curved wavefronts and r is the radial distance from the common beam axis. If the radii of wavefront curvature are R_1 and R_2 the phase difference $2\phi(r)$ is given by

$$2\phi(r) = r^2 |R_1 - R_2| / 2\lambda R_1 R_2 = 2k^2 r^2, \quad (3)$$

where k is a constant of the optical geometry. On replacing $\phi(r)$ by $k^2 r^2$, and writing the intensity distribution for the 01* mode as $I_0(1)$, the fringe intensity derived from eq. (2) is

$$I = I_0(1) \cos^2(\theta + k^2 r^2). \quad (4)$$

Starting from the dark centre, dark fringes are given by $\cos(\theta + k^2 r^2)$ equal to zero, from which they are defined by

$$kr = [(n + 1/2)\pi - \theta]^{1/2}, \quad (5)$$

$$n = \dots -2, -1, 0, 1, 2 \dots$$

Examination shows two distinct spiral branches, one for n odd and one for even. Thus two-start spiral

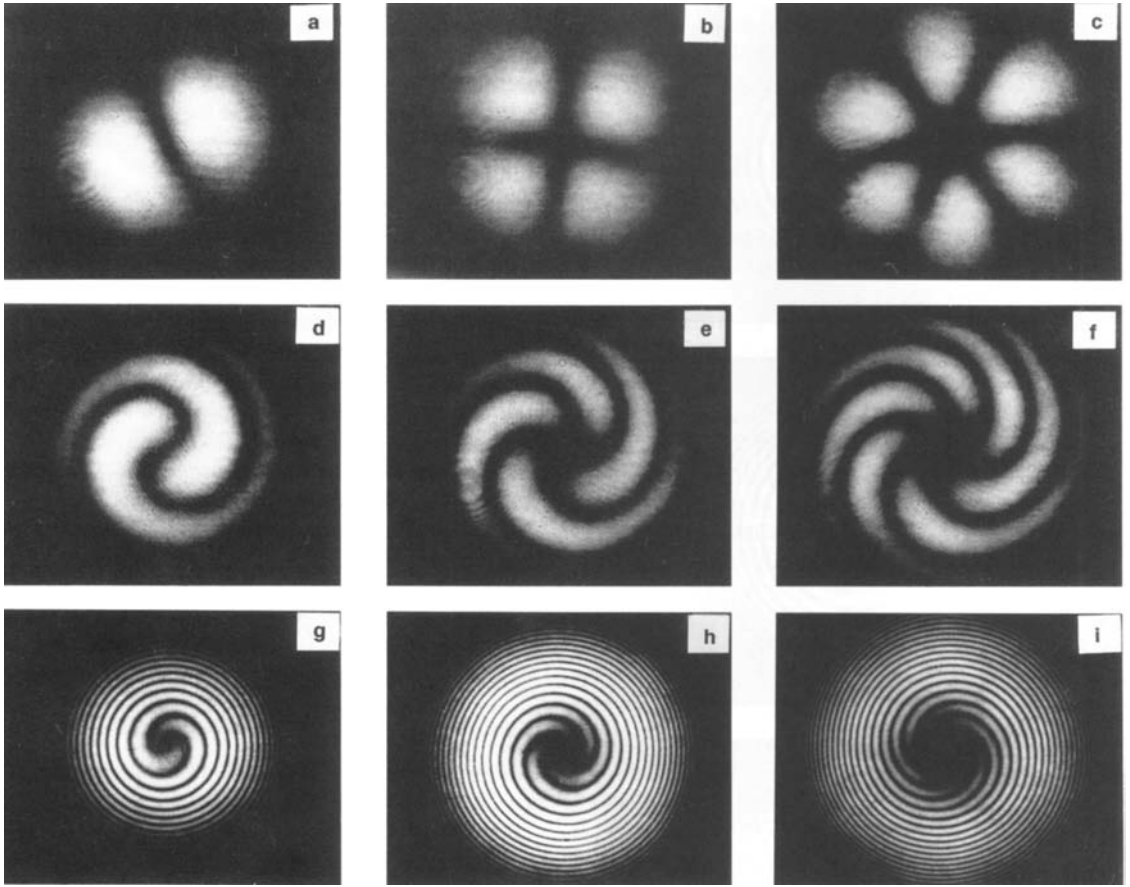


Fig. 3. Spiral fringe patterns with the laser operating on doughnut modes of different order: 01^* ((a), (d), (g)); 02^* ((b), (e), (h)) and 03^* ((c), (f), (i)). The beam curvatures are well matched in the upper row resulting in fringes that closely approximate the intensity patterns of individual $TEM_{01,02,03}$ spot modes. In contrast the beam curvature difference for the middle row is small (less than one wavelength) and for the lower row is several wavelengths. Note the multi-start spirals (two, four and six, respectively) and also the differing handedness of (g) and (i) (clockwise) from which the absolute helical sense can be established.

fringes are superimposed on the intensity distribution $I_0(1)$ of the single-start helical mode. Corresponding expressions for the higher-order modes are similarly derived; for the m th order (m -start) helix the fringe distribution is $I_0(m) \cos^2(m\theta + \phi)$ thus giving a $2m$ -start spiral. Computer-generated contour plots of these expressions show patterns in excellent agreement with figs. 2 and 3. Such multistart cophasal helices obviously require that the pitch of each individual helix must be $m\lambda$. Detailed examination of the fringes in fig. 3 shows also that for d,

e, f and h the spirals are anti-clockwise and for g and i clockwise. This reflects the fact that such modes have two possible states (of roughly equal probability) of left- and right-handed helicity.

In a further investigation the first few resonator modes have been calculated for a spherical-mirror open resonator, closely similar to the argon laser cavity, with the introduction of circular pinhole apertures and circular absorbers. The model is based on resonator matrix codes for CO_2 lasers [25] and has been adapted for Laguerre-gaussian TEM_{lp} basis

modes at 488 nm. For the calculations closest to the experimental arrangement a fixed pinhole aperture of 2.0 mm diameter was used and all basis modes up to $l=7$ and $p=7$ were included; the resultant round trip losses versus absorber diameter are shown in fig. 4. In practice it may be expected that the laser will operate on the lowest-loss resonator mode. Thus over a certain range of absorber/pinhole diameter one particular circular doughnut will be preferred; in effect modes of lower and higher order are significantly blocked by the absorber and pinhole, respectively. Not surprisingly the exact shapes of the predicted loss curves (and their crossing points) are rather sensitive to the choice of pinhole diameter; the calculation is also simplified by assuming that the absorber and pinhole lie at a mirror plane. The curves in fig. 4 should therefore not be interpreted too strictly. Nevertheless they agree qualitatively with the observations and show clearly the transitions (marked by the arrows) from TEM_{00} (very small absorber) through the first three doughnut modes.

In summary, single, pure 1- 2- and 3-start optical helices have been generated in an argon ion laser with a low Fresnel number cavity in reasonable accord with resonator mode calculations. The modes are stable over a wide range of power and it is likely that yet higher order modes and considerably higher power could be generated by suitable discharge and cavity design. A novel interferometric arrangement has been developed that immediately manifests the

helical character of the beams, and characterises both the state of helicity and the multi-start order number. Possible applications of helical modes have recently been listed by Heckenberg et al. [20], and include small particle levitation and focusing of atomic beams. Transfer of angular momentum [23] and rotational sensing are other possibilities. The stable, well-defined, multi-start helices presently demonstrated in an argon ion laser, together with the potential high power, should be valuable to such applications.

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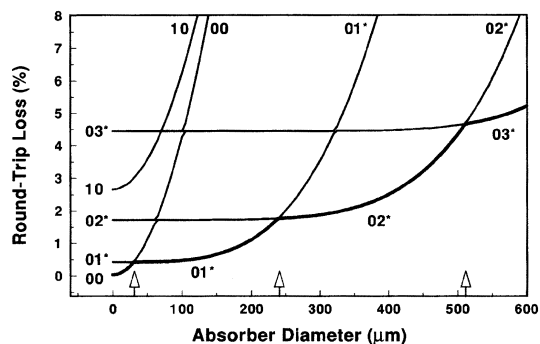


Fig. 4. Calculated round-trip loss versus absorber diameter for various resonator modes and with a 2.0 mm diameter pinhole aperture. The transitions (marked by arrows) from the TEM_{00} mode through the first three doughnut modes agree reasonably well with the observations.

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Astigmatic laser mode converters and transfer of orbital angular momentum

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We present the design of a mode converter which transforms a Hermite–gaussian mode of arbitrarily high order to a Laguerre–gaussian mode and vice versa. The converter consists of two cylindrical lenses and is based on appropriate use of the Gouy phase. We demonstrate mode conversion experimentally and consider where the concomitant transfer of orbital angular momentum is localized.

1. Introduction

A paraxial light beam of circular polarization is known to carry *spin* angular momentum of $+\hbar$ or $-\hbar$ per photon, for σ^+ or σ^- polarized beams, respectively. Interaction of such a light beam with a birefringent plate may lead to a mechanical torque, as was first demonstrated by Beth [1]. Recently we have shown theoretically that *orbital* angular momentum of light is also a useful concept for a paraxial light beam, particularly for a Laguerre–gaussian beam, which has a $(\exp i l \phi)$ azimuthal dependence [2]. Explicit calculation based on Maxwell's equations shows that such a beam, when linearly polarized, carries $l\hbar$ orbital angular momentum per photon. When the beam is circularly polarized, it carries $(l \pm 1)\hbar$ as total angular momentum per photon. We have speculated that conversion of a paraxial beam with specific orbital angular momentum into another beam, with a different orbital angular momentum, will give rise to a torque on the converter. In the previous paper [2] the properties of the converter were only briefly alluded to. In this paper we discuss the design of the mode converter in detail, and report experimental demonstration of its ability to transform modes. We also analyze theoretically how the converter takes up the change in orbital angular momentum of the light beam.

The first part of our paper is, in fact, a generali-

zation of previous work by Tamm on gaussian mode conversion [3,4]; we extend the results obtained by Tamm for low-order modes to modes of arbitrary order. There is also a strong connection with recent work by Abramochkin and Volostnikov [5]; they also deal with the influence of astigmatism on gaussian modes. They consider cases in which the astigmatism does not conserve the gaussian mode character of the incoming beam, in that the transverse intensity pattern of the outgoing beam changes upon propagation. We deal with suitable astigmatic elements which conserve the mode character and thus operate as mode *converters*; this restriction greatly simplifies the theoretical discussion.

2. Mode decomposition

In this section we introduce expansion formulas for a Hermite–gaussian (HG) and a Laguerre–gaussian (LG) mode which will turn out to be essential for an understanding of the mode converter. We use the following definitions for the amplitude of the Hermite–gaussian (HG) and Laguerre–gaussian (LG) laser modes which propagate along the z axis

$$u_{nm}^{\text{HG}}(x, y, z) = C_{nm}^{\text{HG}}(1/w) \exp[-ik(x^2 + y^2)/2R] \\ \times \exp[-(x^2 + y^2)/w^2] \exp[-i(n + m + 1)\psi] \\ \times H_n(x\sqrt{2}/w) H_m(y\sqrt{2}/w), \quad (1)$$

$$\begin{aligned}
u_{nm}^{\text{LG}}(r, \phi, z) &= C_{nm}^{\text{LG}} (1/w) \exp(-ikr^2/2R) \exp(-r^2/w^2) \\
&\times \exp[-i(n+m+1)\psi] \exp[-i(n-m)\phi] \\
&\times (-1)^{\min(n,m)} (r\sqrt{2}/w)^{|n-m|} \\
&\times L_{\min(n,m)}^{|n-m|}(2r^2/w^2), \quad (2)
\end{aligned}$$

with

$$R(z) = (z_R^2 + z^2)/z, \quad (3)$$

$$\frac{1}{2}kw^2(z) = (z_R^2 + z^2)/z_R, \quad (4)$$

$$\psi(z) = \arctan(z/z_R). \quad (5)$$

$H_n(x)$ is the Hermite polynomial of order n , $L_p^l(x)$ is the generalized Laguerre polynomial [6], k is the wave number, and z_R is the Rayleigh range (half the confocal parameter) of the mode. We introduce $N = n + m$ as the order of the mode. Normalization of the amplitude such that $\iint dx dy |u|^2 = 1$ yields

$$C_{nm}^{\text{HG}} = \left(\frac{2}{\pi n! m!} \right)^{1/2} 2^{-N/2}, \quad (6)$$

$$C_{nm}^{\text{LG}} = \left(\frac{2}{\pi n! m!} \right)^{1/2} \min(n, m)!. \quad (7)$$

Note that the indices we use for the LG mode differ from those normally used. The radial index p normally used is $\min(n, m)$, the minimum of n and m ; the azimuthal index l is $n - m$. Our notation brings advantages in the context of the present paper: we shall show that a mode converter can transform a HG_{nm} mode into a LG_{nm} mode or vice versa.

By using relations between Hermite and Laguerre polynomials (see e.g. refs. [2,5]) one can show that a LG mode can be decomposed into a set of HG modes of the same order:

$$u_{nm}^{\text{LG}}(x, y, z) = \sum_{k=0}^N i^k b(n, m, k) u_{N-k,k}^{\text{HG}}(x, y, z), \quad (8)$$

with real coefficients

$$\begin{aligned}
b(n, m, k) &= \left(\frac{(N-k)! k!}{2^N n! m!} \right)^{1/2} \\
&\times \frac{1}{k!} \frac{d^k}{dt^k} [(1-t)^n (1+t)^m] \Big|_{t=0}. \quad (9)
\end{aligned}$$

The factor i^k in eq. (8) corresponds to a $\pi/2$ relative

phase difference between successive components. Perhaps surprisingly, a HG mode whose principal axes make an angle of 45° with the (x, y) axes (a ‘diagonal’ mode) can be decomposed, using relations between products of Hermite polynomials [2,5], into exactly the same constituent set:

$$\begin{aligned}
u_{nm}^{\text{HG}}\left(\frac{x+y}{\sqrt{2}}, \frac{x-y}{\sqrt{2}}, z\right) \\
= \sum_{k=0}^N b(n, m, k) u_{N-k,k}^{\text{HG}}(x, y, z), \quad (10)
\end{aligned}$$

with the same real coefficients $b(n, m, k)$ as above. In this expansion, however, the successive components are in phase. In fig. 1 some examples of a mode decomposition of order 2 are given in diagrammatic form. In table 1 the coefficients $b(n, m, k)$ are given for modes up to order 3. For completeness we note that the relationship between the LG and HG modes can also be established via operator algebra [7] and by direct comparison with a 2D quantum harmonic oscillator [8].

3. Astigmatic Gouy phase

From eqs. (8), (10) it is clear that in order to perform the conversion from a HG mode to a LG mode one has to rephase the terms in the decomposition. This can be done by exploiting the Gouy phase $\psi(z)$

(nm)

$$\begin{aligned}
02 \quad \bigcirc &= \frac{1}{2} \begin{array}{|c|} \hline \text{---} \\ \hline \end{array} + \frac{i}{\sqrt{2}} \begin{array}{|c|} \hline \text{---} \text{---} \\ \hline \end{array} - \frac{1}{2} \begin{array}{|c|} \hline \text{---} \text{---} \\ \hline \end{array} \\
11 \quad \odot &= \frac{1}{\sqrt{2}} \begin{array}{|c|} \hline \text{---} \\ \hline \end{array} + \frac{1}{\sqrt{2}} \begin{array}{|c|} \hline \text{---} \text{---} \\ \hline \end{array} \\
20 \quad \bigcirc &= \frac{1}{2} \begin{array}{|c|} \hline \text{---} \\ \hline \end{array} - \frac{i}{\sqrt{2}} \begin{array}{|c|} \hline \text{---} \text{---} \\ \hline \end{array} - \frac{1}{2} \begin{array}{|c|} \hline \text{---} \text{---} \\ \hline \end{array} \\
02 \quad \begin{array}{|c|} \hline \text{---} \text{---} \\ \hline \end{array} &= \frac{1}{2} \begin{array}{|c|} \hline \text{---} \\ \hline \end{array} + \frac{1}{\sqrt{2}} \begin{array}{|c|} \hline \text{---} \text{---} \\ \hline \end{array} + \frac{1}{2} \begin{array}{|c|} \hline \text{---} \text{---} \\ \hline \end{array} \\
11 \quad \begin{array}{|c|} \hline \text{---} \text{---} \\ \hline \end{array} &= \frac{1}{\sqrt{2}} \begin{array}{|c|} \hline \text{---} \\ \hline \end{array} - \frac{1}{\sqrt{2}} \begin{array}{|c|} \hline \text{---} \text{---} \\ \hline \end{array} \\
20 \quad \begin{array}{|c|} \hline \text{---} \text{---} \\ \hline \end{array} &= \frac{1}{2} \begin{array}{|c|} \hline \text{---} \\ \hline \end{array} - \frac{1}{\sqrt{2}} \begin{array}{|c|} \hline \text{---} \text{---} \\ \hline \end{array} + \frac{1}{2} \begin{array}{|c|} \hline \text{---} \text{---} \\ \hline \end{array}
\end{aligned}$$

Fig. 1. Examples of the decomposition of HG_{nm} and LG_{nm} modes of order 2.

Table 1

The coefficients $b(n, m, k)$ which occur in eqs. (8), (10).

n	m	$k=0$	1	2	3
0	0	1			
0	1	$1/\sqrt{2}$	$1/\sqrt{2}$		
1	0	$1/\sqrt{2}$	$-1/\sqrt{2}$		
0	2	$1/2$	$1/\sqrt{2}$	$1/2$	
1	1	$1/\sqrt{2}$	0	$-1/\sqrt{2}$	
2	0	$1/2$	$-1/\sqrt{2}$	$1/2$	
0	3	$1/\sqrt{8}$	$\sqrt{3/8}$	$\sqrt{3/8}$	$1/\sqrt{8}$
1	2	$\sqrt{3/8}$	$1/\sqrt{8}$	$-1/\sqrt{8}$	$-\sqrt{3/8}$
2	1	$\sqrt{3/8}$	$-1/\sqrt{8}$	$-1/\sqrt{8}$	$\sqrt{3/8}$
3	0	$1/\sqrt{8}$	$-\sqrt{3/8}$	$\sqrt{3/8}$	$-1/\sqrt{8}$

of a gaussian mode which appears in eqs. (1), (2), i.e. the phase shift that the beam undergoes when going through a waist as compared to that of a plane wave. Tamm has recognized that one can use the difference in Gouy phase between an astigmatic HG₀₁ and HG₁₀ mode to convert a first-order LG mode to a HG mode [3]. Here we show that an arrangement similar to that used by Tamm can transform modes of arbitrary order.

For an isotropic (i.e., non-astigmatic) gaussian beam the Gouy phase appears in eqs. (1), (2) as

$$(n+m+1) \psi(z), \quad (11)$$

with $\psi(z) = \arctan(z/z_R)$ for a waist at position $z=0$. For an astigmatic beam the situation is different. Consider first an astigmatic HG beam which has its nodal lines parallel to the axes of the astigmatism. Such a beam can be produced by passing a HG beam through a cylindrical lens aligned along the axes of the mode pattern. The amplitude of this mode can be considered separately in the two transverse planes (x, z) and (y, z); in each plane the beam is characterized by the z -coordinate and the Rayleigh range of the waist. The resulting Gouy phase has two contributions, one from each transverse direction [9,10], and may be written as

$$(n+1/2) \psi_x(z) + (m+1/2) \psi_y(z), \quad (12)$$

with

$$\psi_x = \arctan[(z - z_{0x})/z_{Rx}], \quad (13)$$

$$\psi_y = \arctan[(z - z_{0y})/z_{Ry}], \quad (14)$$

where z_{0x} and z_{0y} are the positions of the waists and z_{Rx} and z_{Ry} the corresponding Rayleigh ranges in the (x, z) and (y, z) planes, respectively. For an isotropic HG beam the two waists coincide and have equal diameters, so that eq. (12) reduces to eq. (11). It follows that isotropic HG beams of the same order $n+m$ have the same Gouy phase. But for the astigmatic HG beam ψ_x and ψ_y are different functions of z so that the relative phase of HG modes of the same order, but with different n and m , is a function of z .

It should be realized that neither a LG beam, nor a HG beam when passed through a cylindrical lens at an *arbitrary* angle relative to the mode pattern, can be described in the same way, since the amplitude of these beams is not separable in x and y . For such beams (and also for more general beams) one should decompose the transverse pattern into HG modes oriented along the axes of the lens. Since in an astigmatic beam the relative phase of these HG components changes upon propagation, the transverse pattern of these astigmatic beams will change accordingly. This applies to the cases considered in ref. [5].

4. Mode converter

In order to exploit the Gouy phase to construct a mode converter, the beam should be made astigmatic in a confined region only, while it is isotropic outside this region. When the beam is passed through

this region, a definite phase difference will be introduced between the HG components which are oriented along the axes of astigmatism. Consider for example an astigmatic beam for which the waists coincide but have different Rayleigh ranges, z_{Rx} and z_{Ry} , respectively (fig. 2a). At the position where the two transverse radii of the astigmatic beam are equal, a cylindrical lens may be placed to match the radii of curvature of the beam such that the beam outside the lens is no longer astigmatic (fig. 2b). When the same cylindrical lens is also placed on the other side of the waist and if the input beam is properly mode-matched, the beam is astigmatic only between the two lenses (fig. 2c). The condition that the transverse radii of the beam (given by eq. (4)) are equal at the position of the lens $z = \pm d$ leads to

$$\frac{z_{Rx}^2 + d^2}{z_{Rx}} = \frac{z_{Ry}^2 + d^2}{z_{Ry}}, \quad (15)$$

and the condition that the input beam is mode-matched is fulfilled if the focal distance f of the cylindrical lenses satisfies (see eq. (3)),

$$\frac{1}{f} = \frac{1}{R_x(d)} - \frac{1}{R_y(d)} = \frac{d}{z_{Rx}^2 + d^2} - \frac{d}{z_{Ry}^2 + d^2}. \quad (16)$$

It is useful to introduce a parameter p such that

$$p = \sqrt{\frac{1 - d/f}{1 + d/f}}. \quad (17)$$

Equations (15), (16) then lead to

$$z_{Rx} = dp, \quad (18)$$

$$z_{Ry} = d/p. \quad (19)$$

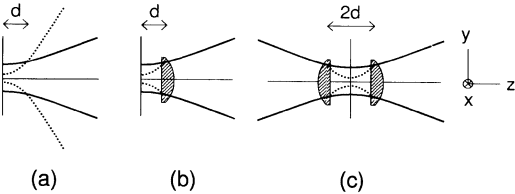


Fig. 2. Sketch of a symmetric mode converter. The dashed curve denotes the gaussian beam envelope in the (x, z) plane, and the solid curve that in the (y, z) plane. (a) An astigmatic waist at $z=0$. (b) A cylindrical lens matches the radii of curvature at $z=d$. (c) Two cylindrical lenses act as a converter on a mode-matched beam.

The change in Gouy phase $\psi(z)$ of a HG mode oriented along the axes of the lenses, when passed through this region, is, using eq. (12),

$$\Delta\psi = (n+m+1)(\Delta\psi_x + \Delta\psi_y)/2 + (n-m)(\Delta\psi_x - \Delta\psi_y)/2, \quad (20)$$

with

$$\Delta\psi_x = \psi_x(d) - \psi_x(-d) = 2 \arctan(d/z_{Rx}) \quad (21)$$

and analogously for $\Delta\psi_y$.

We now consider a 'diagonal' HG mode which is passed through the converter of fig. 2c and expand the input mode into HG modes of the same order $n+m$ oriented along the lens axes (eq. (10)). The successive terms in this expansion differ by two in the value of $(n-m)$. Therefore the total phase difference which is introduced *between* successive terms is

$$\theta = 2[\arctan(d/z_{Rx}) - \arctan(d/z_{Ry})] = 2[\arctan(1/p) - \arctan p]. \quad (22)$$

The phase difference is thus determined by the parameter p only and ranges from 0 to π .

If this phase difference is set equal to $\pi/2$ the system introduces a factor i^k in front of each term in the expansion of eq. (10), so that the HG mode is converted into a LG mode with the same indices n, m (eq. (8)). The condition $\theta = \pi/2$ is fulfilled if $p = -1 + \sqrt{2}$, which leads to

$$d = f/\sqrt{2}. \quad (23)$$

Mode-matching (eq. (19)) requires that the input beam has a Rayleigh range (cf. fig. 2c)

$$z_{Ry} = f + d = (1 + 1/\sqrt{2})f. \quad (24)$$

We will call this converter, which converts a diagonal HG mode to a LG mode, a ' $\pi/2$ converter', referring to the value of θ . Of course the argument can also be reversed: the $\pi/2$ relative phase different between the components of the LG mode can be removed with a $\pi/2$ converter, so that a HG mode is produced.

The converter with $\theta = \pi$, the ' π converter', implies $p = 0$, and therefore

$$d = f, \quad z_{Rx}/f = 0, \quad z_{Ry}/f = \infty, \quad (25)$$

which corresponds to a confocal configuration of the

cylindrical lenses with a collimated incident beam. Note that the ideal π converter exists only in the geometrical optical limit. In practice, i.e. in a wave-optical description, $z_{R,x}$ and $z_{R,y}$ are always finite, so that $\theta = \pi - \epsilon$ instead of π . In a diffraction-limited system, ϵ can be made arbitrarily small by making the system (and incident-beam diameter) sufficiently large. In the geometrical optical limit, the lens system exchanges the left and the right side of the beam, so that a diagonal HG mode u_{nm}^{HG} is converted to u_{mn}^{HG} , and a LG mode u_{nm}^{LG} is converted to u_{mn}^{LG} , which has an azimuthal dependence of the opposite sign.

The $\pi/2$ and π converters are compared in fig. 3; here we have varied the distance between the cylindrical lenses, keeping their focal length constant. It illustrates that a $\pi/2$ converter generally requires a tightly focussed input beam, whereas a π converter operates on a collimated beam.

Obviously, such converters can also be constructed by using two cylindrical *mirrors*. Another possibility would be to exploit the astigmatism furnished by an off-axis configuration of spherical lenses and/or mirrors.

5. Experiments

For testing the mode converters described in sect. 4, we used a HeNe laser consisting of a gain tube of 35 cm length with Brewster windows (Spectra Physics 120S) in a two-mirror cavity (fig. 4). As we wanted this laser to operate in a higher-order trans-

verse mode, we had to make the Fresnel number of the cavity as large as possible. Since the bore of the gain tube ($\varnothing$ 1.8 mm) is the effective aperture in the cavity, this implies a small beam diameter at the Brewster windows. From eq. (4) it follows that the beam radius at a distance d from the waist has a minimum for $z_R = d$. Therefore we chose the position of the waist in the middle of the tube, with a Rayleigh range of about half the length of the tube ($z_R \approx 18$ cm). This was obtained with two commercial HeNe laser mirrors, one with 600 mm radius of curvature, placed at 525 mm from the middle of the gain tube, and the other with 437 mm radius of curvature, placed at 306 mm from the middle of the tube. The first mirror was a high-reflector for 633 nm, the second mirror was an output coupler with 1.2% transmission. In order to force the laser to operate in a higher-order Hermite-gaussian mode, two metal wires of 20 μ m diameter were carefully positioned inside the cavity in front of the high-reflector, one vertical and the other horizontal, both perpendicular to the beam. The wires force the laser to operate in the higher-order HG mode which has nodal lines at the position of the wires. Any mode of order 0 up to 3 could be made this way, with output powers between 0.5 and 5 mW.

We first tested the operation of the $\pi/2$ converter by mode-matching a 'diagonal' HG laser beam to such a converter by means of two spherical lenses. The cylindrical lenses of the converter had a focal distance of 19 mm, while the Rayleigh range of the input beam and the distance between the lenses were chosen according to eqs. (23), (24). A lens with a short focal length projected the output mode on to a screen, and this output pattern was photographed. The results shown in fig. 5 illustrate the conversion of several input HG modes to corresponding LG modes. When the $\pi/2$ converter was rotated around the propagation axis of the beam, an incoming HG beam was transformed to a HG mode when the converter was at 0 or 90 degrees and to a LG mode when the converter was at ± 45 degrees (fig. 6a). In another experiment the HG laser mode was converted into a LG mode with a fixed $\pi/2$ mode converter and then passed through a rotatable $\pi/2$ converter, which transformed the mode back to a HG mode. The output mode was photographed with the second converter at various angles (fig. 6b). The output is a HG

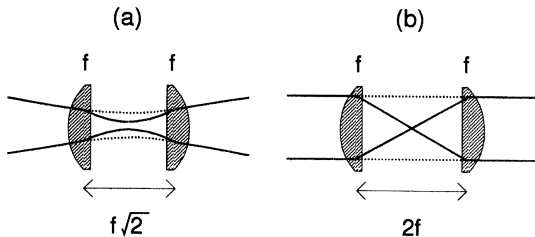


Fig. 3. Comparison of (a) a $\pi/2$ converter and (b) a π converter. Both converters consist of two identical cylindrical lenses of focal length f ; they focus in the plane of the paper. The distance between the lenses is $f\sqrt{2}$ for the $\pi/2$ converter and $2f$ for the π converter. Dashed lines indicate the propagation of the beam in the other transverse direction.

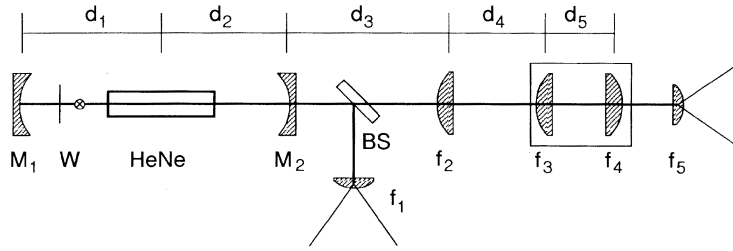


Fig. 4. Experimental arrangement used to demonstrate the operation of the $\pi/2$ mode converter. Two mutually perpendicular wires (W) are placed perpendicular to the mode axis inside a HeNe laser consisting of two mirrors (M_1 , $R=600$ mm and M_2 , $R=437$ mm) and a HeNe gain tube (HeNe). With a beam splitter (BS) part of the output is split off and projected onto a screen with a lens of short focal length (f_1). The beam is mode-matched with lens f_2 into a $\pi/2$ converter built with two cylindrical lenses (f_3 and f_4). The output is projected onto a screen with a lens of short focal length (f_5). Dimensions are: $d_1=525$ mm, $d_2=306$ mm, $d_3=225$ mm, $d_4=176$ mm, $d_5=27$ mm; $f_1=f_5=20$ mm, $f_2=160$ mm, $f_3=f_4=19$ mm (cylindrical).

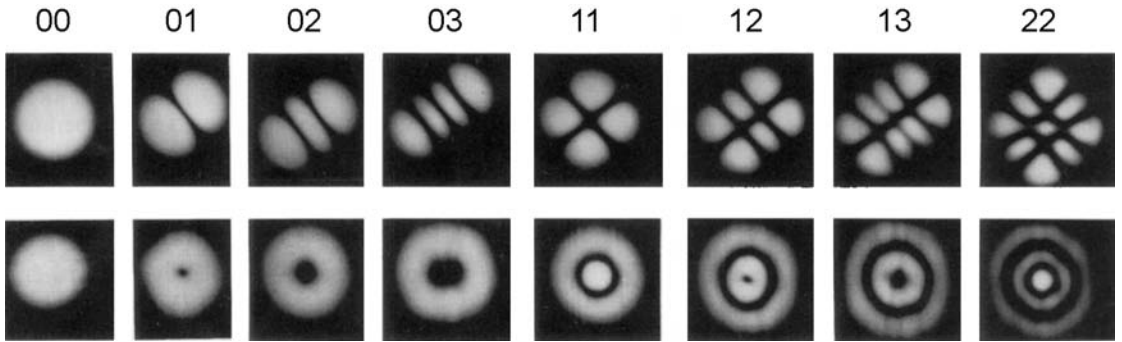


Fig. 5. Experimental results obtained with the $\pi/2$ converter. The top row shows the input HG_{nm} mode; the bottom the output LG_{nm} mode, where n, m is indicated above the modes.

mode which is always at 45 degrees with respect to the axes of the lenses of the converter.

To test the π converter we mounted such a converter in the rotatable mount and passed a collimated HG beam through it. The output is again a HG mode which rotates twice as fast as the converter, as shown in fig. 7.

To interpret these results it is helpful to keep in mind the resemblance between decomposition of a mode pattern on the one hand and decomposition of polarization on the other hand, as pointed out in ref. [2]. A quarter-wave plate converts linearly to circularly polarized light by introducing a $\pi/2$ phase difference between the linearly polarized compo-

nents and is therefore analogous to the $\pi/2$ mode converter, which converts a HG mode into a LG mode by introducing a $\pi/2$ phase difference between the HG components. A half-wave plate converts left-handed to right-handed circularly polarized light by introducing a π phase difference and is therefore similar to the π converter.

Using this analogy, the experiment of fig. 6a is similar to passing a linearly polarized beam through a rotating quarter-wave plate. Depending on the orientation of the plate, this would result in a circularly or linearly polarized beam. The experiment of fig. 6b is similar to passing a circularly polarized beam through a rotating quarter-wave plate. In this case

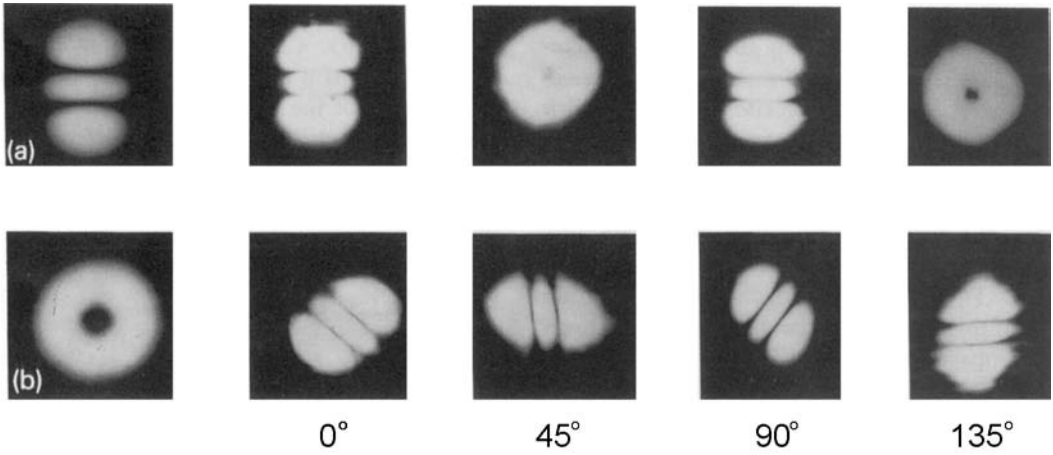


Fig. 6. Experimental results obtained with the $\pi/2$ converter at different angles, (a) with an input HG_{02} mode and (b) with an input LG_{02} mode. The left-most pattern in each row is the input, on the right are the output patterns with the angle of the converter indicated below.

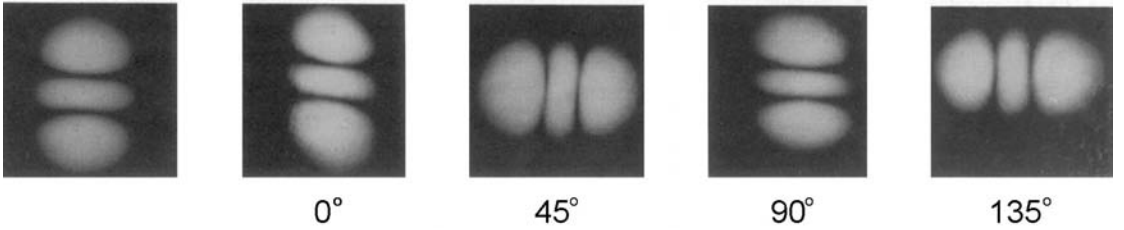


Fig. 7. Experimental results obtained with the π converter at different angles. On the left is the input HG_{02} mode, on the right the output patterns.

the output would be a linearly polarized beam whose polarization axis is at 45 degrees with respect to the axes of the plate. The last experiment (fig. 7) is similar to passing a linearly polarized beam through a half-wave plate. This would result in a linearly polarized beam whose polarization axis rotates twice as fast as the plate.

Note that in the paraxial approximation the polarization decomposition is two-dimensional, spanned by two polarization vectors. The mode decomposition has dimension $N+1$ for a mode of or-

der N since such a mode can be expressed as a superposition of $N+1$ modes of the same order.

6. Angular momentum transfer

Now that we have studied in detail how a HG mode may be converted to a LG mode we remind the reader that it has been established previously that a LG beam carries orbital angular momentum whereas a HG beam does not [2]. It may thus be expected that

the transformation leads to a mechanical torque on the converter. We are preparing an experiment to investigate this issue. Here we wish to address theoretically the question where precisely, that is at which cylindrical lens, the transfer of angular momentum will take place. We consider a linearly polarized mode with vector potential $\mathbf{A} = \mathbf{e} u(x, y, z) \exp(-ikz)$ in the Lorentz gauge (with $\mathbf{e}$ the polarization vector), where $u(x, y, z)$ is a solution of the paraxial wave equation. The time-averaged angular momentum density of this mode has a component in the direction of propagation given by [2]

$$\begin{aligned} M_z &= i\omega\epsilon_0/2 \left[\left(xu^* \frac{\partial u}{\partial y} - xu \frac{\partial u^*}{\partial y} \right) \right. \\ &\quad \left. - \left(yu^* \frac{\partial u}{\partial x} - yu \frac{\partial u^*}{\partial x} \right) \right] \\ &= -\omega\epsilon_0 (\partial\alpha/\partial\phi) |u|^2, \end{aligned} \quad (26)$$

with $\alpha = \arg u$. The angular momentum density per unit length in the direction of propagation is

$$L_z = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dx dy M_z. \quad (27)$$

For a HG mode we deduce from eq. (1) the phase of u

$$\alpha = -kr^2/2R(z) - (n+m+1)\psi(z) \quad (28)$$

and substitution in eq. (26) yields that the angular momentum of such a mode is zero. For a LG mode we deduce from eq. (2)

$$\begin{aligned} \alpha &= -kr^2/2R(z) \\ &\quad - (n+m+1)\psi(z) - (n-m)\phi, \end{aligned} \quad (29)$$

so that the ϕ derivative leads to an angular momentum proportional to $n-m$. It has been shown previously that it is in fact equal to $(n-m)\hbar$ per photon [2]. As the mode is linearly polarized, the intrinsic spin angular momentum is zero and the angular momentum found here is orbital angular momentum.

The $\pi/2$ converter transforms a HG mode without orbital angular momentum to a LG mode with orbital angular momentum. It might be anticipated that the transfer of angular momentum takes place at both lenses, but it turns out that this is not the case. For any mode $u(x, y, z)$ which is a solution of the par-

axial wave equation, the angular momentum is given by eqs. (26), (27). When this mode is passed through a lens it acquires an extra phase χ which is a function of the transverse coordinates^{#1}, so that the mode function directly after the lens can be written as

$$u' = u \exp[i\chi(x, y)]. \quad (30)$$

In ref. [11] it is shown that

$$L_z(u') = L_z(u) + \delta L_z, \quad (31)$$

with

$$\begin{aligned} \delta L_z &= -\omega\epsilon_0 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dx dy \left(x \frac{\partial \chi}{\partial y} - y \frac{\partial \chi}{\partial x} \right) \\ &\quad \times |u(x, y, z)|^2. \end{aligned} \quad (32)$$

For a cylindrical lens we have $\chi = -\frac{1}{2}(k/f)x^2$, so that

$$\delta L_z = -\omega\epsilon_0 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dx dy (k/f)xy |u(x, y, z)|^2. \quad (33)$$

This may be normalized against the energy in the beam to yield the angular momentum transferred to the lens per photon. For a LG mode, as $|u_{nm}^{LG}|^2$ is symmetric around the z axis, δL_z is zero. A diagonal HG mode with $m \neq n$, however, has unequal intensities in the four xy quadrants, which leads to $\delta L_z \neq 0$. The conclusion is that in the $\pi/2$ converter, with a HG mode ($m \neq n$) incident on one side and a LG mode exiting from the other, the angular momentum conversion is expected to take place *only* at the lens which sees the input HG mode. It follows that the astigmatic HG mode just after the first lens already contains the orbital angular momentum of the output LG mode. The function of the second lens is to change the astigmatic beam into a pure LG mode. For the same reason, if a LG mode is incident on a $\pi/2$ converter and is converted into a HG mode, the transfer of orbital angular momentum is expected to take place at the second lens.

In fact, for a macroscopic lossless dielectric object such as a lens the radiative force is a conservative force ("gradient force") which is derived from the

^{#1} Writing the operation of the lens as a phase factor is only correct for a lens whose principal planes coincide, i.e. a thin lens.

dielectric polarization energy as the potential [12]. Dielectric matter is drawn to positions where the electric field strength is largest (as an example, when the plates of a capacitor are dipped into a dielectric fluid, the fluid is drawn into the volume between the plates). This explains why the transfer of angular momentum, that is the mechanical torque, occurs at the 'HG side' of the $\pi/2$ converter only. For a cylindrical lens in a HG beam the dielectric polarization energy clearly depends on the angle between the cylinder axis and the HG axes x and y . The cylindrical lens will tend to align with the maximum optical intensity inside the dielectric. A radiative torque on the lens results unless the polarization energy is minimum or maximum, that is unless the cylinder axis is aligned with x or y . For a cylinder lens in a LG beam the radiative torque is zero as the polarization energy is independent of its orientation.

It follows from eq. (33), and also from the energy arguments above, that the change in orbital angular momentum may in principle be made arbitrarily high by choosing the lens arbitrarily strong ($f \rightarrow 0$). It follows similarly that the orbital angular momentum may be changed by astigmatic elements other than a cylinder lens (e.g. a prism). Also, for similar reasons, a LG mode with an angular sector removed with a suitable mask *will* transfer orbital angular momentum to a cylindrical lens. However, all such operations will not lead to a gaussian mode at the output and thus do not correspond to a well-defined mode conversion.

The π converter seems an anomaly in this context. If the perfect π converter has a LG mode incident at the first lens, a LG mode of opposite handedness will leave the second lens. According to the arguments given above it appears that there is no orbital angular momentum exchange on either of the lenses, although the orbital angular momentum of the beam changes sign. As noted in sect. 4, however, a perfect π converter does not exist. If a perfect LG mode is incident on a practical converter, the output mode will always, however slightly, differ from a perfect LG mode, which invalidates the analysis given above. An appropriate analysis of this converter has been given by Van Enk and Nienhuis [11].

7. Conclusions

We have shown that it is possible to build a mode converter with two cylindrical lenses which converts a Hermite–gaussian mode or arbitrary order into a Laguerre–gaussian mode of the same order and vice versa. The conversion is described using a mode analysis, based on the decomposition of a LG mode and a diagonally oriented HG mode into Hermite–gaussians. Two converters are introduced: the $\pi/2$ converter which converts a HG to a LG mode or vice versa, and the π converter which exchanges the indices of the incoming mode and thereby converts a LG mode into one with opposite azimuthal dependence. We have demonstrated these conversions experimentally and have shown the analogy between the mode converters and half- and quarter-wave plates. The transfer of orbital angular momentum, occurring when a HG mode without orbital angular momentum is converted into a LG mode with orbital angular momentum, is shown to take place at the lens onto which the HG mode is incident.

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Observation of the Dynamical Inversion of the Topological Charge of an Optical Vortex

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We report what is believed to be the first detailed experimental observation of the dynamic inversion of the topological charge of an optical vortex under free-space propagation. The vortex self-transformation occurs through continuous deformation of the noncanonical strength of the corresponding screw wave front dislocation, and is mediated by the occurrence of an extremely sharp turn in a Berry vortex trajectory, which observed at a Freund critical foliation appears as an edge-line dislocation orthogonal to the propagation direction, at a crucial point of the light evolution.

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Topological deformation of fields is at the heart of the description of a vast variety of natural phenomena ranging from cosmology, to oceanography, chemistry, and bio-sciences. Its most extreme, yet ubiquitous manifestation occurs when the field is so strongly folded that it bends back upon itself to form a topological dislocation, or defect [1–5]. Screw dislocations, or vortices, are a common defect type. The multiplicity of the folding around the defect determines its topological charge. The net charge within a region of space must be conserved under continuous evolution provided that no charges enter or leave the region [6]. Therefore, dynamic vortex creation typically occurs through the nucleation of twin pairs of opposite charge. However, here we uncover a striking mechanism that inverts the topological charge of a single vortex under propagation. This vortex self-transformation is mediated by the occurrence of an extremely sharp turn in a Berry vortex trajectory [7,8], which in the experimental plane of observation appears as an edge-line dislocation orthogonal to the light propagation direction, at a crucial point of the light evolution. Observations were performed with vortices nested in light beams [5–14]. The phenomenon can have direct implications to quantum photon states with angular momentum [15,16], to superfluidity [17,18], and to weakly interacting Bose-Einstein condensates (BEC's) [19,20], and is believed to be generic to diverse vortex entities.

The key ingredient of the new phenomenon is the existence of vortices with intrinsic spatial structure [6], hereafter referred to as noncanonical vortices. In the case of optical beams, vortices are spiral phase ramps around a dark spot where the phase of the wave is undefined and thus the light intensity vanishes. The slowly varying field envelope E of a single-charge vortex nested in a paraxial light beam propagating along the Z axis has the form

$$E = \{x - x_1(z) + iA(z)[y - y_1(z)]\}F(x, y, z), \quad (1)$$

where (x_1, y_1) stands for the vortex location in the X - Y plane and F is the host beam. The value of the complex quantity A determines the noncanonical strength of the vortex: Only when $A = \pm 1$ does the vortex have a traditional, or canonical, structure. Vortices are located at the complex zeros of E . In the case of canonical vortices, the lines in the transverse plane where the real and imaginary parts of E vanish cross at right angles. When $A \neq \pm 1$, the intensity pattern of the light beam E has no radial symmetry around the vortex core, and at the location of the phase dislocation the real and imaginary parts of E cross each other at general angles. The light intensity and phase structure of a single-charge noncanonical vortex with strength $A = 1 + i$ are shown in Fig. 1. The sign of the topological charge of the vortex, given by the circulation of the gradient of the phase of E around the dislocation, is given by the sign of $\text{Re}(A)$. Therefore, a family of different noncanonical vortices that carry the same topological charge can be constructed by varying the complex value of A while keeping the sign of $\text{Re}(A)$ fixed.

However, the dynamical evolution followed by the vortices, including the value of its topological charge, can depend critically on the noncanonical strength of the dislocations. In particular, self-inversion of the topological charge of a noncanonical vortex can occur. Furthermore, in contrast to traditional views, such phenomenon can take place in a variety of physical settings, including properly tailored light propagation in vacuum, counter-intuitive though it might seem. The principle of the experimental realization of the dynamic inversion of the topological charge of a vortex is illustrated by the paraxial evolution of a single-charge vortex nested in the elliptical Gaussian beam $F = C(z)\exp[ix^2/q_x(z) + iy^2/q_y(z)]$, where $C(z)$ is the complex beam amplitude and $q_x(z)$ and $q_y(z)$ are the complex beam widths. In free space, the beam widths evolve as $q_x = 2(z - z_x) - iw_{0x}^2$, $q_y = 2(z - z_y) - iw_{0y}^2$, where z_x and z_y stand for the locations

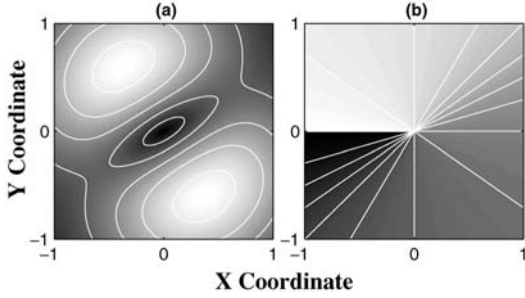


FIG. 1. Core of a noncanonical vortex with strength $A = 1 + i$. (a) Amplitude. (b) Phase. Contour lines are spaced 0.11 dimensionless units in (a), and $\pi/8$ units in (b).

of the beam waists along the propagation axis Z and w_x , w_y are the corresponding waists. Under such conditions, one finds that the noncanonical strength of the vortex varies during propagation according to the expression [21]

$$A(z) = A_0 \frac{2(z - z_x) - iw_{0x}^2}{2(z - z_y) - iw_{0y}^2}, \quad (2)$$

where A_0 is a constant. The vortex moves during propagation and it is located at the transverse coordinates that satisfy

$$\frac{x_1(z)}{q_x} + iA_0 \frac{w_{0y}^2}{w_{0x}^2} \frac{y_1(z)}{q_y} = \frac{x_{1,0}}{iw_{0x}^2}. \quad (3)$$

When $z_x = z_y$ and $w_{0x} = w_{0y}$, $A(z)$ is constant during propagation and hence so is the topological charge of the vortex. Nevertheless, as shown in Fig. 2a, this is not the case when the noncanonical character of the vortex is taken into account, and the charge of the vortex inverts sign. When the vortex is located exactly on axis (i.e., $x_{1,0} = 0$), the inversion of the vortex charge occurs through a vortex trajectory [7] orthogonal to the light propagation direction (Fig. 2c). In the experimental plane of observation, which corresponds to Freund's *critical foliation* for this case [8], such trajectory appears as an edge-line dislocation. When either $x_{1,0} \neq 0$ or the beam is not exactly elliptical, as is the case in any experiment, the vortex trajectory is predicted to feature two extremely sharp turns towards and from the transverse infinity (Fig. 2d). An analogous behavior as above is obtained in more complex linear and non-linear media, where, in general, the vortex evolution can be traced by solving numerically the paraxial wave equation, and accurate approximations of the evolution of $A(z)$ can be derived using multiple-scale expansion techniques around the vortex core [21,22]. As an illustrative example with relevance to evolution of vortices in trapped weakly interacting Bose-Einstein condensates [19,20], shown in Fig. 2b, is the evolution of an initially canonical vortex in an asymmetric graded-index (GRIN) linear medium (the GRIN medium plays the role of a magnetic trap for the BEC). In this case, the charge of the noncanonical vortex is predicted to change periodically along propagation.

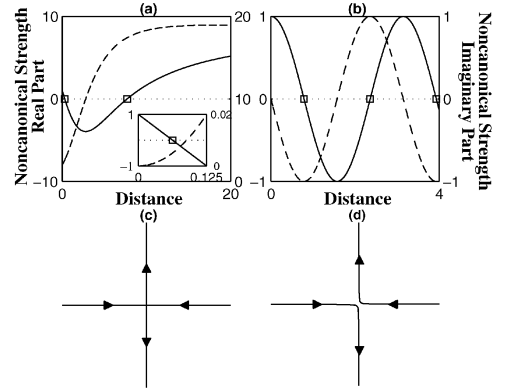


FIG. 2. Predicted evolution of the noncanonical strength of a vortex under different propagation conditions. Continuous and dashed lines show the evolution of the real and imaginary parts of A , respectively. Axes are in dimensionless units. The squares highlight the critical propagation distances where the inversion of the topological charge of the vortex occurs. (a) Evolution of a noncanonical vortex with $A(0) = 1 + 2i$, nested in an elliptical Gaussian beam with $w_{0x} = 1$, $w_{0y} = 3$. Inset: Predicted evolution of the noncanonical strength of the vortex under the conditions of our experimental setup. (b) Evolution of an input canonical vortex nested in an elliptical beam propagating in a GRIN medium with refractive index $n(x, y) = 2x^2 + 8y^2$. The widths of the Gaussian beam are the stationary modes of the medium: $w_x = 1$, $w_y = 1/\sqrt{2}$. (c), (d) Sketch of the Berry trajectories followed by on-axis and slightly off-axis vortices, respectively, when they undergo a topological charge inversion.

The experimental realization of an evolving noncanonical vortex can be performed in different settings. In an effort to highlight the universality of the phenomenon and to rule out complexity that might make the origin of the observations controversial, we performed our experiment to realize the evolution shown in Fig. 2a using an astigmatic optical scheme in free space. A linearly polarized, collimated light beam produced by a He:Ne laser at a wavelength 632.8 nm was split into two beams. Single-charge screw wave front dislocations were nested in one of the beams by using computer-generated holograms [11]. We used off-axis holograms, which were designed to generate traditional canonical vortices, to optimize the separation of the resulting Fresnel diffraction orders. The other beam was used as a reference wave. The beam after the holographic mask was spatially filtered to extract the diffraction order containing the single-charge vortex and then passed through a lens of 10 cm focal length. The wave front of the resulting beam was analyzed by superposing the beam with the vortex nested and the reference wave tilted slightly relative to the propagation axis. The interference pattern arising exhibits a characteristic fork at the position of the dislocation and its topological charge is visually evaluated. The pattern of output light was analyzed by a charge-coupled device (CCD) camera. Noncanonical vortex dynamics were generated by operating the system with a

cylindrical lens. The left and central rows of Fig. 3 show the observed outcome at different planes after the lens. As the ellipticity of the beam increases while it focuses, the vortex core becomes increasingly eccentric. At a crucial instant of the evolution the vortex converts itself into what, according to Figs. 2c and 2d, is believed to be a transverse vortex trajectory which inside the finite experimental accuracy always appears as an edge-line dislocation. Subsequently the quasiedge is observed to transform itself into a vortex of opposite topological charge relative to the input value. The orientation of the edge line in the plane is dictated by the value of the noncanonical strength. Notice that the inversion of the helicity of the spiral phase front of the vortex field is fully consistent with the imaging properties of the lens. We verified that the charge of a traditional canonical vortex does not vary by operating the system with a spherical lens (Fig. 3, right row).

We note the one-to-one map existing between the astigmatic scheme employed in the experiment and the evolution displayed in Fig. 2a. We stress that the observed inversion of the topological charge of the vortex does not occur at the surface of the lens, i.e., it does not involve a wave front dislocation induced by external discontinuities. We also stress that it occurs in free space, thus it is not related to any instabilities mediated with nonlinear effects or vortex solitons [10]. Rather, on the light of the relationship between general screw and edge dislocations studied by Nye [9], and of the generic vortex trajectories reported by Freund [8], it is believed to be a genuine and fundamental dynamic self-effect of noncanonical vortices.

Because of the associated spiral wave front, paraxial light beams with nested vortices carry orbital angular mo-

mentum [12] that can be transferred to trapped suitable material particles causing them to rotate [5]. The total angular momentum can also contain a spin contribution, associated with polarization [13]. In the case of canonical vortices nested in symmetric beams, the orbital angular momentum, in units of $\hbar\omega$, is given by $L = mN$, where m is the topological charge of the vortex and N is the number of photons. In our experiment $m = 1$, thus for the beam incident upon the lens $L/N = 1$. We verified that in spite of the inversion of the topological charge of the vortex, the angular momentum of the beam is conserved during its evolution. Such angular momentum could be modified by passing the noncanonical vortex through additional optical elements [14], but otherwise in free space L is constant so that $L/N = 1$ everywhere. Therefore, contrary to traditional views, beyond the point where the charge of the vortex is inverted, the orbital angular momentum carried by the beam and the topological charge of the vortex nested in it have opposite signs. This fact has a simple interpretation. The beam incident upon the cylindrical lens is an almost pure Laguerre-Gaussian mode. The astigmatic lens redistributes the incoming orbital angular momentum into a superposition of Laguerre-Gaussian modes carrying different charges (Fig. 4), while keeping the total value constant. Therefore, the density of angular momentum is dynamically redistributed across the beam during its evolution.

We now consider the implications of the above to quantum photon states. The Laguerre-Gaussian modes form a complete Hilbert set and can thus be used to represent the quantum photon states within the paraxial theory of light propagation. The quantum angular momentum

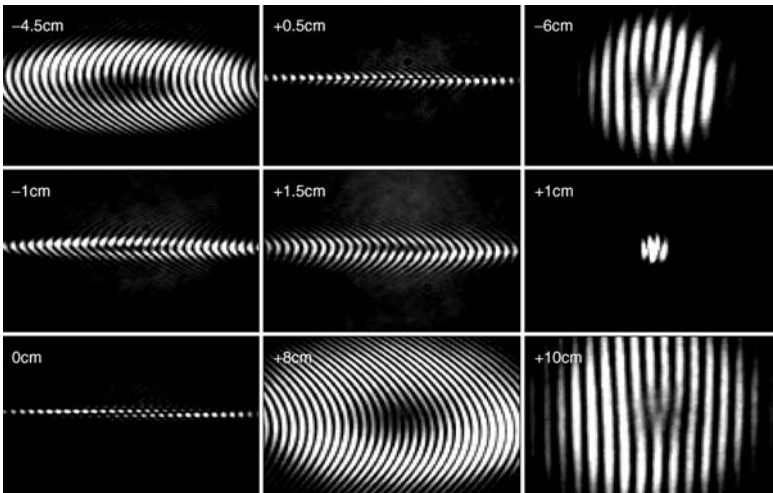


FIG. 3. Observed evolution of the topological charge of a vortex during a noncanonical evolution. The pictures show the evolution on propagation of the interference between the vortex beam and a reference beam tilted relative to the propagation axis. The images were acquired by a CCD camera placed at different positions beyond the lens. Labels inside the pictures stand for the location of the CCD camera relative to the lens focal plane. Left and center rows: Experimental realization of the noncanonical evolution generated by an input canonical vortex nested in a radially symmetric Gaussian beam with a 5 mm waist, passed through a cylindrical lens with a 10 cm focal length. Right row: Observed light pattern under identical conditions but with a spherical lens.

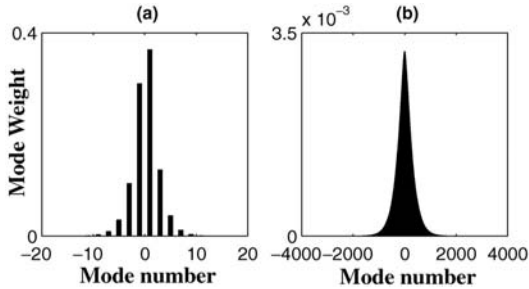


FIG. 4. Decomposition into normal modes with a well-defined angular momentum. Modes are numbered according to their topological charge, or azimuthal index. (a) Mode decomposition for the beam parameters of Fig. 2a. (b) Mode decomposition for the light beam with the noncanonical vortex nested used in the experimental observations, revealing that it contains 10^3 modes.

number carried by the photon is then represented by the charge, or azimuthal index m , of the corresponding mode. The photons incident upon the lens are in an almost pure angular momentum state $|m = 1\rangle$, but after the lens the photon state is a superposition of orbital angular momenta $|m\rangle$ (Fig. 4). The outcome is that, even though a full quantum average over many realizations yields the classical expectation $\langle L \rangle / n = 1$, with n the mean photon number, the quantum angular momentum number obtained in most of the realizations is not $m = 1$. We thus notice the direct implications to the generation and teleportation of quantum entangled photons with angular momentum [15,16] by parametric down-conversion of light in quadratic nonlinear crystals pumped with noncanonical vortex beams [23].

We finally point out that our observations here showing vorticity transformations of complex fields through noncanonical vortex trajectories and edge dislocations may provide new insight on the issue of how vorticity enters a different physical system governed by similar evolution equations. In particular, they may illuminate an old issue in superfluidity [17,18], namely, how do quantized vortices enter into weakly interacting Bose-Einstein condensates trapped in a container that is stirred or rotated [19,20]. Our observations on the self-transformation of noncanonical vortices open up a new broad scenario where general dislocations, including edge dislocations, might evolve dynamically from the edge of the container towards the interior where the vortices are stabilized by repulsive many-body nonlinearities. They also suggest the possibility of dynamic charge inversion of a weakly interacting Bose-Einstein condensate vortex under the action of an asymmetric trapping potential.

In conclusion, we emphasize that the observed self-inversion of the topological charge of the vortex occurs through its dynamic propagation, and it does not involve discontinuities or nonlinearities or the nucleation of twin vortex pairs. Our observations reported here were performed with vortices nested in light beams, but the phe-

nomenon is believed to be a generic effect of noncanonical vortices, thus the results have implications to all systems employing noncanonical vortex entities.

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Orbital angular momentum exchange in cylindrical-lens mode converters

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Abstract

Cylindrical-lens mode converters (Beijersbergen M W, Allen L, van der Veen H E L O and Woerdman J P 1993 *Opt. Commun.* **96** 123–32) are used to transform between Hermite–Gaussian and Laguerre–Gaussian modes with a resulting transfer of angular momentum to the light beam and a corresponding torque on the lenses. By numerically analysing both the total and local angular momentum of the light beam, we explain the origin of this torque and confirm that is not evenly distributed between the lenses. We also confirm that any vortex contained within the beam may change sign even when the orbital angular momentum of the beam remains constant.

Keywords: Orbital angular momentum density, cylindrical-lens mode converter, vortex inversion

It is now well understood that light beams can carry an orbital angular momentum about the beam axis associated with helical wavefronts. Allen *et al* [1] showed that any beam, of which the Laguerre–Gaussian mode is perhaps the best known example, which has an $\exp(i\ell\phi)$ helical phase structure carries an orbital angular momentum of $\ell\hbar$ per photon.

One method for producing, or transforming, pure modes is to use a pair of cylindrical lenses to convert from high-order Hermite–Gaussian to a Laguerre–Gaussian mode of the same mode order. This form of mode converter has been fully described by Beijersbergen *et al* [2]. Two forms of the mode converter are particularly useful. For a beam with Rayleigh range $(1 + 1/\sqrt{2})f$, and for lenses separated by $\sqrt{2}f$, a $\pi/2$ converter transforms any Hermite–Gaussian mode with indices m, n to a Laguerre–Gaussian mode with indices $l = m - n$ and $p = \min(m, n)$. The other, arguably notional, is a π converter, where two identical cylindrical lenses, of focal length f , are separated by $2f$. For a collimated beam, a π converter forms a mirror image of the beam such that, for a Laguerre–Gaussian mode, the azimuthal mode index l is transformed to $-l$. As these mode converters introduce phase shifts of $\pi/2$ or π between orthogonal modes, they are analogous to waveplates which introduce phase shifts between orthogonal polarization. The analogy extends to geometrical construction of a Poincaré sphere [3] and mathematical representations in the form of the Jones matrices [4].

In general, the passage of a light beam through either of these mode converters results in a change in the azimuthal mode index of the constituent Laguerre–Gaussian modes and

hence in a change of the orbital angular momentum of the light beam. Clearly, such a change in angular momentum results in a torque on one, or both, of the lenses. It is the purpose of this paper to examine the physical origin of this torque and to identify how it is divided between the lens pair.

To examine the nature of the light beam at each point in the optical system we employ a plane wave decomposition of the type used by Sziklas and Siegman [5] and used recently to look at imperfections within mode converter design [6]. This allows the calculation of the field in any subsequent plane for an arbitrary field in any plane. Once the amplitude of the wave is known, the momentum density can be calculated from [7]

$$\mathbf{p} = \frac{\varepsilon_0}{2} (\mathbf{E}^* \times \mathbf{B} + \mathbf{E} \times \mathbf{B}^*)$$

$$= i\omega \frac{\varepsilon_0}{2} (u^* \nabla u - u \nabla u^*) + \omega k \varepsilon_0 |u|^2 \mathbf{z} + \omega \sigma \frac{\varepsilon_0}{2} \frac{\partial |u|^2}{\partial r} \Phi \quad (1)$$

where $u \equiv u(r, \phi, z)$ is the complex scalar function describing the distribution of the field amplitude and σ describes the degree of polarization of the light; $\sigma = \pm 1$ for right- and left-hand circularly polarized light, respectively, and $\sigma = 0$ for linearly polarized. For a Laguerre–Gaussian mode, $u(r, \phi, z)$ is given by

$$u(r, \phi, z) = \frac{C_{pl}^{LG}}{w(z)} \frac{z_R}{(z_R^2 + z^2)} \left[\frac{r\sqrt{2}}{w(z)} \right]^l L_p^l \left[\frac{2r^2}{w^2(z)} \right]$$

$$\times \exp\left(\frac{-r^2}{w^2(z)}\right) \exp\left(\frac{-ikr^2 z}{2(z^2 + z_R^2)}\right)$$

$$\times \exp(-il\phi) \exp\left(i(2p + l + 1) \tan^{-1} \frac{z}{z_R}\right) \quad (2)$$

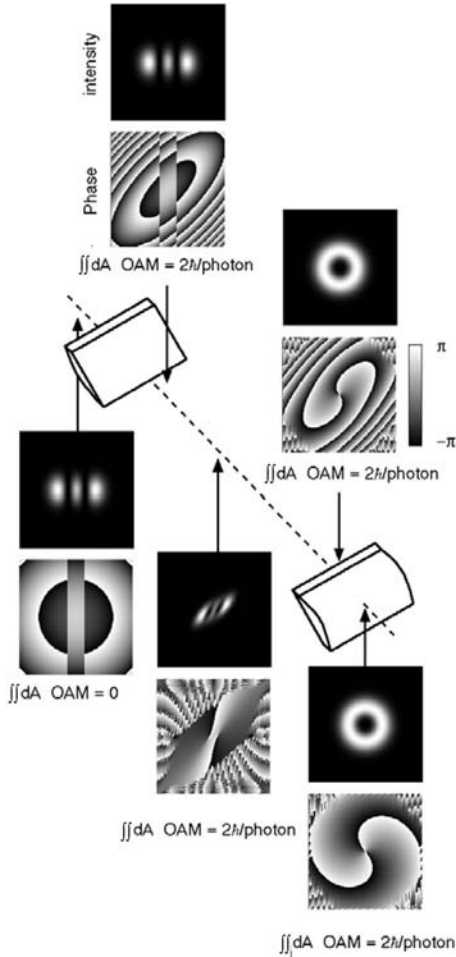


Figure 1. A $\pi/2$ converter comprising two cylindrical lenses of focal length f separated by $\sqrt{2}f$. An input Hermite–Gaussian mode of Rayleigh range $(1 + 1/\sqrt{2})f$ is converted to a Laguerre–Gaussian. Numerical methods are applied to calculate the intensity and phase structure at any point in the system. The angular momentum transfer takes place exclusively at the left-hand lens. (Note that, far from the beam axis in regions of low intensity, the numerical calculation of the phase is subject to noise.)

where C_{pl}^{LG} is the normalization constant, L_p^l is a generalized Laguerre polynomial, $w(z)$ is the radius of the beam at position z and z_R is the Rayleigh range. In addition to the azimuthal mode index, l , there is a second mode index, p , which describes the number of radial nodes in the field. Note that all modes of the same mode order, $N = (2p + l)$, have the same Gouy phase.

The cross product of the momentum density (equation (1)) with the radius vector $\mathbf{r} \equiv (r, 0, z)$ yields an angular momentum density. The angular momentum density in the z direction depends upon the Φ component of $\mathbf{p}$, such that

$$j_z = rp\phi. \quad (3)$$

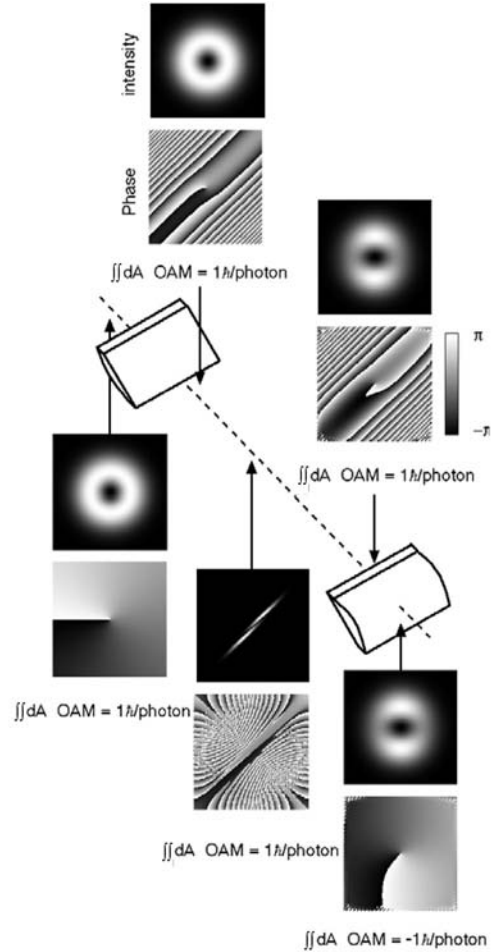


Figure 2. A π converter comprising two cylindrical lenses of focal length f separated by $2f$. A collimated Laguerre–Gaussian mode with azimuthal index l is converted to a Laguerre–Gaussian with index $-l$. Numerical methods are applied to calculate the intensity and phase structure at any point in the system. Despite the reverse in vortex sign which occurs about the centre position, the angular momentum transfer takes place at the right-hand lens. (Note that, far from the beam axis in regions of low intensity, the numerical calculation of the phase is subject to noise.)

The expression for j_z can be numerically evaluated to give both a local value of the angular momentum density and an integrated value over the whole beam. Figure 1 shows a schematic of a $\pi/2$ converter and the numerically calculated intensity, phase and total orbital angular momentum in different planes.

As argued by Beijersbergen *et al* [2], we see that the angular momentum transfer occurs only at the lens where the Hermite–Gaussian mode enters or leaves the mode converter. The reasoning behind their observation was that the mechanism for such a momentum transfer lies in the gradient force that acts on any dielectric object, producing a force directed towards the high intensity region of the beam. A beam with a cylindrically

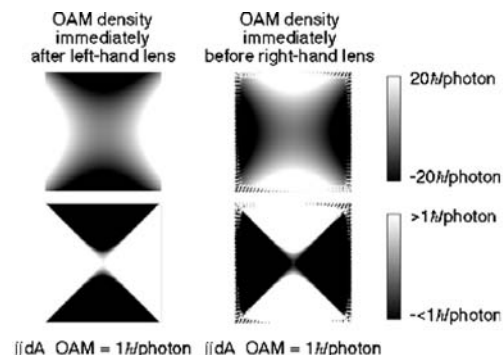


Figure 3. The local orbital angular momentum per photon, calculated in the planes immediately after and before the left and right lenses, respectively. For these conditions, the density reaches $\pm 20\hbar$ (upper plots). When integrated over the whole beam cross section the total orbital angular momentum in both planes corresponds to an average of $\hbar$ per photon. However, careful inspection (lower plots) in the region of the vortex shows that its sign has changed from $+1$ to -1 .

symmetric intensity cannot therefore create a torque upon a transparent object. In contrast, the rectangularly symmetric Hermite–Gaussian modes, when rotated by 45° with respect to the axis of the cylindrical lens, do create a torque on the lens which acts to align the optically thick axis of the lens with the long axis of the mode.

A physical understanding of the π converter is somewhat more difficult. In the case of a collimated Laguerre–Gaussian input mode we expect the converter to transform the azimuthal mode index from l to $-l$, thereby introducing an angular momentum exchange of $2\hbar$ per photon. However, as both the input and output Laguerre–Gaussian beams have a circularly symmetric intensity, how can either exert a torque on the lenses? The resolution of this paradox was suggested by both Beijersbergen *et al* [2] and van Enk and Nienhuis [8]. Due to the diffractive spreading of the beam between the lenses, even a perfectly aligned mode converter still results in a residual ellipticity of the output Laguerre–Gaussian mode. The slightly elliptic ‘Laguerre–Gaussian’ mode has its major axis at 45° to the lens axis and hence does exert a torque. Figure 2 shows a schematic of a π converter and the numerically calculated intensity, phase and total orbital angular momentum at different positions. The slight ellipticity of the transformed beam is evident in the calculated intensity distributions. In general, the degree of ellipticity is reduced by using a beam of larger diameter, where the diffractive spreading between the lenses is reduced. Indeed the larger diameter means that the degree of ellipticity needed to produce the required torque is smaller.

As can be seen from the calculated phase distributions, figure 2, as the beam propagates between the lenses the sign of the optical vortex is reversed. The inversion of the topological

charge by cylindrical focusing has recently been observed by Molina-Terriza *et al* [9] and might be thought counter-intuitive, as the vortex within the optical field is often described as carrying the orbital angular momentum. However, orbital angular momentum is not linked directly to vortices [10] but rather to an azimuthal phase term. Figure 3 shows the calculated orbital angular momentum density immediately after and immediately before the first and second lenses, respectively. If integrated over the whole of the beam, the total orbital angular momentum between the lenses remains constant. We do, however, note that in the vicinity of the beam axis the optical angular momentum density reverses sign, as does the sign of the optical vortex.

The numerical calculations we have performed within this work are compatible with the reasoning of Beijersbergen *et al* [2] and van Enk and Nienhuis [8]. It confirms that the inversion of the topological charge upon cylindrical focusing observed by Molina-Terriza *et al* [9] still occurs, even when the overall orbital angular momentum of the beam is conserved.

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Laser beams with screw dislocations in their wavefronts

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Coherent optical fields with wavefront dislocations of various orders have been produced experimentally and studied during the passage of a laser beam through a multimode waveguide and during diffraction by some holograms which have been synthesized.

Wavefront dislocations¹ or optical vortices have attracted interest as some of the new entities which exist in optical fields with a complex spatial structure or in laser cavities with a large Fresnel number.² When there is a screw dislocation, the wavefront is a common helical surface (right-handed or left-handed) with a singularity. When this singularity is circumvented, there is a phase shift of some multiple of 2π . The wavefront surface may have a singularity only where the modulus of the complex field amplitude E vanishes, i.e., where the real and imaginary parts are simultaneously zero. First-order dislocations (those for which the change in phase is 2π) have been observed in the speckle field of scattered coherent light.¹

Our purpose in the present study was to produce, and learn about the properties of, regular optical fields with screw dislocations, including dislocations of higher orders.

To study the structure of the field of an isolated dislocation, we used an experimental apparatus (Fig. 1) consisting of a Mach-Zehnder interferometer with a length of braided multimode optical fibers in one arm. The beam from a helium–neon laser with a Gaussian mode is focused into one of the fibers of the braid. At the exit from this fiber we observe a pattern consisting of two or three spots of irregular shape. A diverging lens is placed in the reference arm of the interferometer in order to equalize the divergences of the interfering beams. By rotating the semitransparent exit mirror of the interferometer, we are able to observe an interference pattern in various regions of the beam emerging from the waveguide. The customary annular interference pattern is observed at the light intensity maxima. If there is instead a dislocation at the center of the interference region, the pattern becomes a helix (Fig. 1b), clearly demonstrating the existence of a helical wave surface. At the center of the helix there is always an intensity zero of the beam emerging from the waveguide.

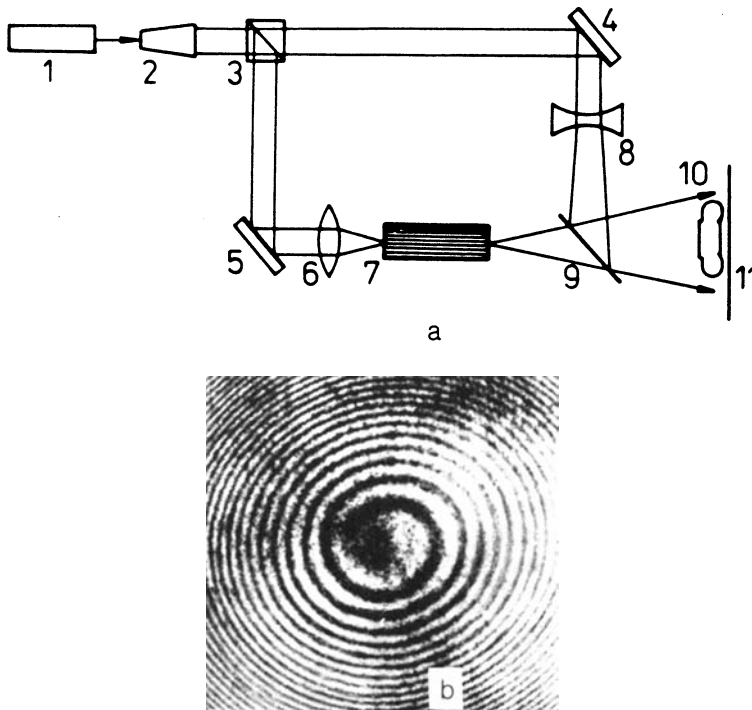


Figure 1. a: Experimental setup. 1 – He–Ne laser; 2 – beam expander; 3 – beam splitter; 4, 5 – mirrors; 6 – objective; 7 – braided optical fibers; 8 – diverging lens; 9 – half-silvered mirror; 10 – camera; 11 – screen for observing the interference of the beams. b: Interference pattern formed by the spherical reference wave and a wave with a screw wavefront dislocation.

When the interfering beams are incident on a screen at a small relative angle, we observe a splitting or disappearance of an interference fringe, similar to the event described in Ref. 1 for dislocations in speckle fields.

The dislocations observed experimentally in the setup in Fig. 1a had only the plus first or minus first order (the change in phase upon a circumvention was $+2\pi$ or -2π , respectively). The numbers of right-handed and left-handed screw dislocations were the same. It is extremely unlikely that dislocations of higher orders would be observed in this setup, since such an observation would require the intersection at one point of more than two pairs of lines on which the conditions $\text{Re } E = 0$ and $\text{Im } E = 0$ hold. In the interference pattern of the beams this situation would correspond to a splitting of the interference fringe into four or more new fringes. In order to produce beams with dislocations of higher orders, we accordingly synthesized some amplitude holograms to simulate the interference field of a plane wave and a wave carrying a dislocation of the desired order. After a numerical calculation was carried out, and the results displayed on a monitor, reduced copies were made on photographic film with a period of 0.1 mm and a size of 2×2.5 mm. Figure 2 shows the gratings which were synthesized.

The properties of the beams diffracted by these holograms were studied in a setup similar to that shown in Fig. 1a, in which the braided fiber and the focusing objective were replaced by this grating. A beam expander was not used in this case. An annular structure of the beams was clearly observed in the diffraction orders. With increasing

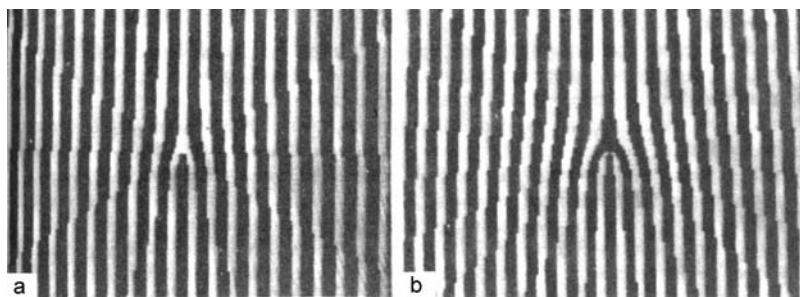


Figure 2. a – The hologram which was synthesized in order to produce beams with a first-order dislocation; b – the same, for a second-order dislocation.

order of the dislocation, the intensity dip at the center of the beam acquired progressively sharper edges. For the dislocations of higher orders, the pattern of the interference with the reference wave takes the form of several nested helices, in a number equal to the order of the dislocation. The helices corresponding to positive and negative diffraction orders wind in opposite directions. The transverse structure of the beams is not substantially altered as they propagate. This result means that the conclusion that there is an instability of optical fields with higher-order dislocations¹ requires refinement.

We have also observed an increase in the order of the dislocation in diffracted beams of higher orders (in precise correspondence with the order of the diffraction), but the quality of the beams produced in this manner turned out to be extremely low.

In summary, this study has demonstrated that it is possible to produce coherent light beams with a regular transverse structure which have screw wavefront dislocations. Beams with high-order dislocations have been produced for the first time with the help of some holograms which have been synthesized.

We wish to thank SG Odulov and VV Shkunov for useful discussions.

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Laser beams with phase singularities

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Phase singularities in an optical field appear as isolated dark spots and can be generated in active laser cavities or by computer generated holograms. Detection and categorization of these singularities can easily be achieved either by interferometry or Fourier transform pattern recognition using a computer generated hologram.

1. Introduction

Most laser beams have essentially spherical wavefronts and any deviation from sphericity constitutes a degradation of beam quality. However, the TEM_{01}^* ‘doughnut’ mode, often observed in high-power lasers, can exhibit a *helical* wavefront structure, associated with a phase singularity on the beam axis [1]. This occurs when the frequency-degenerate TEM_{01} and TEM_{10} modes oscillate simultaneously in phase quadrature.

The electric field will have the form

$$\begin{aligned} E_{01}^* &= E_0[(x \pm iy)/\omega] e^{-(x^2+y^2)/\omega^2} e^{-ikr^2/2R} e^{i(kz+\Phi)} \\ &= E_0(r/\omega) e^{\pm i\theta} e^{-(x^2+y^2)/\omega^2} e^{-ikr^2/2R} e^{i(kz+\Phi)} \end{aligned} \quad (1)$$

where Φ is the Guoy phase shift, R is the wavefront radius of curvature, ω is the spot size and r and θ are polar coordinates in the X - Y plane. Notice that during any complete circuit around the axis the phase changes by $\pm 2\pi$, expressing the helical form of the wavefronts, and that the field goes to zero on the axis, as it must where the phase is undefined.

The doughnut mode is a stable cavity mode because it is a linear combination of Hermite–Gaussian TEM_{01} and TEM_{10} modes and propagates in a self-similar way in free space and through optical systems for the same reason. In fact other more complex combinations containing phase singularities can be produced from higher-order modes. According to resonator theory [2] Hermite–Gaussian modes TEM_{mn} with the same total $m + n$ (or Laguerre–Gaussian modes TEM_{pl} with the same total $2p + l$) are frequency degenerate and can form such combinations. The singularities show up as multiple isolated irradiance zeros in the modal spot pattern.

One example is the TEM_{02}^* hybrid or ‘optical leopard’ [3] which can be constructed from the Gaussian–Hermite modes TEM_{02} and TEM_{20} . The pattern has a central irradiance peak surrounded by four smaller peaks and four zeros where there are two positive and two negative singularities diagonally opposed. Figure 1a shows the irradiance distribution and Fig. 1b the form of the wavefronts with four interconnected helices.

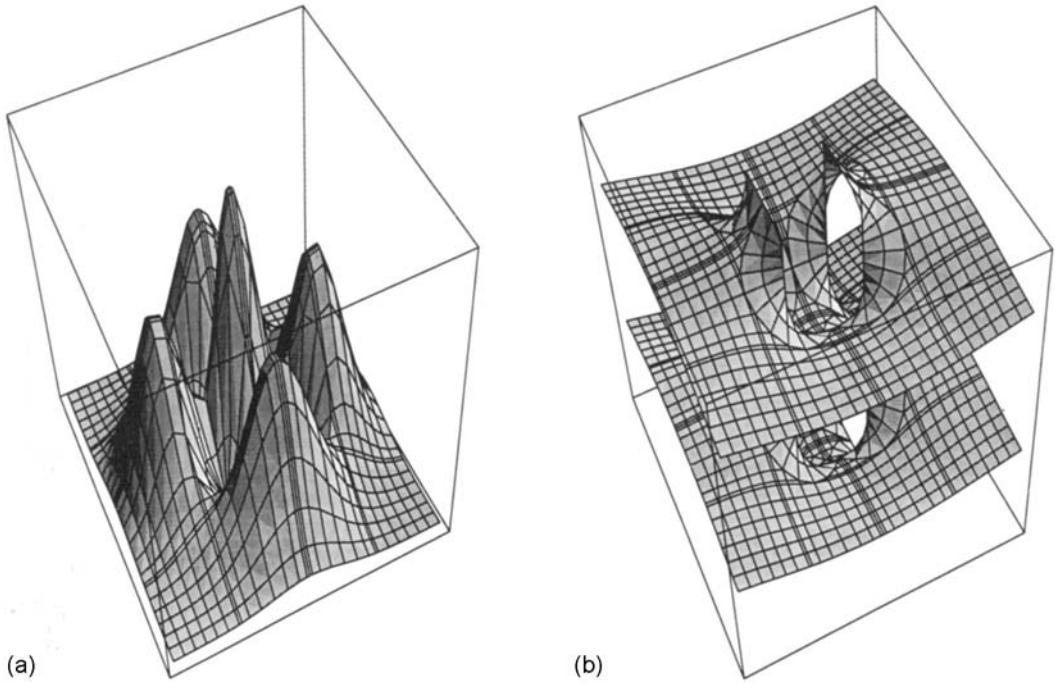


Figure 1 TEM_{02}^* or 'optical leopard'. (a) Irradiance distribution (vertical) with spatial position (horizontal). (b) Form of wavefronts. In the diagram, phase increases in the vertical direction. Two surfaces of constant phase differing by 2π , corresponding to a plane wave at large distances from the beam axis, are connected around the singularities.

Other examples contain higher-order singularities. The simplest cases are the higher-order doughnuts (which are just Gaussian–Laguerre TEM_{0n} modes)

$$E_{0n}^* = E_0(r/\omega)^n e^{-(x^2+y^2)/\omega^2} e^{\pm in\theta} e^{-ikr^2/2R} e^{i(kz+\Phi)} \quad (2)$$

The parameter n is often referred to as the 'charge' of the singularity, with zero charge corresponding to a Gaussian TEM_{00} beam. The irradiance profiles of several higher-order doughnuts are shown in Fig. 2. The wavefronts form multistart helices.

Of course, in a real laser, astigmatism in the cavity often removes the frequency degeneracy between such modes, but Brambilla *et al.* [3] have shown theoretically and experimentally that, so long as the astigmatism is not too severe, a cooperative frequency-locking process can occur, leading to a range of stable patterns.

However, it is important to remember that not all dark spots in patterns are necessarily phase singularities. If the frequency degeneracy of the contributing modes is broken, a rapidly time-varying pattern will result, the time average of which may still contain dark spots [1]. In some applications, involving only average irradiances, that will not matter, but in others it will be important. We show below how such cases can be distinguished experimentally.

2. Production of beams with singularities

There are several ways in which beams with singularities might be produced. As mentioned above, thanks to the process of cooperative frequency locking [3] such modes can arise

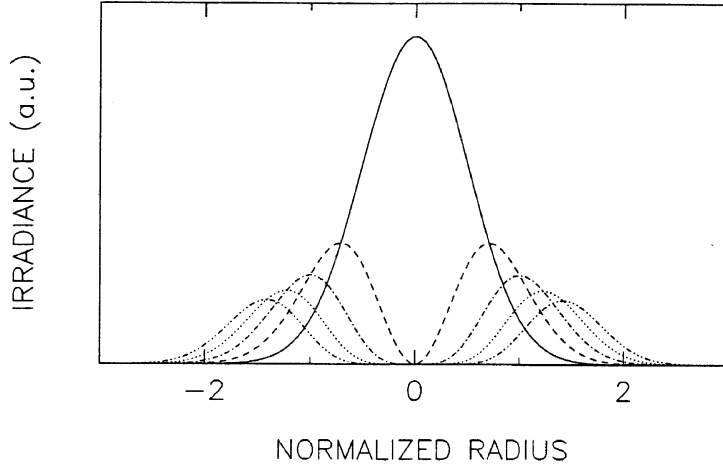


Figure 2 Irradiance profiles for Gaussian and doughnut modes up to charge 4. Each has the same total power.

spontaneously within a laser and in some cases a considerable degree of control can be exercised by the experimenter. Alternatively, a hologram can be used to convert part of the output of an existing laser into the desired beam. We consider both approaches below.

The 525 nm Na₂ vapour laser pumped by the Ar-ion laser [4] is ideal for studies of combination-mode formation because its velocity-selective optical pumping mechanism leads to very narrow gain linewidths so that only one ‘family’ of transverse modes with a certain $m + n$ or $2p + l$ sum can oscillate at a time, and the otherwise dominant TEM₀₀ Gaussian can be suppressed. We have used a system very similar to that used by Brambilla *et al.* [3] to generate a range of patterns and, as described below, to study methods to detect and classify modes with singularities.

A less efficient but a more flexible way to produce modes with singularities is through the use of computer-generated holograms. In our first experiments we used on-axis holograms which have the form of spiral Fresnel zone plates [5], but these suffer the same problem as Gabor’s original holograms, i.e. lack of separation between reconstructed beam and incident beam. It turns out that even crude off-axis binary holograms are quite effective, as will now be explained.

A hologram is really just a recording of the interference pattern between a field of interest and some simple reference field. For the relatively simple fields involved in modes with singularities it is possible to calculate the form of such patterns and plot them out. Let us take as an example a charge-one doughnut (Equation 1), at a beam-waist ($R \rightarrow \infty$) for simplicity. Consider the interference pattern on a screen in the X - Y plane when a plane reference beam

$$R = R_0 e^{ik_x x + ik_z z} \quad (3)$$

is incident at an angle $\phi = \sin^{-1}(k_x/k)$. The irradiance on a screen at $z = 0$ will be

$$\begin{aligned} I &= |(R_0 e^{ik_x x} + E_0(r/\omega) e^{i\theta} e^{-r^2/\omega^2})|^2 \\ &= R_0^2 + E_0^2(r/\omega)^2 e^{-2r^2/\omega^2} + 2R_0 E_0(r/\omega) e^{-r^2/\omega^2} \cos(k_x x - \theta) \end{aligned} \quad (4)$$

It is the last term which expresses the interference pattern. A photographic recording of this

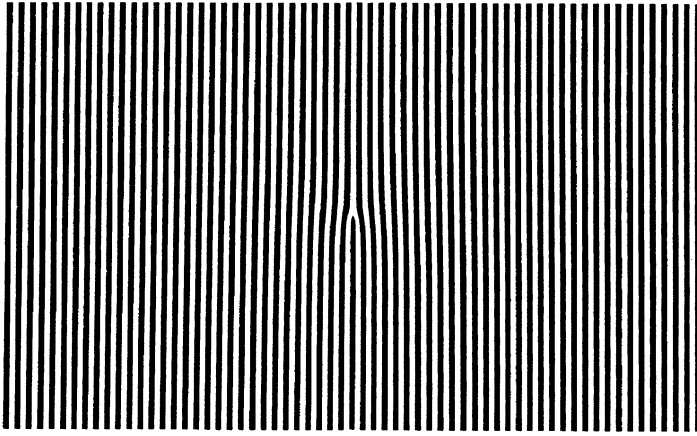


Figure 3 Pattern for off-axis binary hologram for charge-one singularity. The origin is at the tip of the fringe defect.

pattern can now act as a hologram capable of 'reconstructing' the original doughnut (and its complex conjugate) when illuminated by a wave given by Equation 3.

In what follows it will be convenient to work with a simplified pattern which ignores the amplitude variation of the doughnut beam and retains only the important phase information in the form of a spatially varying transmissivity

$$T = \frac{1}{2}(1 - \cos(k_x x - \theta)) \quad (5)$$

Now consider the effect of illuminating this pattern with a Gaussian beam propagating along the axis. Just after the hologram the field will be

$$E_T = T A_0 e^{-r^2/\Omega_0^2} \quad (6)$$

where A_0 is the central amplitude and Ω_0 is the spot size of the beam, assumed plane at this point. Substituting for T , we find

$$E_T = (A_0/2) e^{-r^2/\Omega_0^2} - (A_0/4) e^{-r^2/\Omega_0^2} e^{i(k_x x - \theta)} - (A_0/4) e^{-r^2/\Omega_0^2} e^{i(-k_x x + \theta)} \quad (7)$$

This field can be recognized as consisting of a zero-order beam propagating along the axis, and two (conjugate) first-order diffracted beams, each of them containing a singularity of opposite charge.

In fact it is much easier to print binary holograms than ones incorporating the sinusoidal variations in optical density implicit in Equation 5. Thus we actually use a 'square wave' transmissivity function which can be expressed in the form

$$T = \frac{1}{2} - \sum_{n=1}^{\infty} \text{sinc}(n\pi/2) \cos[n(k_x x - \theta)] \quad (8)$$

Its appearance is shown in Fig. 3. It has the appearance of a grating with a defect where a stripe branches.

When this is illuminated, the output field will contain terms of the form

$$E_n = (A_0/4) \text{sinc}(n\pi/2) e^{-r^2/\Omega_0^2} e^{i(nk_x x - n\theta)} \quad (9)$$

Each is of course a diffraction order from the 'grating' and can be recognized as being

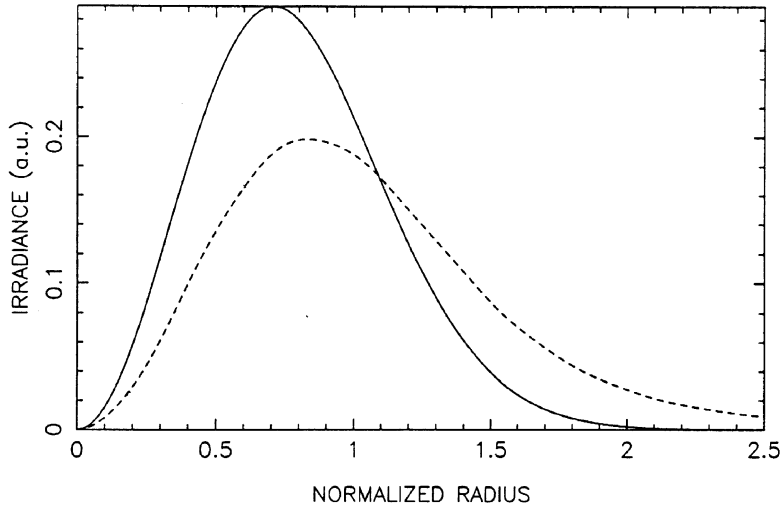


Figure 4 Comparison of the far-field spatial profile of the TEM_{01}^* mode (full line) with a first-order beam from a hologram illuminated by a Gaussian beam with the same parameters (dashed line). The TEM_{01}^* mode has the same power as the zeroth term of the series in Equation 13.

closely related to an n th-order doughnut propagating at an angle

$$\phi_n = \sin^{-1}(nk_x/k) \simeq n \sin^{-1}(k_x/k) \quad (10)$$

to the axis. Actually, Equation 9 has the form of a charge- n singularity embedded in a Gaussian beam and as such will not propagate in a self-similar way. To investigate the far-field spatial profile of this beam, we can decompose Equation 9 in terms of the orthogonal set of Gaussian-Laguerre modes:

$$\begin{aligned} \psi_{pl} = & \sqrt{\left(\frac{2p!}{\pi(p+l)!}\right)} \left(\frac{\Omega_0}{\Omega_1}\right) \left(\frac{\sqrt{2}r}{\Omega_1}\right)^{|l|} L_p^{|l|} \left(\frac{2r^2}{\Omega_1^2}\right) \\ & \times e^{-r^2/\Omega_1^2} e^{i l \theta} e^{-i k r^2/2R} e^{i(2p+l+1)\Phi_1} e^{i(nk_x x + k_z z)} \end{aligned} \quad (11)$$

where Ω_1 is the spot size, R_1 is the radius of curvature of the beam and Φ_1 is the Guoy phase shift. As the beam propagates, the parameters Ω_1 , R_1 and Φ_1 are related to the waist spot size Ω_0 via the standard Gaussian-Laguerre propagation laws [6]. The amplitudes of the terms of the series expressing the field at the hologram are given by

$$\begin{aligned} E_{pl} &= 0 \quad \text{if} \quad l \neq n \\ E_{pn} &= \frac{A_0}{4} \text{sinc}(n\pi/2)\Omega_0^2 \sqrt{\left(\frac{\pi p!}{2(p+n)!}\right)} \frac{n}{2} \frac{\Gamma(p+n/2)}{p!} \end{aligned} \quad (12)$$

In the far-field the spatial amplitude distribution is

$$E_{\text{far}} = \sum_{p=0}^{\infty} E_{pn} \psi_{pn} \quad (13)$$

where $\Phi_1 = \pi/2$. Figure 4 shows that the far-field amplitude distribution for the first-order beam closely resembles a TEM_{01}^* doughnut of slightly increased spot size. The helical structure of the wavefronts is identical. Similar results are obtained for higher orders. Thus,

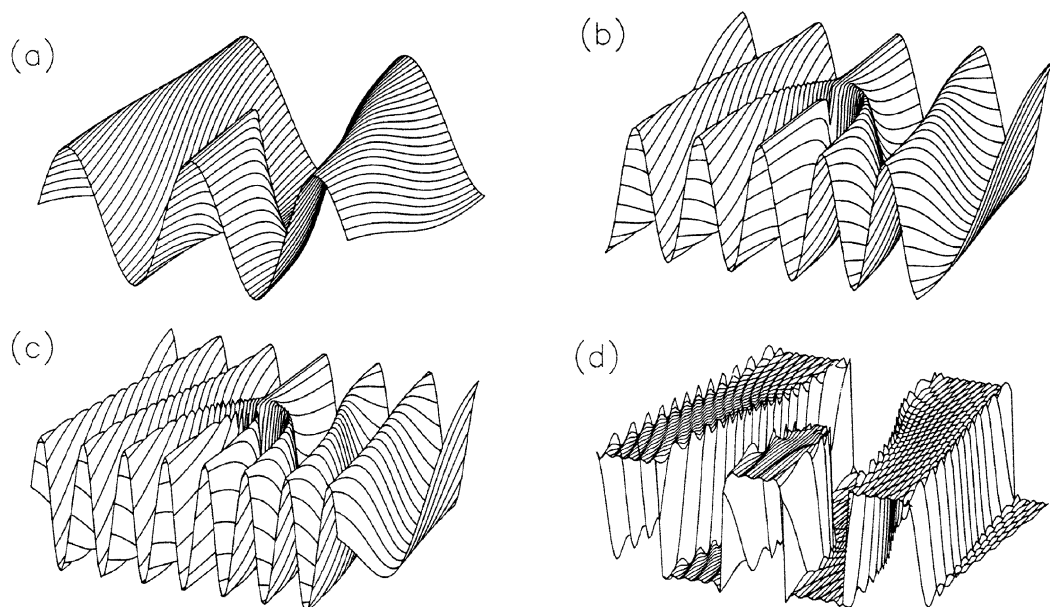


Figure 5 Decomposition of binary hologram pattern into harmonic components: (a–c) first, second and third components; (d) synthesis of first 30 components.

although the beams produced by our holograms are not strictly TEM_{0n}^* doughnuts, in the far-field they are, for all practical purposes, equivalent and for simplicity we refer to them as such.

The way in which the binary pattern produces multiply charged singularities can be understood by reference to Fig. 5, where the first three harmonic components of the very centre of the pattern in Fig. 3 are plotted. Each of Figs 5a to 5c can be recognized as the hologram for a successively higher-order singularity. Figure 5d shows the square-wave pattern synthesized from 30 components.

Figure 3 showed an example of a computed binary off-axis hologram pattern. This was printed by a laser printer on A4 paper and reduced by photographing onto half of a 35-mm slide. Figures 6a and 6b show charge-one and charge-two doughnuts produced as first and second orders, using a HeNe laser for illumination. The patterns, like all subsequent experimental ones, were recorded by an Electrim EDC-1000 CCD camera. About 5% of the incident power was coupled into the first order. This could be improved considerably by converting the present hologram into a phase hologram if facilities were available. Although the charge-two pattern is somewhat distorted, it shows clearly the larger diameter and narrower bright annulus expected. In fact, a symmetric grating like ours would be expected to produce a dim second-order term at best as it should effectively be a 'missing order'. We have also used holograms with four fringe defects to generate an 'optical leopard' pattern and still more complex beam shapes could be produced. It would even be possible to generate combinations involving transverse modes of different $m + n$ (or $2p + l$) sum (e.g. $\text{TEM}_{00} + \text{TEM}_{01}^*$) which could not be generated at a single frequency in a laser oscillator. These would suffer some change of shape during propagation owing to differential Guoy phase shift, but perhaps that could be put to some use.

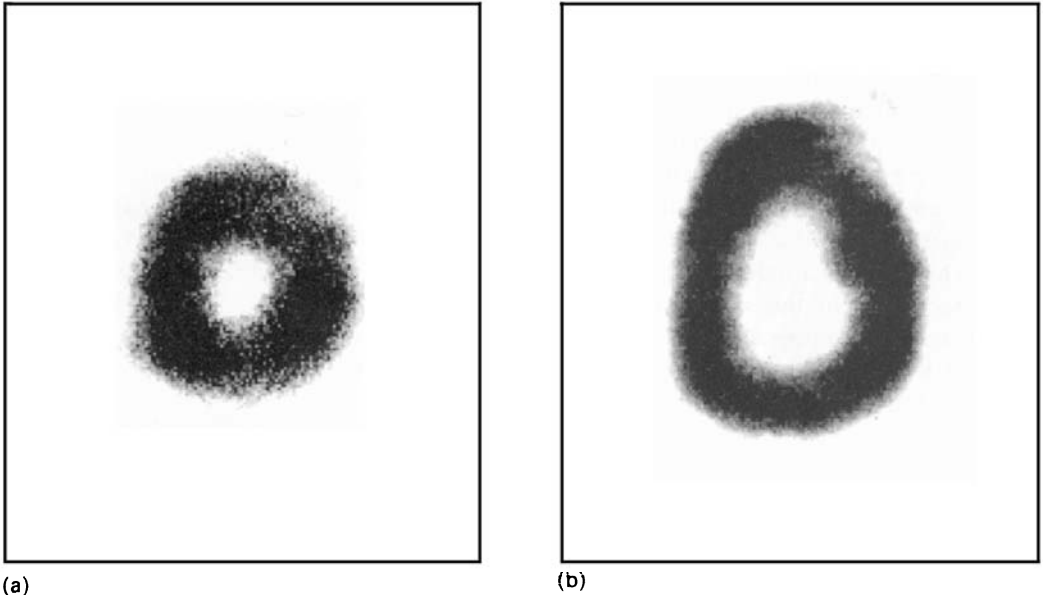


Figure 6 Doughnut beams produced by hologram using HeNe laser light: (a) charge one; (b) charge two.

3. Detection of phase singularity modes

In their studies of doughnut modes, Tamm and Weiss [7] used an astigmatic imaging technique to determine the charges of the singularities, but a more versatile approach is to measure the phase structure of the beam by interference, as had been done by Vaughan and Willetts [1]. Ideally, one should have available a coherent plane wave to act as reference, and in the vicinity of each singularity a fringe pattern with a defect, like that in Fig. 3, would be observed. Exactly this behaviour is demonstrated in Fig. 7, where interference patterns

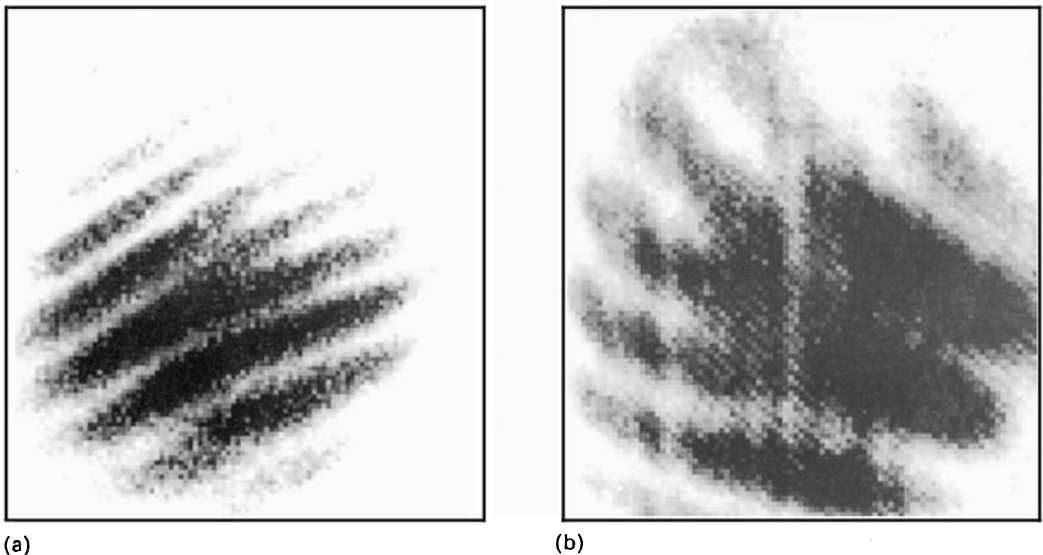


Figure 7 Interference of doughnuts of Fig. 6 with plane wave: (a) charge one; (b) charge two.

for the two holographically produced doughnuts shown in Fig. 6 are displayed. In the charge-one case, a bright fringe (dark in this print) splits into two; in the charge-two case a fringe splits into three.

When the singularity mode pattern is produced in a laser, it is not so easy to obtain a coherent plane wave for use as reference, and a simple split-beam technique is useful [1]. The beam of interest is split in a Mach-Zehnder interferometer and recombined with sufficient misalignment to produce straight fringes and to displace the two patterns so that the singularities in one fall in relatively uniform areas of the other where the phase varies only slowly. The result is a defect in the combined fringe pattern at each singularity position. Opposite charges in the same pattern fork in opposite directions, and corresponding singularities behave oppositely at the two points where they appear. Although the interference pattern becomes complicated, this works well when the pattern does not contain too many singularities [8].

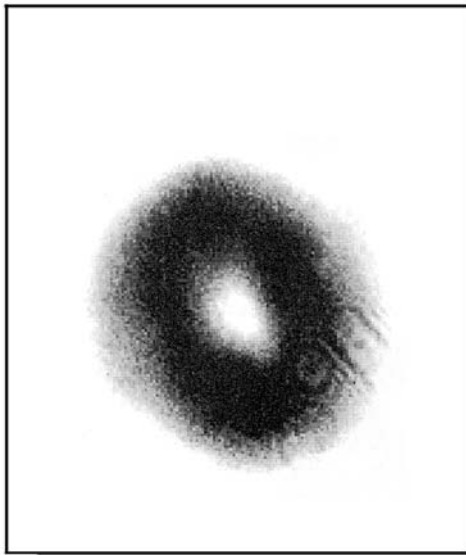
In interpreting these patterns it is important to keep in mind that the visibility of the fringes will be small near the singularity as the irradiance there is small, and that the exact form of the fringe splitting depends on the position of the fringes.

Figure 8a shows a slightly asymmetric doughnut produced by the Na_2 laser and Fig. 8b shows the corresponding split-beam interference pattern, indicating the presence of a charge-one phase singularity in the beam. For comparison Fig. 8c shows a split-beam interference pattern for an 'unlocked' doughnut, i.e. one where cavity astigmatism has broken the frequency degeneracy of the constituent modes. This shows the characteristic 'sideways shift' of the fringes inside a circular region as explained by Vaughan and Willetts [1]. The loss of frequency degeneracy leads to mode beating at a frequency of a few megahertz which can be detected with a photodiode.

An alternative means of detecting and classifying phase singularities in a beam is by optical Fourier transform recognition techniques where the holograms discussed above can be used as matched filters [9].

This has been demonstrated using the arrangement shown in Fig. 9. The Fourier transform of an input pattern is formed at the focal plane of a lens where the hologram for a pattern of interest, say a charge-one singularity, is placed. The transmitted field is Fourier transformed again by a second lens. At the output plane three beams can be distinguished — a central magnified image of the field at the input plane, and two 'first-order' fields. (Owing to the binary nature of our holograms, higher-order fields also appear but are generally too weak to be useful.) One first-order field gives the cross-correlation between the input field and the field used to make the hologram, and the other the convolution of the input field with the hologram field [9]. In this case it is more helpful to realize that this is also the cross-correlation between the input field and the conjugate of the hologram field. Thus, if a hologram for a charge-one singularity is used as a filter, a bright spot will appear in one field at each point where a positive charge-one singularity appears in the input field, and in the other field a bright spot will appear at each point where a negative charge-one singularity appears in the input.

This is shown in Fig. 10a where the singularity in a charge-one doughnut produced by the Na_2 laser is recognized. At the centre of the picture is the image of the input doughnut, flanked by the recognition fields (which are slightly magnified as a result of lens aberrations). The intense spot on the right indicates a charge-one singularity. Figure 10b shows the result when the input field is a 'leopard'. Here the diagonal placement of the four charge-one singularities, two of each sign, shows up clearly. An unlocked doughnut gives



(a)



(b)



(c)

Figure 8 Doughnuts produced by Na_2 laser. (a) Phase-locked doughnut. (b) Split-beam interference pattern for (a). Note two forks in the fringe pattern, indicating the presence of a singularity. (c) Split-beam interference pattern for unlocked doughnut.

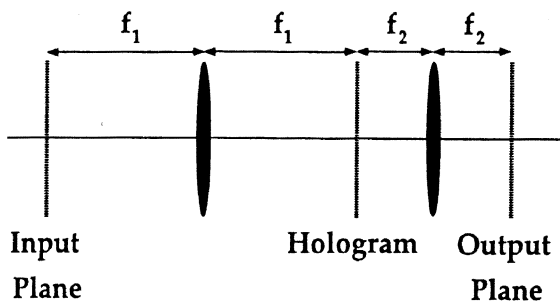


Figure 9 Arrangement for Fourier transform recognition of singularities in beams.

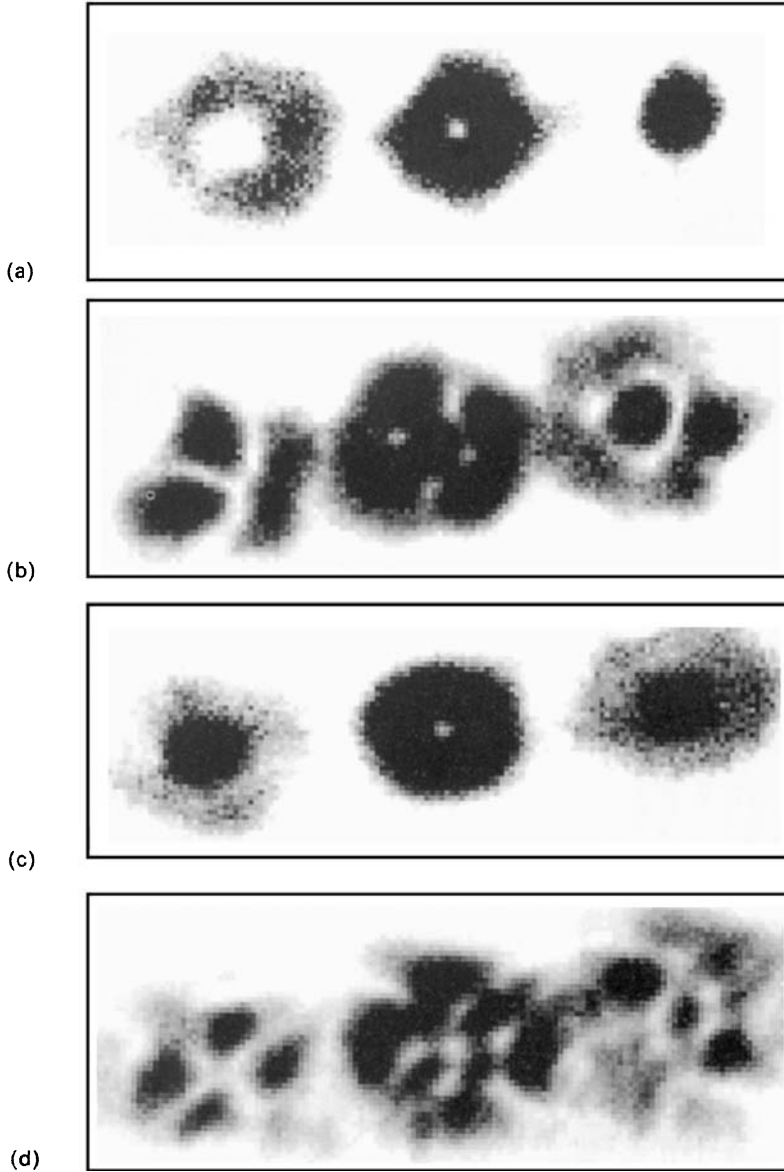


Figure 10 Fourier transform recognition of singularities. In each case an image of the input field appears at the centre, flanked by recognition fields for ± 1 charged singularities: (a) Charge-one doughnut from the Na_2 laser. (b) 'Leopard' from the Na_2 laser. (c) Unlocked $m + n = 1$ doughnut from Na_2 laser. (d) Seven-spot beam from Na_2 laser.

the result shown in Fig. 10c, as the actual pattern is rapidly changing, spending part of each beat period as a doughnut of each sign [1], so that bright spots appear in the recording on both sides. A more complex pattern from the Na_2 laser, with seven dark spots, is analysed in Fig. 10d, which shows that four charge-one singularities of one sign form a square with the three others of opposite sign lying along a line. Such a pattern, called a 'seven-hole' by Brambilla *et al.* [3], would be difficult to analyse using split-beam interference.

Although there is clearly room for improvement, the optical Fourier recognition technique has the advantage of needing only a single beam and providing output in the form of readily recognized spots rather than 'forks' in an interference pattern. Indeed, it is interesting to note that, presumably owing to the local rotational symmetry and self-similarity of the regions surrounding singularities, the technique is less affected by variations in scale-size or orientation than is, say, recognition of alphanumeric characters.

4. Applications

This work arose originally out of an interest in pattern formation and optical turbulence but it is clear that a 'technology' now exists for the production, detection, and classification of beams with phase singularities associated with isolated irradiance zeros. Many lasers can be run on the TEM_{01}^* mode, and well-designed phase holograms could convert the light from almost any laser into such a form, or a more complex pattern [10].

In some cases the helical phase structure is of importance, as in switching helicities as a means of information processing [7], but in many other cases any beam with a central minimum in irradiance (thus including unlocked doughnuts) could be of use. Note that the structure we have been discussing is entirely independent of polarization. Such a shape could be advantageous when it is desired to launch a beam through a reflecting telescope with a central obstruction. A higher-order doughnut would be appropriate for this.

Another area of potential application is in light-particle interactions [11]. The use of doughnut beams has been suggested for small-particle levitation [12] and more recently for the focusing of atomic beams [13].

Finally, it should be pointed out that speckle patterns formed by laser beams passing through inhomogeneous media have been shown to contain many phase singularities [14] presumably produced by random structures approximating the holograms discussed above.

5. Conclusions

We have shown that laser beams with phase singularities can be generated in lasers or can be produced from normal Gaussian beams using computer-generated holograms. The presence of the singularities can be detected and they can be classified either by reference to defects in interference patterns or by optical Fourier transform pattern-recognition techniques using computer-generated holograms as matched filters. Possible applications of the simplest of such beams, the TEM_{01}^* doughnut, include efficient launching of single-mode beams through telescopes and atom and particle trapping.

Acknowledgement

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Topological charge and angular momentum of light beams carrying optical vortices

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We analyze the properties of light beams carrying phase singularities, or optical vortices. The transformations of topological charge during free-space propagation of a light wave, which is a combination of a Gaussian beam and a multiple charged optical vortex within a Gaussian envelope, are studied both in theory and experiment. We revise the existing knowledge about topological charge conservation, and demonstrate possible scenarios where additional vortices appear or annihilate during free propagation of such a combined beam. Coaxial interference of optical vortices is also analyzed, and the general rule for angular-momentum density distribution in a combined beam is established. We show that, in spite of any variation in the number of vortices in a combined beam, the total angular momentum is constant during the propagation.
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INTRODUCTION

Light beams possessing phase singularities, or wave-front dislocations [1,2] have been studied intensively in linear and nonlinear optics [3–21], to reveal their basic properties, and for the sake of possible applications. At the singularity the phase becomes undetermined and the wave amplitude vanishes, resulting in a “dark beam” within a light wave. Phase singularities appear on wave fronts in diffuse light scattering, when a light wave has a form of speckle field, and each speckle has one screw wave-front dislocation in the vicinity [3], also known as an optical vortex [4]. At present, several different techniques are used to generate “singular beams:” synthesized holograms [5,6], phase masks [7–9], active laser systems [10–13], and low-mode optical fibers [14]. Recently the generation of phase singularities was observed as a result of light wave-front deformations caused by self-action in nonlinear media [15,16].

The general result of these investigations is the origin of a new chapter of modern optics and laser physics—singular optics, which operates with terminology and laws quite new to traditional optics. Phase singularities are topological objects on wave-front surfaces, and possess topological charges which can be attributed to the helicoidal spatial structure of the wave front around a phase singularity. This structure is similar to a crystal lattice defect, and therefore was at first known as a wave-front screw dislocation [1,2]. The interference of a wave possessing such wave-front screw dislocation with an ordinary reference wave produces a spiral fringe pattern [12,17,18], or, in the case of equal wave-front curvatures, radial fringes [19]. The number of fringes radiating from the center of the interference pattern equals the modulus of the topological charge, and the direction of the spiraling is determined by the sign of the charge and relative curvature of the wave fronts.

Optical vortices embedded in a host light beam behave in some degree as charged particles. They may rotate around the beam axis, repel and attract each other, and annihilate in

collision [20,21]. Other types of wave-front defects also may occur, such as edge or mixed screw-edge dislocations [22]. The possible transformations between edge and screw dislocations (in laser mode terms, 01 and doughnut modes) may be performed by modal converters [11,23,24].

The light field of a singular beam carries angular momentum [23] which may be transferred to a captured microparticle causing its rotation in a direction determined by the sign of the topological charge [25]. In nonlinear optics, so-called vortex solitons, which are singular beams in nonlinear medium, are a subject of growing interest [26,27].

However, singular beams demonstrate unusual properties even in linear optics, in free-space propagation. Optical phase singularities as morphological objects (tears of wave front) are robust with respect to perturbations. For instance, addition of a small coherent background does not destroy a vortex, but only shifts its position to another place where the field amplitude has a zero value. For vortices with multiple charge this operation will split an initially m -charged vortex into $|m|$ single-charge vortices [21]. Intuition suggests that the total topological charge would be conserved in a beam propagating in free space [1,2]. Our goal is to analyze the main properties of beams containing phase singularities in a general way, and demonstrate the limitations on the topological charge conservation principle for real beams, both theoretically and in experiment. On the basis of the present study, we establish the rules of angular-momentum transformations in light beams with phase singularities.

WAVE EQUATION SOLUTIONS POSSESSING PHASE SINGULARITIES

In this section we demonstrate some particular solutions possessing phase singularities of the scalar wave equation for a uniform isotropic medium

$$\nabla^2 E = \frac{1}{c^2} \frac{\partial^2 E}{\partial t^2}, \quad (1)$$

where E is the wave amplitude, c is the speed of light, and t is time. The existence of these solutions for a monochromatic light wave was first emphasized by Nye and Berry [1]. The derived complex amplitude of a light wave with frequency ω , wavelength λ , and wave vector $k = 2\pi/\lambda$ oriented along z axis has the form

$$E(\rho, \varphi, z, t) \propto \begin{cases} \rho^{|m|} \exp(im\varphi + ikz - i\omega t), & (2a) \\ \rho^{-|m|} \exp(im\varphi + ikz - i\omega t), & (2b) \end{cases}$$

where ρ , φ , and z are cylindrical coordinates. Both solutions have a phase depending on the azimuth angle φ multiplied by an integer m (positive or negative) called the topological charge of the phase singularity (optical vortex). The wave-front (equal phase surface) forms in space part of a helicoidal surface given by the equality $m\varphi + kz = \text{const}$. After one round trip of the wave front around the z axis there is a continuous transition onto the next (or preceding) wave-front sheet separated by $m\lambda$, which results in a continuous helicoidal wave-front surface. The topological charge attributed to this wave-front structure is positive for a right-screw helicoid ($m > 0$), and vice versa. In the case $|m| > 1$, the wave-front structure is composed from $|m|$ identical helicoids nested on the z axis and separated by the wavelength λ .

The solutions in forms (2a) and (2b) cannot describe any real wave because of the radial amplitude dependence which grows proportionally to ρ for Eq. (2a), and tends to infinity when $\rho \rightarrow 0$ for Eq. (2b). To avoid unwanted amplitude growth, we may combine the solution in Eq. (2a) with a Gaussian beam. Using the paraxial approximation of the scalar wave equation in the form

$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial E}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 E}{\partial \varphi^2} - 2ik \frac{\partial E}{\partial z} = 0, \quad (3)$$

we obtain the corresponding solution for a ‘‘singular’’ wave in a Gaussian envelope carrying an optical vortex with charge m [17]:

$$E(\rho, \varphi, z) = E_s \frac{\rho_s}{w_s} \left(\frac{\rho}{w_s} \right)^{|m|} \exp\left(-\frac{\rho^2}{w_s^2}\right) \exp[i\Phi_s(\rho, \varphi, z)], \quad (4)$$

where E_s is the amplitude parameter, and ρ_s is the beam waist parameter. The phase Φ_s is

$$\Phi_s(\rho, \varphi, z) = -(|m| + 1) \arctan \frac{2z}{k\rho_s^2} + \frac{k\rho^2}{2R_s(z)} + m\varphi + kz, \quad (5)$$

the transversal beam dimension is

$$w_s = \sqrt{\rho_s^2 + (2z/k\rho_s)^2}, \quad (6)$$

and the radius of the wave front curvature is

$$R_s(z) = z + k^2 \rho_s^4 / 4z. \quad (7)$$

The amplitude distribution in a transverse cross section of the beam has a form of an annulus, and the waist parameter ρ_s is connected with the radius of maximum amplitude at $z = 0$ by a relation

$$\rho_{\max} = \rho_s \left(\frac{|m|}{2} \right)^{1/2}, \quad (8)$$

The maximum amplitude value of a singular wave at $z = 0$, $\rho = \rho_{\max}$, therefore amounts to

$$E_{sm} = E_s \left(\frac{|m|}{2e} \right)^{|m|/2}. \quad (9)$$

The phase singularity disappears when $m = 0$, and solution (4) becomes an ordinary Gaussian beam:

$$E(\rho, z) = E_g \frac{\rho_g}{w_g} \exp\left(-\frac{\rho^2}{w_g^2}\right) \exp[i\Phi_g(\rho, z)], \quad (10)$$

where the propagating parameters for Gaussian beam correspond to Eqs. (5)–(7) with $m = 0$.

As an example we show that a solution in form (2b) having an amplitude singularity at $\rho \rightarrow 0$ may be used to create a solution with only a phase singularity. For this reason we take a similar solution with an amplitude singularity within Gaussian envelope,

$$E(\rho, \varphi, z) = E_s \frac{\rho_s}{\rho} \exp\left(-\frac{\rho^2}{w_s^2}\right) \exp[i\Phi_s(\rho, \varphi, z)], \quad (11)$$

where the phase Φ_s is

$$\Phi_s(\rho, \varphi, z) = \frac{k\rho^2}{2R_s(z)} + \varphi + kz. \quad (12)$$

The combination of solutions (11) and (2b) for $m = 1$ removes the amplitude singularity and gives a wave which has a phase singularity at $\rho = 0$:

$$E(\rho, \varphi, z) = E_s \frac{\rho_s}{\rho} \{1 - \exp[-\rho^2/w_s^2 + ik\rho^2/2R_s(z)]\} \times \exp(i\varphi + ikz). \quad (13)$$

The amplitude of a wave created this way is zero at the center ($\rho = 0$) and decreases on periphery $\propto 1/\rho$. Other possible solutions of the scalar wave equation are Bessel beams [28] and Bessel-Gauss beams [29] carrying optical vortices.

OPTICAL VORTICES IN COMBINED BEAMS

In any practical realization of singular beams by use of synthesized holograms or special optical elements, a small coherent background is always present in a singular beam. The origin of this background may be a scattering in the direction of the singular beam propagation or readout beam diffraction by the fundamental spatial frequency of an imperfect hologram. This background causes splitting of an optical vortex with charge $|m| > 1$ into $|m|$ single-charged vortices [21]. Another case is the interference between a singular wave and copropagating reference wave which is used in analyzing the value and sign of the topological charge of the phase singularity [18,21]. We shall now generalize the problem of coherent coaxial addition of singular beams carrying optical vortices with different charges (including the vortex-free wave, $m = 0$). Our goal is to establish principles of to-

pological charge addition and subtraction, and revise the existing knowledge about topological charge conservation in a light beam propagating in free space.

Without loss of generality, we shall take a coaxial Gaussian wave as a coherent background for a singular wave. By varying the amplitude E_g and waist parameter ρ_g of the Gaussian beam, we may examine its influence in near and far zones, as plane or spherical waves in limiting cases. The sum of the singular beam with the coaxial Gaussian beam (or another singular beam) we shall call a combined beam.

The presence of a coherent background changes the position of a vortex which was initially localized at the center of singular beam Eq. (4). To find the positions of the vortices in the combined beam, we need to write two equations, one giving the radius of the zero-amplitude point (amplitudes of the singular and Gaussian waves are equated), and another giving the angular coordinate φ , which corresponds to the destructive interference between the singular and Gaussian waves:

$$E_g \frac{\rho_g}{w_g} \exp\left(-\frac{\rho^2}{w_g^2}\right) = E_s \frac{\rho_s}{w_s} \left(\frac{\rho}{w_s}\right)^{|m|} \exp\left(-\frac{\rho^2}{w_s^2}\right),$$

$$\Phi_g(\rho, z) = \Phi_s(\rho, \varphi, z) \pm \pi. \quad (14)$$

Equations (14) are the basis for analysis of vortex behavior in a combined beam. The first equation is easy to analyze to show the number of possible amplitude zeros in a combined beam.

To simplify the calculations, we suppose both beams have a waist at $z=0$, and use a normalized transverse coordinate $r=\rho/\rho_s$ and distance $\xi=z/L_R$, where L_R is the Rayleigh length of the singular beam, $L_R=k\rho_s^2/2$. The first equation of system (14) may be rewritten as

$$r^{|m|} = \frac{E_g}{E_s} C(\xi) \exp(\alpha r^2), \quad (15)$$

where

$$C(\xi) = (1 + \xi^2)^{|m|/2} \left(\frac{1 + \xi^2}{1 + \xi^2/\kappa^4} \right)^{1/2}, \quad (16)$$

$$\alpha = \frac{1}{1 + \xi^2} - \frac{1}{\kappa^2 + \xi^2/\kappa^2}, \quad (17)$$

and κ is the ratio of waist parameters $\kappa = \rho_g/\rho_s$.

If $\alpha \leq 0$, Eq. (15) has only one root ($|m|$ times degenerate), as the left side is a function of r growing from zero to infinity, and the right side is a function decreasing from E_g/E_s to zero. The condition $\alpha \leq 0$ corresponds to the following relations for κ and ξ :

$$\kappa \geq 1, \quad \xi \geq \kappa, \quad (18a)$$

$$\kappa \leq 1, \quad \xi \leq \kappa, \quad (18b)$$

and

$$\kappa = 1,$$

$$\alpha = 0, \quad 0 < \xi < \infty. \quad (18c)$$

All zeros of amplitude in the combined beam are located at the same distance from the center. The number of zeros, n , is equal to $|m|$ and each amplitude zero is a center of a single vortex. The total topological charge is conserved in the combined beam when $\alpha \leq 0$.

Another situation occurs when $\alpha > 0$. The function of r in right side of Eq. (15) grows from E_g/E_s to infinity. Analysis shows three possibilities: no real roots of Eq. (15), two roots (each $|m|$ times degenerate), and one root ($2|m|$ times degenerate). The case of one root corresponds to the touching of lines representing the left and right sides of Eq. (15). This condition is determined by taking derivatives on r , which in combination with Eq. (15) gives a solution

$$\alpha r^2 = \frac{|m|}{2}, \quad (19)$$

which gives the critical ratio between the amplitude of the Gaussian beam E_g and maximum amplitude of the singular beam E_{sm} , Eq. (9):

$$\left(\frac{E_g}{E_{sm}} \right)_{cr}^2 = \eta_{cr}^2 = \frac{1 + \xi^2/\kappa^4}{1 + \xi^2} \left(1 - \frac{1 + \xi^2}{\kappa^2 + \xi^2/\kappa^2} \right)^{-|m|}. \quad (20)$$

When the amplitude ratio $\eta = E_g/E_{sm}$ is higher than η_{cr} , no vortices exist in the combined beam, as Eq. (15) has no real roots. The resulting topological charge is zero. If the ratio η is smaller than η_{cr} , additional $|m|$ single-charged vortices appear with charge opposite to the original m -charged vortex, and the resulting total charge is zero again.

As the parameter α changes its sign during propagation of the combined beam both for cases $\kappa > 1$ and $\kappa < 1$, we may expect a variation in the number of vortices n in a combined beam propagating in free space, which means a change of the topological charge of the beam. Only the case $\kappa = 1$ will conserve the initial topological charge unchanged from $\xi = 0$ to infinity, independent of the beam amplitude ratio.

Figure 1 demonstrates the variation of the number of vortices in a combined beam during propagation along the ξ axis. The solid curve dividing the diagram is the dependence of η_{cr} vs ξ for a particular value of κ , $\kappa = 0.5$ [Fig. 1(a)] and $\kappa = 2$ [Fig. 1(b)]. The part of the diagram above the curve corresponds to zero number of vortices in a combined beam. In the case $\kappa < 1$ [Fig. 1(a)], the combined beam conserves topological charge until $\xi = \kappa$ ($\alpha \leq 0$). This area on the diagram is separated by a vertical line. Outside this region the number of vortices may vary between zero and $2|m|$, depending on the ratio η , with total topological charge equal to zero. A similar situation occurs for $\kappa > 1$, but now conservation of initial topological charge of singular beam will apply at $\xi > \kappa$ ($\alpha \leq 0$).

Amplitude profiles of Gaussian and singular beams are plotted in Fig. 2 for different distances ξ . The choice of parameters $m = 1$, $\kappa = 2$, and $\eta = 1.05$ corresponds to the diagram shown in Fig. 1(b). Two points of intersection exist in a region $0 < \xi < 0.655$ (two oppositely charged vortices in the combined beam), no intersections in the region $0.655 < \xi$

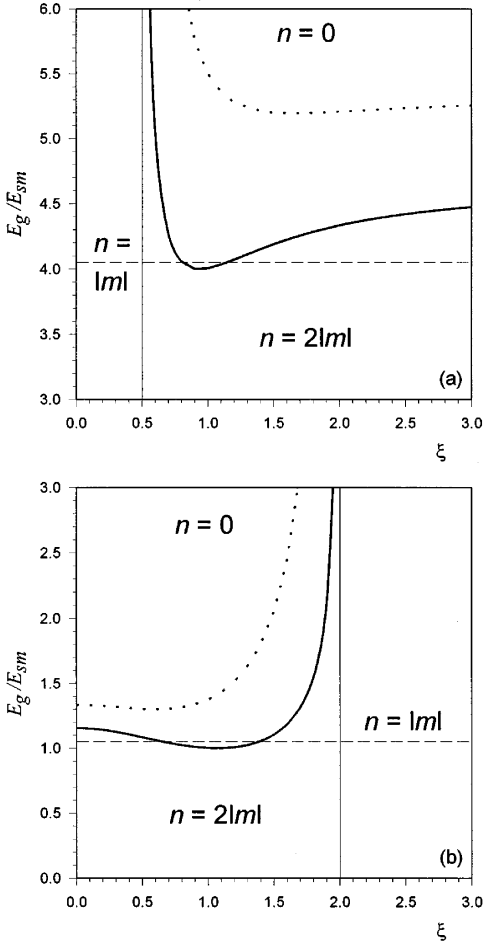


FIG. 1. Diagrams (a) and (b) show the number of vortices in a combined beam. (a) The solid curve is the dependence of the critical amplitude ratio $(E_g/E_{sm})_{cr}$ on the normalized distance ξ plotted for $\kappa=0.5$ and $|m|=1$. The area above the curve contains no vortices. The region $\xi < \kappa$ separated by the vertical line contains $|m|$ vortices, and the part below the curve corresponds to $2|m|$ vortices in a combined beam. The horizontal dashed line corresponds to a particular value of the amplitude ratio, $\eta=4.05$. This line crosses in turn regions with $|m|$, $2|m|$, 0 , $2|m|$ vortices. The total topological charge is m , while $\xi \leq \kappa$, and zero outside. (b) The same as (a), but for the case $\kappa > 1$. The solid curve is a plot of η_{cr} for $\kappa=2$. The horizontal dashed line corresponds to the amplitude ratio $\eta=1.05$. For higher m values, diagrams (a) and (b) are very similar: dotted lines are plots of η_{cr} for $|m|=2$.

< 1.39 (no vortices in the beam), two intersections and two oppositely charged vortices for $1.39 < \xi < 2$, and finally one intersection and thus a single vortex for $\xi > 2$.

In a practical situation we may not have information about the initial beam amplitude ratio and waist parameters ρ_g and ρ_s , or the optical paths of beams from waist to observation plane may be different. The only experimental measured values at the plane of observation are intensity

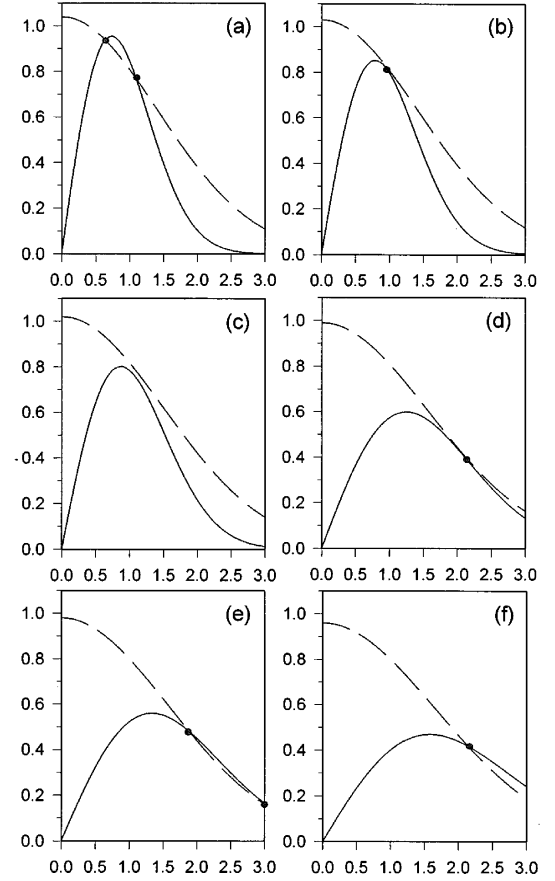


FIG. 2. Amplitude profiles of Gaussian (dashed line) and singular (solid line, $|m|=1$) waves with amplitude ratio $\eta=1.05$ and $\kappa=2$, for different normalized distances ξ . (a) $\xi=0.3$, curves intersect in two points shown by circles. (b) $\xi=0.655$, touching of the curves. (c) $\xi=1$; no intersections. (d) $\xi=1.39$, once again touching. (e) $\xi=1.5$, two intersections. (f) $\xi=2$, only the one intersection. Horizontal axis: normalized transverse coordinate r ; vertical axis: wave amplitudes $E(r)/E_{sm}=1$.

distributions of the Gaussian and singular beams. The amplitudes $A_g(\rho)$ and $A_s(\rho)$ and transversal parameters w_g and w_s may be calculated from energy distributions, and m is also easy to determine. The amplitudes necessary for existence of vortices in a combined beam may be written in this case as

$$A_g \exp\left(-\frac{\rho^2}{w_g^2}\right) = A_s \left(\frac{\rho}{w_s}\right)^{|m|} \exp\left(-\frac{\rho^2}{w_s^2}\right) \quad (21a)$$

or

$$\left(\frac{\rho}{w_s}\right)^{|m|} = \frac{A_g}{A_s} \exp\left(\frac{\rho^2}{w_s^2} - \frac{\rho^2}{w_g^2}\right), \quad (21b)$$

where A_g is the amplitude of the Gaussian beam at maximum, and A_s is connected with the maximum amplitude of

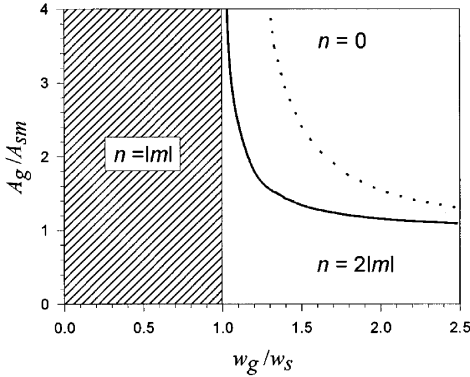


FIG. 3. Diagram showing the number of vortices in a beam combined from arbitrary Gaussian and coaxial singular beams. The horizontal coordinate is the transverse size ratio, the vertical coordinate is the amplitude ratio. The dashed area corresponds to the case of conservation of the total topological charge in the combined beam (the singular beam is wider than the Gaussian). The solid curve at the right side is the critical amplitude ratio for $|m|=1$, and the dotted curve is for $|m|=3$. The area under the curves corresponds to $2|m|$ vortices in the combined beam. No vortices are in the combined beam with parameters in the area above the curves.

the singular wave A_{sm} by the same relation as Eq. (9). Taking into account the previous analysis, we may formulate the following consequences important for experimental observation of vortex transformations in a combined beam: If the singular beam is broader than the Gaussian beam at the observation plane, $w_s > w_g$, Eq. (21b) has one $|m|$ -times degenerate root, the total topological charge is conserved, and the original m -charged vortex breaks into $|m|$ vortices. If the Gaussian beam is broader than the singular, $w_g > w_s$, the total topological charge of the combined beam is zero. The number of vortices in a combined beam is determined by the relation between amplitudes. If $A_g/A_{sm} > (A_g/A_{sm})_{cr}$, the combined beams will contain no vortices. If $A_g/A_{sm} < (A_g/A_{sm})_{cr}$, an additional $|m|$ vortices will appear in the combined beam, as discussed above. The critical ratio $(A_g/A_{sm})_{cr}$ is

$$\left(\frac{A_g}{A_{sm}}\right)_{cr} = \left(\frac{w_g^2}{w_g^2 - w_s^2}\right)^{|m|/2}. \quad (22)$$

The number of vortices in the combined beam for this practical situation may be determined from the diagram shown as Fig. 3. The hatched area corresponds to the case $w_s > w_g$ when the topological charge remains the same as in the singular beam. In the right part of the diagram a solid curve is plotted according to Eq. (22) for $m=1$ and a dotted curve for $m=3$. The region under the curves corresponds to the appearance of $|m|$ additional vortices and zero total topological charge. No vortices exist in the region above the corresponding curves.

To determine the position of vortices on a plane r, φ we need to use the second equation of system (14). Alternatively, we may calculate directly the position of the vortices as points of intersection of lines representing the zeros of the

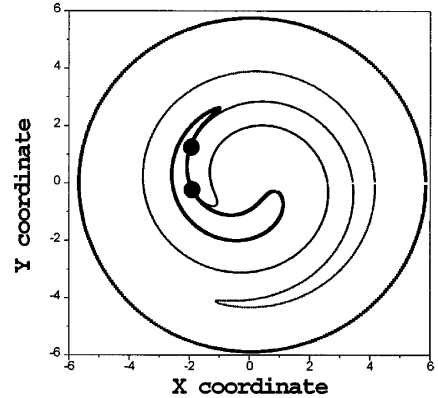


FIG. 4. Phase map shows the lines $\text{Re}(E)=0$ (thick line) and $\text{Im}(E)=0$ (thin line) in (x,y) coordinates normalized on ρ_s at a cross section $\xi=1.405$ of a combined beam with parameters $m=1$, $\kappa=2$, and $\eta=1.05$. Positions of vortices are shown by dots (intersections of zero lines). The original vortex is shifted from the center, and an additional vortex with opposite charge is located at the second intersection point.

real and imaginary parts of the complex amplitude $E(r, \varphi, \xi)$ of the combined beam: $\text{Re}(E)=0$ and $\text{Im}(E)=0$ [3,15,30,31]. Figure 4 shows these lines in x,y coordinates (normalized on ρ_s) at cross section $\xi=1.405$ of a combined beam with parameters $m=1$, $\kappa=2$, and $E_g/E_{sm}=1.05$. The positions of vortices are shown by dots (intersections of zero lines). The original vortex is shifted from the center and an additional vortex with opposite charge is located at the second intersection point. Both lines $\text{Re}(E)=0$ and $\text{Im}(E)=0$ are closed.

Figure 5 exhibits transformations of zero-amplitude lines and corresponding vortices map for a combined beam with $m=3$, $\kappa=2$, and $\eta=1.75$. At near field ($\xi=0.5$) all vortices are suppressed by Gaussian wave with larger amplitude and waist parameter. The faster transverse spread of singular beam leads to the equalizing of amplitudes at combined beam periphery and appearance of three pairs of opposite charged vortices. Three vortices move away from the beam and disappear when $\xi=2$. Finally, when $\xi > 2$, combined beam contains three single vortices located symmetrically with the conservation of the initial topological charge of the singular beam.

The trajectories of vortices within the cross section of the combined beam are shown in Fig. 6 for two main situations, $\kappa > 1$ and $\kappa < 1$. The combined beam ($m=1$, $\kappa=2$, and $\eta=0.5$) starts with two vortices [Fig. 6(a)] which move on their trajectories as shown in Fig. 6(a) in opposite directions, and the negative vortex leaves the beam at $\xi=2$. For another initial amplitude ratio $\eta=1.05$, the behavior of the vortices is somewhat different. Two vortices annihilate in collision at $\xi \approx 0.655$ [Fig. 6(b)], and then the beam does not contain any singularity until $\xi \approx 1.39$. Born as a pair, two new vortices with opposite charges repel each other and finally one (negatively charged) disappears at infinity ($r \rightarrow \infty$) at $\xi=2$, and the remaining one carries the initial charge of the singular beam. In the case $\kappa < 1$ the combined beam ($m=1$, $\kappa=0.5$, and $\eta=4.05$) starts with only the primary vortex (shifted from the center), and an additional negative vortex enters the beam at

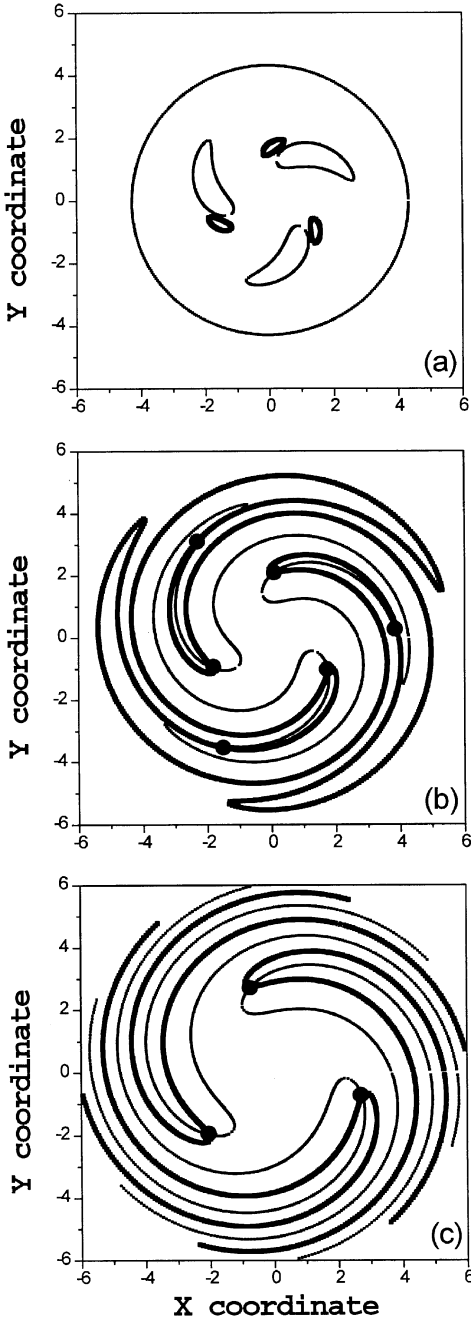


FIG. 5. Phase maps for different distances ξ of a combined beam propagation with parameters $m=3$, $\kappa=2$, and $\eta=1.75$. (a) $\xi=0.5$, lines $\text{Re}(E)=0$ and $\text{Im}(E)=0$ do not intersect, and no vortices exist in the combined beam. (b) $\xi=1.25$, lines $\text{Re}(E)=0$ and $\text{Im}(E)=0$ intersect in points shown by dots, and six single-charged vortices (three pairs) exist in the beam. (c) $\xi=2.5$, three intersection points are shown by dots, and three vortices are located symmetrically around the beam center.

$\xi=0.5$ coming from infinity [Fig. 6(c)]. After several round trips it meets the prime vortex, and both annihilate in collision at $\xi \approx 0.81$. A new vortices pair originates at $\xi \approx 1.16$, and exists within the beam up to infinity.

The analysis of vortex behavior in a beam combined from two coaxial singular beams may be performed in a similar manner. However, the combined beam may possess some new features in this case. First, both beams have zero amplitude at the center. If $|m_1| \neq |m_2|$, the combined beam will have at the center a vortex with the smaller absolute value of the charge. On the beam periphery, depending on the ratios of amplitudes and sizes, the number of vortices may vary between zero, $|m_1 - m_2|$, and $2|m_1 - m_2|$. In the particular case when $|m_1| = |m_2|$, there are two situations: $m_1 = m_2$ and $m_1 = -m_2$. When charges have the same sign, waves add coherently producing circular interference fringes [32], and the only m -charged vortex is located at the beam center. When vortices have opposite charges, they compete with each other for the central position. For small ρ , we may neglect the Gaussian envelope of the beams, and write the amplitude at the core as

$$E(\rho, \varphi) \propto \rho^{|m|} [A_{s1} e^{im\varphi} + A_{s2} e^{-im\varphi}], \quad (23)$$

which determines the phase near the core as $\arctan[(A_{s1} - A_{s2})/(A_{s1} + A_{s2}) \tan(m\varphi)]$. This means that the vortex with larger host wave amplitude will win, but becomes anisotropic [30]. Finally, if $A_{s1} = A_{s2}$, vortices annihilate each other. The interference pattern displays $2|m|$ fringes radiating from the center.

EXPERIMENTAL OBSERVATION OF VORTICES IN COMBINED BEAM

The setup for our experiment is shown in Fig. 7. The linearly polarized output beam of a 10-mW He-Ne laser operating on the TEM₀₀ mode is split first at beamsplitter BS1. The directly transmitted beam is diffracted at the holographic grating, and the first diffracted order (singular beam) is selected with the iris aperture. The holographic gratings are phase holograms restoring charge -1 and 3 singular beams, both blazed for greatest efficiency into first order [25]. The reflected beam from beamsplitter BS1 is further split at BS2 into the “background” Gaussian beam (reflected beam) and reference wave (transmitted beam). Lenses $L1-L4$ are used to control the sizes and radii of curvature of the Gaussian and reference wave. Lens $L2$ is finely controlled with a translation stage to allow fine adjustments of the wave-front curvature. Polarizing beamsplitters (PBS’s) are used to control the relative intensities of the singular and Gaussian beams. BS3 recombines the singular and Gaussian beams, and BS4 interferes with the combined beam and the reference wave. (Double reflection of the singular beam from BS3 and mirror M does not change the sign of its topological charge.) Finally, lens $L5$ creates a magnified image of the beam on the screen, recorded with a charge-coupled device (CCD) camera.

The experimental technique consisted, first, of an alignment of the singular and Gaussian beams to make them coaxial. The resulting interference pattern, observed in the far field, was then used to determine the relative difference in curvature of wave fronts between the two beams. This was

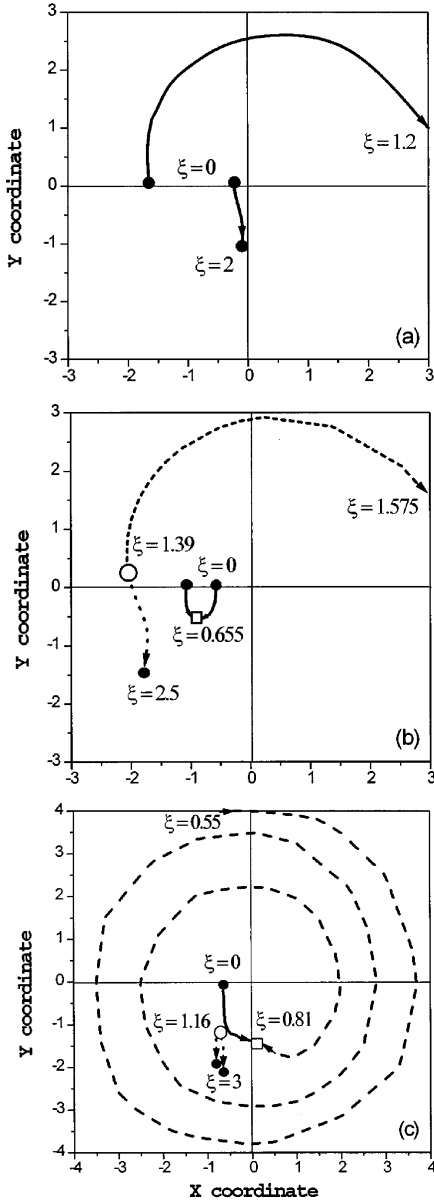


FIG. 6. Vortex trajectories in a combined beam cross section. (a) For $m=1$, $\kappa=2$, and $\eta=0.5$. The combined beam starts with two oppositely charged vortices shown by dots. The negative vortex leaves the beam at $\xi=2$. (b) Two vortices in a beam with $\eta=1.05$ annihilate in collision at $\xi \approx 0.655$. A pair of two new vortices with opposite charges appear at $\xi \approx 1.39$. They repel each other and finally one (negatively charged, dashed line) disappears at infinity ($r \rightarrow \infty$) at $\xi=2$. (c) Combined beam ($m=1$, $\kappa=0.5$, $\eta=4.05$) starts with only the primary vortex (shifted from the center), and an additional negative vortex enters the beam at $\xi=0.5$ (dashed line) coming from infinity. After several round trips it meets the prime vortex, and both annihilate in collision at $\xi \approx 0.81$. New vortices' pair originates at $\xi \approx 1.16$ (dotted lines).

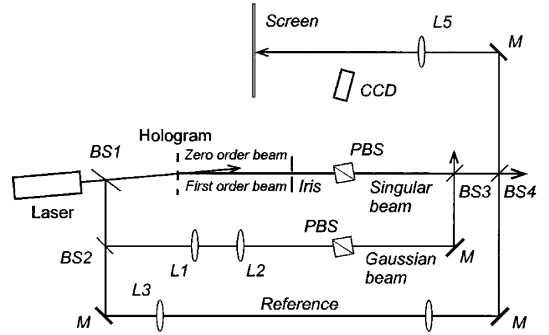


FIG. 7. Sketch of the experimental setup. The He-Ne laser beam splits into three channels. In the first channel a singular beam is created by the hologram, the second channel carries the Gaussian wave, and the reference wave is formed in the third channel. The interference patterns are observed on a screen.

adjusted using the position of lens $L2$, so that there was at most one round of dark fringe in the pattern. At this point, the wavefronts of two beams were effectively matched at the observation screen.

The profiles of energy distribution in the Gaussian and singular beams were obtained from separate CCD images, and the relative adjustment in beam intensities using PBS's could then be carried out to obtain the necessary number of amplitude graph intersection points. The corresponding intensity distributions are shown in Figs. 8 and 9 for Gaussian and singular beams (charges -1 and 3). Once the two beams were roughly the required relative intensity and size for additional vortices to be observable, the reference wave was used to interfere the intensity pattern to observe the presence of fringe dislocations, and hence the vortices in the original pattern.

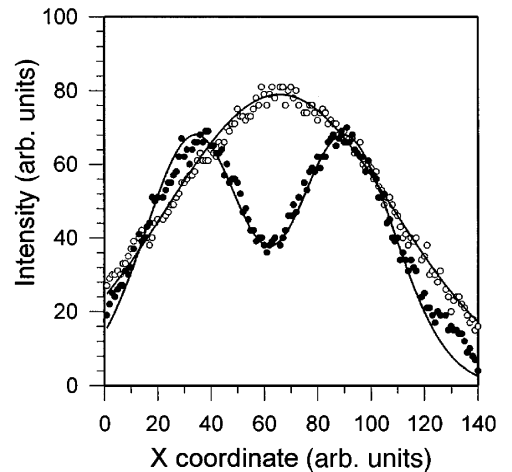


FIG. 8. Experimental intensity distributions for Gaussian ($m=0$, open circles) and singular ($m=-1$, closed circles) beams at the observation plane. Solid lines are numerical fits according to formulas (24).

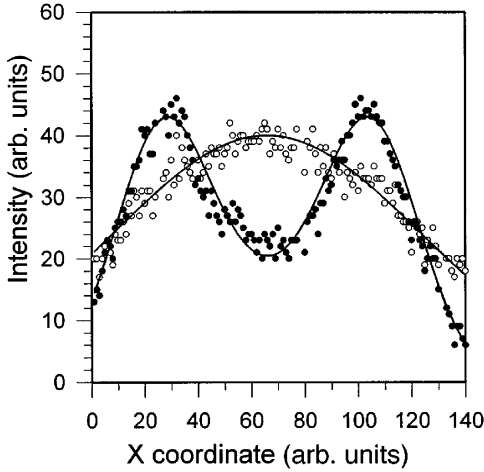


FIG. 9. The same distributions as in Fig. 8, for Gaussian and charge 3 singular beams.

The reference beam was simply adjusted through the choice of lenses $L3$ and $L4$ to be of much larger size than the combined beam cross section. The interference picture produced by the superposition of the reference beam and the combined beam at BS4 could be roughly controlled to give a reasonable number of fringes for sufficient resolution of any fringe dislocations present. To avoid reduction of fringe visibility due to a loss of coherence through multilongitudinal mode operation of the laser, all path differences were made integer multiples of the laser cavity length.

The parameters obtained from a numerical fit of the data were used for a comparison of the experimental results with theoretical predictions. The following functions were used to generate the intensity profiles of Gaussian and singular beams numerically:

$$P_g(x) = \left[A_g \exp \left[-\frac{(x-x_0)^2}{w_g^2} \right] \right]^2,$$

$$P_{-1}(x) = \left[\frac{A_{-1}}{w_{-1}} [(x-x_0)^2 + y_0^2]^{1/2} \exp \left[-\frac{(x-x_0)^2 + y_0^2}{w_{-1}^2} \right] \right]^2,$$

$$P_3(x) = \left[\frac{A_3}{w_3^3} [(x-x_0)^2 + y_0^2]^{3/2} \exp \left[-\frac{(x-x_0)^2 + y_0^2}{w_3^2} \right] \right]^2, \quad (24)$$

where A_g , A_{-1} , and A_3 are the amplitudes of the respective waves (Gaussian, singular charges -1 and 3) and w_g , w_{-1} , and w_3 are the beam sizes. The function $P_g(x)$ is the intensity profile of a Gaussian beam defined in the x, y plane, where the profile is taken along the x direction with $y=0$. $P_{-1,3}$ are the intensity profiles along x axis of singular beams, charges -1 and 3 , at $y=y_0$, in order to take into account a small misalignment of the beam centers, which prevents the central minimum of the singular beam from going to zero. Parameters x_0 and y_0 determine the beam center. The data obtained from the CCD (in relative units) were fitted to functions (24) with fitting parameters: A_g

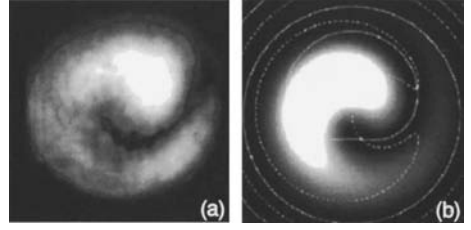


FIG. 10. Experimental (a) and numerically generated (b) intensity distributions in a combined (Gaussian plus $m=-1$ singular) beam cross section. Zero-amplitude lines are shown in (b), when the solid line represents the real part of the complex amplitude, and the dashed line the imaginary part.

$=8.86$, $w_g=85.22$, and $x_0=65.76$ for the Gaussian beam, and $A_{-1}=19.23$, ($A_{-1m}=8.25$), $w_{-1}=45.81$, $x_0=62.25$, and $y_0=16.73$ for the singular (charge -1) beam. The calculated curves are shown in Fig. 8 (solid lines). The ratio of the beam sizes is $w_g/w_s=1.86$, i.e., the Gaussian beam is wider than the singular. The amplitude ratio is $A_g/A_{-1m}=1.07$, and according to the theoretical predictions (see the diagram in Fig. 3) an additional vortex should appear in the combined beam with a sign opposite to the original vortex. The presence of intersection points of the curves also indicates two vortices existence in the combined beam.

For the profiles of the Gaussian and charge 3 singular beam, the fitting parameters were $A_g=6.29$, $w_g=115.92$, and $x_0=66.30$ for the Gaussian beam, and $A_3=15.82$ ($A_{3m}=6.49$), $w_3=41.23$, $x_0=66.05$, and $y_0=33.91$ for the singular beam (Fig. 9). The Gaussian beam is again wider than the singular, and the amplitude ratio corresponds to the case of three additional vortices appearing, with the total topological charge of the combined beam being zero.

For the two superpositions investigated, namely, Gaussian beam with charge -1 singular and Gaussian with charge 3 singular, the intensity patterns were calculated in gray scale and are shown in Figs. 10 and 11. To generate the two-dimensional intensity distribution $I(x, y)$ in a beam combined from Gaussian and charge -1 singular beams numerically, the expression for $I(x, y)$ is represented as follows:

$$I(x, y) = P_g(x, y) + P_{-1}(x, y) + 2\sqrt{P_g(x, y)P_{-1}(x, y)} \times \cos \left(\frac{x^2 + y^2}{R^2} - \arctan \frac{y}{x} + \delta \right), \quad (25)$$

where R is the radius of relative curvature of the Gaussian wave front with respect to the singular wave, δ is an adjustable phase factor, and $\arctan(y/x)$ is the azimuth angle. The intensity distribution for the combined beam with Gaussian and charge 3 singular wave has the same form as Eq. (25), only the azimuth factor becomes $3 \arctan(y/x)$. Figures 10 and 11 also show the experimental patterns of combined beam intensity in the far field and calculated intensity distributions with lines of zeros for the functions $\text{Re}[E(x, y)]=0$ (solid line) and $\text{Im}[E(x, y)]=0$ (dashed line). The points of intersection of the zero lines correspond to the positions of vortices. For the case of the combination of a Gaussian and a

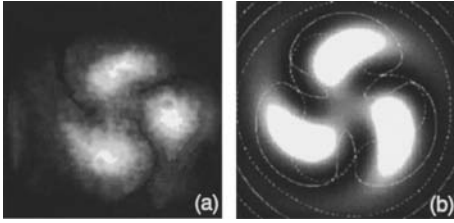


FIG. 11. The same intensity distributions as in Fig. 10, but for Gaussian plus $m=3$ singular combined beam.

charge -1 singular beam, a single dark spiral fringe is seen [Fig. 10(a)]. The curved shape of the fringe is due to slight difference in wave-front curvature. The zero-amplitude lines have two points of intersection, corresponding to two vortices existing within the dark fringe [Fig. 10(b)]. For the Gaussian and charge 3 singular beam, three dark fringes radiating from the center of the pattern are present [Fig. 11(a)]. There are six intersection points of the zero lines in Fig. 11(b), indicating two vortices with opposite charges located in each dark fringe.

To demonstrate the existence of vortices in the combined beams clearly, the interference patterns of the combined beam with the reference wave are shown in Figs. 12 and 13. The results of experimental observations are compared with calculated intensity distributions. For the Gaussian and charge -1 combined beam, both theory and experiment show a dark spiral fringe which begins at the center of the pattern and ends after nearly two turns [Figs. 12(a) and 12(b)]. Thereafter, the interference fringes form rings corresponding well with the theory: the phase has no rotating component at this area. Thus two oppositely charged vortices make the total topological charge zero.

For the Gaussian and charge 3 combined beam, there are three inner interference “forks,” equally separated by 120° and directed clockwise, and three outer “forks” directed anticlockwise (Fig. 13). Hence there are six single vortices altogether, or three pairs of oppositely charged vortices. At the area outside, the fringes again form rings.

The experimental observations clearly demonstrate the appearance of additional vortices in combined beams, changing the total topological charge of a beam. We have thus shown in both theory and experiment that the topological charge of a beam containing phase singularities is not a constant while propagating in free space.

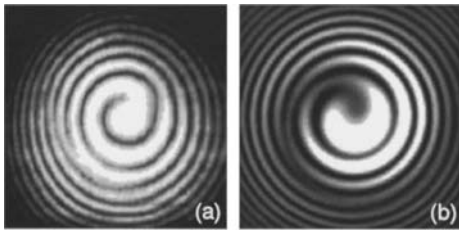


FIG. 12. The interference pattern of the combined beam (Gaussian plus $m=-1$ singular) and reference wave. The primary vortex is located near the center, and an additional vortex is seen with positive charge: (a) experimental picture; (b) numerical simulation.

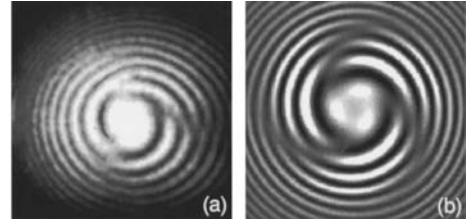


FIG. 13. The interference pattern of the combined beam (Gaussian plus $m=3$ singular) and reference wave. The primary vortex is split into three vortices, and an additional three vortices with negative charge produce, with them, three pairs: (a) experimental picture; (b) numerical simulation.

ANGULAR MOMENTUM OF LIGHT BEAM CARRYING OPTICAL VORTICES

The rotation of phase around the vortex axis causes a nonzero value of angular momentum of a beam [23,33]. The origin of the angular momentum may be easily explained from simple evaluations. As the wave front of a singular beam has a helicoidal shape, the Poynting vector $P(\rho, \varphi, z)$ which is perpendicular to the wave front surface has at each point a nonzero tangential component. In the paraxial approximation this component equals $P_\perp(\rho, \varphi, z) = -mP/k\rho$, and the angular-momentum density in the z -axis direction is $M_z(\rho, \varphi, z) = (\rho/c^2)P_\perp(\rho, \varphi, z)$. As the Poynting vector is proportional to a light wave intensity, $P \propto |E_s(\rho, \varphi, z)|^2$, we may obtain the expression for angular-momentum density of a singular beam:

$$M_z(\rho, \varphi, z) \propto -m|E_s(\rho, \varphi, z)|^2. \quad (26)$$

The time-averaged density of angular momentum directed along the z axis in a combined beam cross section may be calculated in a general form [23,33] as

$$M_z = \frac{i}{2} \omega \epsilon_0 \left[x \left(E^* \frac{\partial E}{\partial y} - E \frac{\partial E^*}{\partial y} \right) - y \left(E^* \frac{\partial E}{\partial x} - E \frac{\partial E^*}{\partial x} \right) \right], \quad (27)$$

where ϵ_0 is the permittivity of free space. The total angular momentum L_z of the beam is an integral over the beam cross section

$$L_z = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} M_z dx dy. \quad (28)$$

The simplest combined beam is a sum of singular [Eq. (4)] and Gaussian [Eq. (10)] beams, and its angular momentum calculated according to Eq. (27) is

$$\begin{aligned} M_z(\rho, \varphi, z) = & -\omega \epsilon_0 m \{ |E_s(\rho, \varphi, z)|^2 \\ & + |E_s(\rho, \varphi, z) E_g(\rho, z)| \cos[\Phi_s(\rho, \varphi, z) \\ & - \Phi_g(\rho, z)] \}, \end{aligned} \quad (29)$$

which attains a simple form for a pure singular beam ($E_g = 0$):

$$M_z(\rho, \varphi, z) = -\omega \epsilon_0 m |E_s(\rho, \varphi, z)|^2, \quad (30)$$

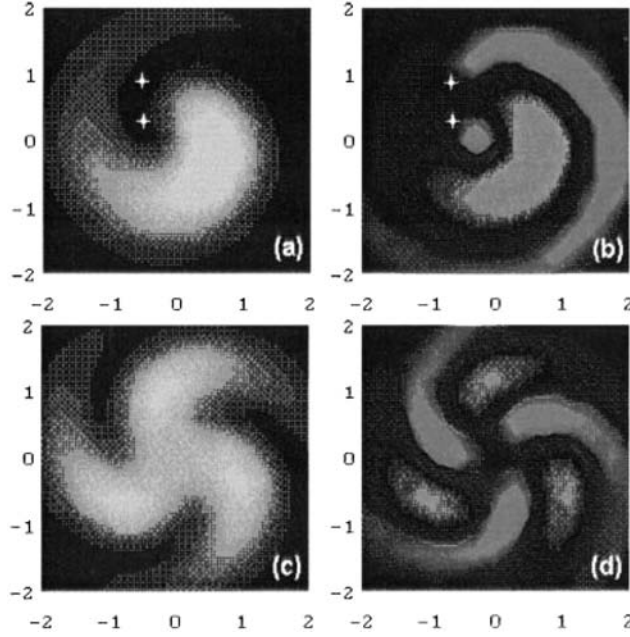


FIG. 14. (Color) Distribution of energy and angular momentum in a combined beam. (a) and (b) $m=1$, $E_g/E_{sm}=1.05$, $\kappa=2$, and $\xi=0.3$. Green is for energy distribution, red for positive angular momentum density $M_z > 0$, and blue for negative angular-momentum density $M_z < 0$. The positions of two oppositely charged vortices are shown by white crosses. (c) and (d) The same distributions for $m=3$, $E_g/E_{sm}=2.5$, and $\kappa=2$. No vortices exist in the combined beam with this amplitude ratio, but a modulation of the angular-momentum density is clearly seen. The transverse coordinates are x and y , normalized on ρ_s .

coinciding with Eq. (26), and

$$L_z = -2\pi\omega\epsilon_0 m E_s^2 \left(\frac{\rho_s}{w_s}\right)^2 \int_0^\infty \rho d\rho \left(\frac{\rho}{w_s}\right)^{2|m|} \exp\left(-\frac{2\rho^2}{w_s^2}\right) \\ = -\frac{\pi m |m|!}{2^{|m|+1}} \omega \epsilon_0 \rho_s^2 E_s^2. \quad (31)$$

The density of the angular momentum of a singular beam M_z is therefore proportional to the beam intensity, and the total angular momentum L_z to the beam energy. In a medium without losses, the total momentum L_z is conserved, as well as the beam energy.

The relation obtained for the angular momentum of a combined beam (29) may be generalized as a law of coaxial addition of N beams carrying optical vortices with topological charges m_i :

$$M_z(\rho, \varphi, z) = -\frac{\omega \epsilon_0}{2} \sum_{i,j}^N (m_i + m_j) |E_i(\rho, \varphi, z) E_j(\rho, \varphi, z)| \\ \times \cos[\Phi_i(\rho, \varphi, z) - \Phi_j(\rho, \varphi, z)]. \quad (32)$$

This law is valid not only for singular waves with a Gaussian envelope, but for all kinds of waves with axially symmetric amplitude distributions.

An interesting consequence following from Eq. (29) is that even a small amount of a singular wave combined with a strong Gaussian wave produces a substantial modulation of

the angular-momentum density. Figure 14 represents the calculation of angular-momentum density of combined beams with parameters $m=1$, $\kappa=2$, and $\eta=1.05$ (a) and $m=3$, $\kappa=2$, and $\eta=2.5$ (b).

For a combined beam with a density of angular-momentum distribution given by Eq. (29), the total angular momentum is equal to that of the singular beam alone, because an interference term gives a zero amount in integral (28). This demonstrates that coherent addition of a wave with zero angular momentum does not affect the resulting angular momentum of the combined beam. Using the general form of the angular-momentum distribution in a combined beam (32), we are able now to calculate the result of coaxial interference of N arbitrary singular beams,

$$L_z = \sum_i^N L_{zi}, \quad (33)$$

which establishes a rule: In a beam combined from N coaxial beams, the angular momenta L_{zi} add arithmetically.

This rule gives interesting consequences for the addition of singular beams with equal and opposite topological charges. Two identical singular beams with equal energy but opposite charges form a vortex-free beam with zero angular momentum, independent of the phase relation between them. Identical beams with equal topological charges of vortices do not obey the general rule (33), because the interference term is a constant in this case, and the energy of the summed

beam is strongly dependent on relative phases of components. Further, adding a vortex-free wave with zero angular momentum cannot change the total angular momentum L_z of a beam, even though it may suppress all the vortices in the combined beam.

To understand the transformations of angular momentum associated with an m -charged vortex in a combined beam where the number of vortices may change and even become zero, we need to imagine the wave front of a combined beam which is not a symmetrical m -pitched helicoid as for a pure singular beam. When all the vortices are suppressed by the strong background wave, the wave front of the combined beam still has folds, but is a smooth surface without defects. The inclination of local "rays" which are perpendicular to the surface produces both positive and negative components of the angular momentum. With an increase of Gaussian background wave amplitude, the folds become smaller, but the beam amplitude grows proportionally, and resulting negative and positive parts of angular momentum have nearly the same value. The difference between the negative and positive components of the angular momentum remains exactly equal to the initial angular momentum of the singular beam, but is now due to small residual amplitude modulation over the combined beam cross section. In the case of multiple vortices localized within a combined beam, the distribution of the angular momentum becomes more complicated, but the resulting total angular momentum remains constant.

CONCLUSIONS

Our analysis of coherent coaxial addition of optical waves carrying (at least one) optical vortices has revealed the general properties of combined beams. We have studied in detail the behavior of vortices in a beam combined from singular

and "background" Gaussian waves in order to check the principle of topological charge conservation. In brief, we obtained the following results.

(1) Addition of a coherent coaxial vortex-free wave to a singular wave with a m -charged vortex may change the number of vortices and the total topological charge in the combined beam. Depending on background wave parameters, the number of vortices in the combined beam may vary between zero, $|m|$, and $2|m|$. The total topological charge is m or zero.

(2) In free-space propagation from $z=0$ to infinity, the total topological charge and number of vortices in a combined beam do change for any choice of background wave parameters, except the case $\kappa=1$ (equal transverse sizes).

(3) We demonstrate for the first time to our knowledge, the possibility, in free-space propagation, of additional vortices appearing from the far periphery of a beam (from infinity), and their disappearance at a beam periphery.

(4) The obtained results are also applicable to the case of combinations of singular beams.

(5) We show that the total angular momentum of a beam is conserved in all cases in free-space propagation, in contrast with the total topological charge. We establish the main rules for addition and subtraction of angular momenta of light beams.

(6) We analyzed the transverse distribution of the angular momentum in a combined beam cross section, and found a strong spatial modulation of the angular-momentum density even in the case of absence of vortices in a combined beam.

ACKNOWLEDGMENTS

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Helical-wavefront laser beams produced with a spiral phaseplate

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Abstract

We demonstrate experimentally that a spiral phaseplate can convert a TEM₀₀ laser beam into a helical-wavefront beam with a phase singularity at its axis. The diffractive-optical effect of the spiral phaseplate is implemented by index matching a macroscopic structure in an optical immersion. We discuss the optical properties of a helical wavefront beam produced this way by means of a mode analysis and by Fraunhofer diffraction calculations.

1. Introduction

The special properties of helical-wavefront or spiral laser beams have recently aroused appreciable interest [1–8]. The complex amplitude u of such a beam has the property

$$u(\rho, \phi, z) \propto \exp(-il\phi), \quad (1)$$

where ρ , ϕ and z are cylindrical coordinates and l is an integer. At the center the phase has a screw dislocation [9], also called a phase singularity, optical vortex, or topological defect of charge l . The equiphase surface is a spiral sheet with a pitch equal to l times the wavelength. Our interest into these beams is due to the fact that fields with such a phase distribution are orbital angular momentum eigenmodes if their intensity distribution is rotationally symmetric [10]. Our ultimate aim is to determine the mechanical consequences of this orbital angular momentum of light [11].

Note that circular polarization of the light beam gives rise to a different type of helicity. The helicity

we discuss here is entirely caused by the shape of the wavefront, not by the polarization. A separation in this sense is only possible in the paraxial approximation [12].

One special set of paraxial spiral beams are the Laguerre-Gaussian (LG) modes. The mode character of these beams expresses itself in the fact that, apart from a change in scale in the transverse plane, the intensity distribution does not change upon propagation in free space.

A helical-wavefront laser beam can be produced with a laser that operates in a LG mode. This can be accomplished by locking the phases of a set of Hermite-Gaussian (HG) transverse modes of a laser [8]. Alternatively, the phase and amplitude distribution of a non-helical laser beam can be changed outside the laser cavity such that a spiral beam is obtained. A cylindrical-lens mode converter may do this if the input beam is a higher-order HG mode. The output is then a pure LG mode. As previously shown, this mode conversion can be fully described in terms of a finite set of Gaussian modes [6,13].

Holography is an alternative and more general way to influence the phase and/or amplitude distribution

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of a field. It has been used to produce beams with a single [14,15] or multiple phase singularities [16]. Depending on the type of holography (phase or amplitude, binary or sinusoidal) a larger or smaller fraction of the incident beam is diffracted into a helical-wavefront beam, that could be a pure LG mode. A modal description of this conversion strongly depends upon the properties of the hologram.

In this paper we introduce a novel way to produce helical wavefront laser beams, using a spiral phaseplate. Phaseplates in general have often been used to influence the amplitude distribution of a beam [17]. Such a thin transparent plate typically has strips or radial sectors that are half a wavelength thicker, obtained by coating or etching, than the rest of the plate. The spatially dependent 180° phase shift can be used to homogenize the phase distribution of a beam that has lobes that are in anti-phase, such as a higher order Hermite-Gaussian beam. Obviously, a phaseplate without spiral symmetry cannot produce a helical wavefront beam. We report here a spiral phaseplate that *can* cause a single screw dislocation to appear in the center of the beam, resulting in a helical wavefront laser beam.

Such a plate has been previously been used to produce a radially polarized laser beam [18]. For that application the function of the plate was interferometric in nature; in the analysis diffraction was neglected, so that the change of the intensity distribution in the far field is ignored. We instead introduce the spiral phaseplate as a means of producing helical-wavefront laser beams. We theoretically analyze the propagation properties of the beam thus produced in two ways. First, we give a paraxial mode analysis with the LG modes as a basis, and next, we discuss Fraunhofer diffraction of these beams. Finally, we show experimentally that it is possible to produce such beams using a spiral phaseplate. For that purpose we developed an immersion variety of the spiral phaseplate that is simple to produce and can be adjusted for a wide range of applications.

2. The spiral phaseplate

A spiral phaseplate is a transparent plate whose thickness increases proportional to the azimuthal angle ϕ around a point in the middle of the plate. The sur-

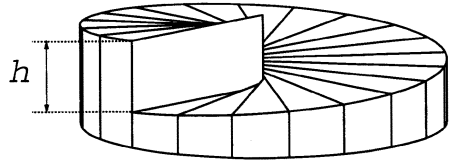


Fig. 1. Sketch of the spiral phaseplate.

face resembles one turn of a staircase, the step height being h (Fig. 1). When a beam is passed through such a plate, the helical surface can be expected to give a helical character to the beam. A rigorous calculation of the operation of the plate would require vector-diffraction theory. However, for beams with small divergence and with a height of the step that is sufficiently small we remain in the paraxial regime, so that the operation of the plate can be considered to be an operation on the phase only. If the incident beam is described by the complex amplitude $u(\rho, \phi, z)$, the amplitude u' directly after the plate is then given by

$$u' = u \exp(-i\Delta l \phi). \quad (2)$$

Here Δl is the height of the step in wavelengths given by

$$\Delta l = \Delta n h / \lambda, \quad (3)$$

where Δn is the difference in refractive index between the plate and its surroundings and λ the vacuum wavelength. Note that the paraxial approximation was not valid in a previously performed microwave experiment in a waveguide, where a similar spiral retarder was used [19]. This violation leads to a coupling between the amplitude and the local polarization, similar to spin-orbit coupling.

The spiral phaseplate may transform a non-helical beam into a helical one, but also change the helicity of a helical beam. When the helicity of a laser beam is modified according to Eq. (2), the phase distribution is changed while the amplitude distribution is not. In most cases the resulting beam is no longer a pure mode that propagates with only rescaling of the intensity distribution, but rather a superposition of modes. The spiral phaseplate is therefore in general not a pure mode converter.

In discussing the operation of the phaseplate we will limit ourselves to the case of an incident LG mode. For these modes we use different indices than used

normally. Our preferred indices n, m make the relation between HG and LG modes more evident [6]. The amplitude u_{nm}^{LG} is defined by

$$u_{nm}^{\text{LG}}(\rho, \phi, z) = C_{nm}^{\text{LG}} \times (1/w) \exp(-ik\rho^2/2R) \exp(-\rho^2/w^2) \times \exp[-i(n+m+1)\psi] \exp[-i(n-m)\phi] \times (-1)^{\min(n,m)} (\rho\sqrt{2}/w)^{|n-m|} L_{\min(n,m)}^{|n-m|}(2\rho^2/w^2), \quad (4)$$

with

$$R(z) = (z_R^2 + z^2)/z, \quad (5)$$

$$\frac{1}{2}kw^2(z) = (z_R^2 + z^2)/z_R, \quad (6)$$

$$\psi(z) = \arctan(z/z_R), \quad (7)$$

where C_{nm}^{LG} is a normalization constant, $k = 2\pi/\lambda$ the wave number, z_R the Rayleigh range, and $L_p^l(x)$ the generalized Laguerre polynomial [6]. The more common indices p, l are related to n, m by $p = \min(n, m)$ and $l = n - m$.

3. Mode analysis

As pointed out, a LG mode with modified helicity is in general no longer a pure mode. Since the LG modes themselves form a complete set of helical modes this is still a useful basis for expanding the modified mode. There is a twofold infinite number of LG basis sets, since there are two free Gaussian beam parameters (e.g. Rayleigh range and z coordinate of the waist). If these are chosen equal to those of the incident beam, the mode decomposition is unique. The mode decomposition of a mode u_{nm}^{LG} whose helicity has been modified by Δl is then defined by the expansion coefficients

$$a_{nm,st} = \langle u_{st}^{\text{LG}} | \exp(-i\Delta l\phi) | u_{nm}^{\text{LG}} \rangle, \quad (8)$$

where the brackets denote integration in the transverse plane (ρ, ϕ) . The relative weight of the modes is given by

$$I_{nm,st} = |a_{nm,st}|^2. \quad (9)$$

As can be seen from the ϕ integral, the only modes present for integer Δl are those that differ by Δl in helicity with the input mode, that is

$$s - t = n - m + \Delta l. \quad (10)$$

Table 1

Mode contents of a LG_{00} mode that has acquired a phase $\exp(-i\phi)$ due to a spiral phaseplate with $\Delta l = 1$. Indicated is the relative intensity in percentages of the LG_{st} modes; the mode index s is on the horizontal and t on the vertical axis

6							
5							0.79
4						1.17	
3					1.92		
2				3.68			
1			9.82				
0		78.5					
(00)	0	1	2	3	4	5	6

The coefficients $I_{nm,st}$ have been calculated numerically by calculating the integrals on a 100×100 mesh with a width and height equal to 8 times the Gaussian radius of the input beam. Table 1 gives the results for a LG_{00} (=TEM₀₀) beam passed through a spiral phaseplate with $\Delta l = 1$; the values in Table 1 represent $I_{00,st}$ as percentage of the incoming TEM₀₀ intensity. Even though the input beam has a maximum at the axis, more than 78% ends up in the LG_{10} mode which is dark at the center. Increasing the order of the input mode gives even higher purity of the output. As an example, the LG_{30} mode is converted for 94% into the LG_{40} mode, leaving only 6% for the other modes.

Table 2 shows a few decompositions for modes passed through a plate with $\Delta l = 2$. A LG_{01} mode is converted into a pure LG_{10} mode, which is obvious since they have the same intensity distribution but inverted phase distributions. This pure mode conversion occurs for all transformations where the phase is inverted, which is the case if

$$\Delta l = -2(n - m). \quad (11)$$

In the other cases the decomposition can contain either a finite or an infinite number of modes.

The coefficients of Tables 1 and 2 can also be deduced analytically. Comparison of some of the coefficients shows that the errors in the numerical calculation are below 10^{-4} . Furthermore, the analytical results show that there is a distinction between even and odd Δl . For odd Δl , the number of modes in the decomposition is always infinite, all individual modes having the expected helicity (Eq. (10)). For even Δl , there are three situations. If Eq. (11) holds, there is only one mode in the output. If Δl is smaller than the

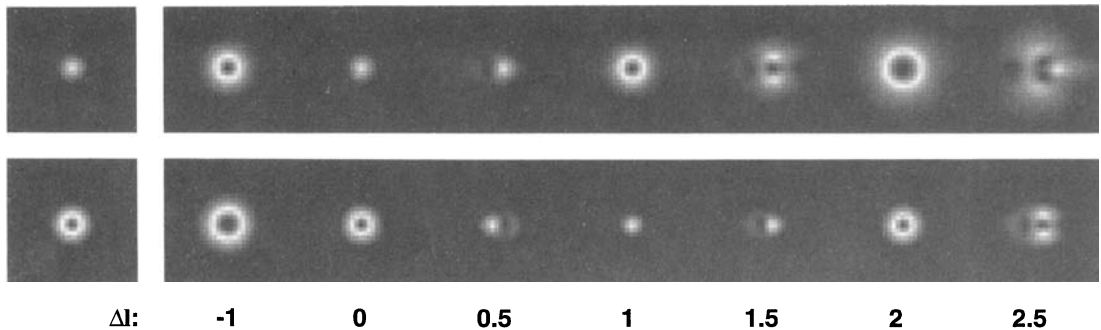


Fig. 2. Far-field calculations of (de)spiralized beams. The top row shows a LG_{00} ($=TEM_{00}$) beam passed through a spiral phaseplate with different heights of the step. The bottom row shows the same for an input LG_{01} mode.

5. Experiments

In Ref. [18] a spiral phaseplate has been described that was made using a dielectric coating, made for operation at $\lambda = 10.6 \mu\text{m}$. The coating thickness was constant in the radial direction, but increased by $6.3 \mu\text{m}$ over a full turn. In order to avoid the complexity of producing a micron-scale structure with a continuously changing thickness, especially so since we aim at a visible wavelength ($\lambda = 633 \text{ nm}$), we have chosen to index-match a plate from acrylic (PMMA, $n = 1.49$) machined with normal tools. A milling tool of 1 mm diameter was used, removing material along radial lines from the edge to the center. This was done repetitively, each time rotating the plate by 5 degrees and moving the tool $10 \mu\text{m}$ down. This results in a step $h = 0.72 \text{ mm}$ or $\Delta l = 577$ at a wavelength of 633 nm. To bring the step down to about one wavelength, the plate was immersed into a liquid with nearly the same index of refraction. At $\lambda = 633 \text{ nm}$, a difference in index of refraction $\Delta n = 8.7 \times 10^{-4}$ will result in a stepsize of one wavelength ($\Delta l = 1$). By tuning the temperature the effective step size could be tuned, due to the difference in temperature coefficient of the refractive index of the immersion liquid and the acrylic.

With this method the phaseplate, and in fact many other diffractive elements, can be produced with ordinary machining. Immersion has often been used to adjust the strength of optical elements, but to our knowledge this is the first use of immersion to produce diffractive optics. Of course it is impossible to also scale down the *transverse* structure to microscopic di-

mensions using immersion. In our plate this is apparent at the center. There should ideally be a infinitely small singular point, but the 1 mm diameter of the tool causes an area of about 1.5 mm diameter to be distorted.

To study the operation of the spiral phaseplate we used a setup as sketched in Fig. 3. The spiral phaseplate is put into a gold-coated brass cell with two BK7 windows, anti-reflection coated on the outside. The cell is filled with index-matching fluid of $n_D = 1.4922$ at 25°C [20]. The temperature of the cell is controlled with a Peltier element. It took about ten minutes for the liquid to adjust from one equilibrium temperature to a 1°C higher temperature.

The input beam is generated by a 633 nm HeNe laser with a Brewster-windowed discharge tube and external mirrors, separated by 83 cm. Laser polariza-

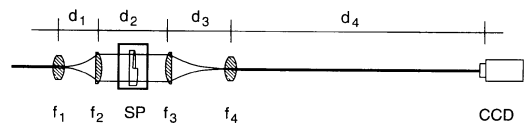


Fig. 3. Experimental setup used to convert a Laguerre-Gaussian mode into a mode with a different helicity using a spiral phaseplate. Input is a LG_{00} or LG_{01} mode, produced with a HeNe laser and a cylindrical-lens mode converter (not shown). A telescope (f_1 , f_2) expands the beam to about 10 mm diameter. This beam passes through the cell in which the spiral phaseplate is immersed. Lens f_3 produces a waist about 20 mm in front of lens f_4 , which projects the far-field image in the waist onto a CCD camera. The focal lengths of the lenses are $f_1 = 20 \text{ mm}$, $f_2 = 200 \text{ mm}$, $f_3 = 300 \text{ mm}$ and $f_4 = 25 \text{ mm}$; the distances are $d_1 = 0.22 \text{ m}$, $d_2 = 0.40 \text{ m}$, $d_3 = 0.34 \text{ m}$ and $d_4 = 1.53 \text{ m}$.

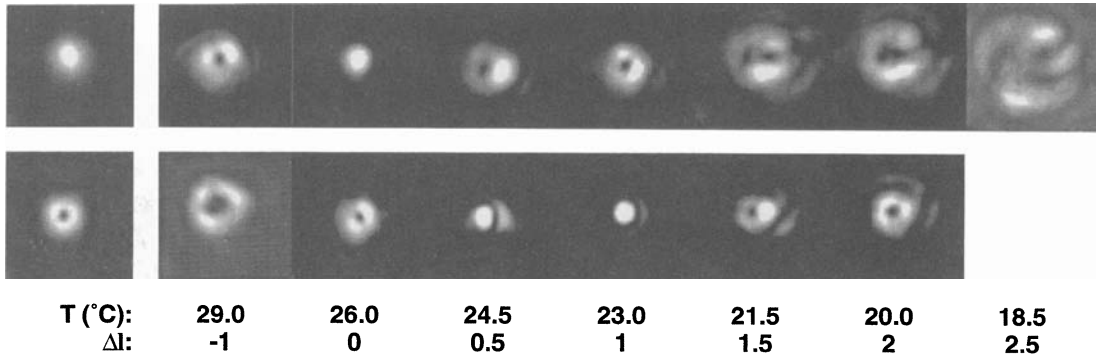


Fig. 4. Experimentally observed far-field patterns of a LG_{00} and LG_{01} laser beam passed through a spiral phaseplate. On the left the far-field image is shown when the phaseplate is removed from the setup; next to it are the far-field images at different temperatures of the cell.

tion is linear and several longitudinal modes may oscillate. Transverse-mode oscillation is normally in a TEM_{00} mode, but can be forced to a HG_{01} mode by means of a thin wire ($20\ \mu\text{m}$ diameter) in the cavity. This HG mode is converted into a LG mode with a cylindrical-lens mode converter outside the laser cavity. This part of the setup is identical to that used in Ref. [6], and produces either a LG_{00} mode ($l = 0$) or a LG_{01} mode ($l = -1$), selected by insertion or removal of the intra-cavity wire. This beam is enlarged to a diameter of about 10 mm and collimated, and is passed through the spiral phaseplate in the cell. The large diameter reduces the influence of the imperfection at the center of the plate. The output beam was focussed with a lens, forming the far-field image in the focal plane. A second lens projects this image enlarged onto a CCD camera. The resulting far-field image was recorded as a function of the temperature of the cell, both for an incident LG_{00} and LG_{01} mode.

In Fig. 4, top row, the results for the LG_{00} mode are shown. The leftmost picture is the far-field image of the beam before inserting the plate into the cell. It is followed by the images with the plate inserted into the cell, at different cell temperatures. At 26°C index-matching is perfect ($\Delta l = 0$). At 24.5°C a dark spot appears next to the maximum, which has moved to the center at 23°C , where it looks like a LG_{01} mode ($\Delta l = 1$). At larger values of Δl two dark spots appear (21.5°C), which merge to a larger, central hole at 20°C ($\Delta l = 2$). At 18.5°C a third dark spot appears.

The bottom row of Fig. 4 shows the results for the

LG_{01} mode, which itself is already a helical mode. At 23°C the dark spot in the center has disappeared, so that the output resembles a LG_{00} mode, which indicates that the helicity of the wavefront has been removed by the spiral phase plate. At 20°C the mode looks like the input mode again, but will now have the opposite helicity ($\Delta l = 2$).

Note that, especially for large Δl , the output shows severe distortion. In an earlier experiment, using a home-made mixture of benzyl alcohol and ethylene glycol as index-matching fluid, this was much more pronounced. The distortion was partly caused by schlieren and other kinds of inhomogeneities. This is more severe for the home-made mixture due to its high viscosity and chemical instability. In the experiments of Fig. 4 the distortion is due to imperfections of the phaseplate (proportional to Δl), temperature inhomogeneities of the liquid (independent of Δl) and spherical aberrations of the imaging optics, which is more severe for large beams. These aspects were not optimized; improvement of the performance should therefore be possible.

6. Conclusion

We have shown that a spiral phaseplate can be used to produce a helical wavefront laser beam. The mode decomposition of a transformed input mode has been determined, and the Fraunhofer diffraction pattern has been calculated. We have shown theoretically and ex-

perimentally that the plate transforms between beams with different helicity. Its design and fabrication has been described. The plate was produced using conventional machining and converted into a diffractive element by immersion. For producing a *solid* device, one could cover the helical surface with a nearly index-matched optical quality epoxy or thermoplastic and flatten the outer surfaces. Finally, since the plate directly transforms the orbital angular momentum of a beam, it might be interesting to study the mechanical consequences of this transformation by suspending the plate by a thin wire so as to form a torsional pendulum [11].

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The generation of free-space Laguerre-Gaussian modes at millimetre-wave frequencies by use of a spiral phaseplate

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Abstract

A spiral phaseplate is used at millimetre-wave frequencies to transform a free-space, fundamental Hermite-Gaussian mode into a Laguerre-Gaussian mode with an azimuthal phase component. A ray optics analysis confirms that the Laguerre-Gaussian beam produced has a total angular momentum equivalent to $\ell\hbar$ per photon.

1. Introduction

Laguerre-Gaussian (LG) modes, like Hermite-Gaussian (HG) modes, form a complete basis set for paraxial light beams. The former exhibit circular symmetry, the latter rectangular [1]. Two indices identify a given mode, and the modes are normally denoted LG_p^ℓ and HG_{mn} . For a Hermite-Gaussian mode m and n are the numbers of nodes in the x and y directions respectively. For a Laguerre-Gaussian mode, ℓ is the number of 2π cycles in phase around the circumference and $(p+1)$ the number of radial nodes. The amplitude, u_p^ℓ of the LG_p^ℓ mode in cylindrical co-ordinates is

$$u_p^\ell(r, \phi, z) \propto e^{-ikr^2/2R} e^{-r^2/w^2} e^{-i(2p+\ell+1)\psi} e^{-i\ell\phi} \\ \times (-1)^p (r\sqrt{2}/w)^\ell L_p^\ell(2r^2/w^2), \quad (1)$$

where R is the wavefront radius of curvature, w is the radius for which the Gaussian term falls to $1/e$ of its on-axis value, ψ is the Gouy phase and L_p^ℓ a generalised Laguerre polynomial. The azimuthal

phase term, $e^{i\ell\phi}$, distinguishes the Laguerre-Gaussian modes from the Hermite-Gaussian modes. This phase term creates helical wavefronts for the Laguerre-Gaussian modes in contrast to the planar wavefronts of the Hermite-Gaussian modes [2]. Laguerre-Gaussian modes possess an orbital angular momentum of $\ell\hbar$ per photon [3], quite distinct from the spin angular momentum associated with the polarisation state. This orbital angular momentum has been recently demonstrated through an optical interaction with microscopic particles [4].

In recent years, the production of Laguerre-Gaussian modes at optical frequencies has attracted considerable interest. Laguerre-Gaussian laser beams may be produced directly [5] or by the conversion of Hermite-Gaussian modes. To date, three different classes of mode converter have been demonstrated. Two of these, spiral phaseplates [6] and computer generated holographic converters [7,8], introduce the azimuthal phase term to a HG_{00} beam. The spiral phaseplate technique has also been employed within a waveguide configuration incorporating a 10 GHz source [9]. In all these devices a screw phase-disloca-

tion, produced on-axis, causes destructive interference leading to the characteristic ring intensity pattern in the far field. The spiral phaseplate may also be used to convert between any two LG_p' modes separated by $e^{i\ell\phi}$ phase terms. In general, the purity of Laguerre-Gaussian modes produced by these methods is limited by the co-production of higher order modes. The other class of converter is the cylindrical-lens mode converter [10] which converts higher order Hermite-Gaussian modes to the corresponding Laguerre-Gaussian mode. Unlike the spiral phaseplate and the holographic converter, this method can produce pure Laguerre-Gaussian modes.

For a Laguerre-Gaussian mode, the orbital angular momentum in the beam is equivalent to $\ell\hbar$ per photon. Consequently, for a fixed power, the angular momentum in the beam is proportional to the wavelength; unlike linear momentum, $\hbar/\lambda$ per photon, where for a fixed power the linear momentum in the beam is wavelength independent.

In this paper we extend the production of free-space, Laguerre-Gaussian modes to millimetre-wave frequencies (~ 100 GHz), where the wavelength is $\sim 10^4$ times that at optical frequencies. Hence, for the same power, the orbital angular momentum is also $\sim 10^4$ time larger, which opens the possibility for observing the transfer of angular momentum to a macroscopic object. We demonstrate the use of a spiral phaseplate to convert the fundamental Hermite-Gaussian to higher order Laguerre-Gaussian modes. The phaseplate is preferable to the cylindrical lens converter because of the relative difficulty of producing high-order, free-space, Hermite-Gaussian beams at millimetre-wave frequencies.

The total angular momentum, J_z , of a Laguerre-Gaussian beam is the sum of orbital and spin angular momenta [3]. Thus for left-hand or right-hand circularly polarised beams $J_z = l \pm 1$. The Hermite-Gaussian mode converted in this work has a well-defined linear polarisation and consequently the total angular momentum in the beam is due entirely to orbital angular momentum.

2. Ray optics analysis of a spiral phaseplate mode converter

The spiral phaseplate (Fig. 1) has one planar and one spiral surface. The spiral surface forms one

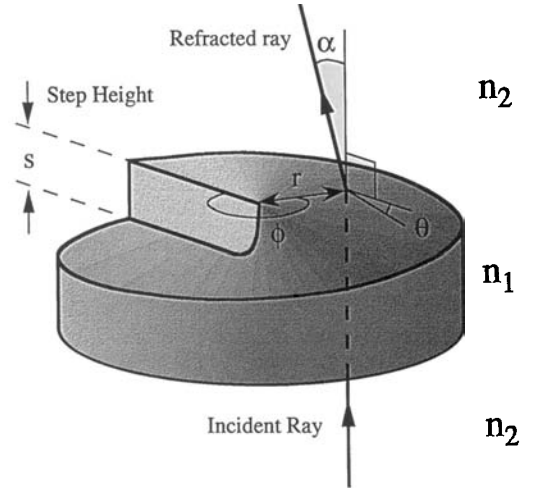


Fig. 1. The spiral phaseplate, showing the refraction of a single ray upon transmission.

period of a helix, with a step discontinuity. Upon transmission through the phaseplate, a beam of wavelength λ is subject to a phase delay, ψ , which depends on the azimuthal angle, ϕ , where

$$\psi = \frac{(n_1 - n_2)s}{\lambda} \phi, \quad (2)$$

n_1 and n_2 are the refractive indices of the phaseplate and surrounding media respectively and s is the physical step height at $\phi = 0$. For a Laguerre-Gaussian mode, the total phase delay around the phaseplate must be an integer multiple of 2π , i.e. $2\pi\ell$. Thus, to produce a Laguerre-Gaussian mode, the physical height of the step in the spiral phaseplate is given by

$$s = \frac{\ell\lambda}{(n_1 - n_2)}. \quad (3)$$

When the step height is not an integer number of wavelengths, the phase of the beam is discontinuous at the step and this is observed as a break in the ring intensity pattern. Beijersbergen et al. [6] have modelled the detuning of the step height through the transition from one Laguerre-Gaussian mode to another. In their small-angle approximation, the converter only changes the phase and not the intensity of the beam. The annular intensity pattern arises from

the far field diffraction of the beam's screw dislocation. However, the beam produced is not a pure mode, but an infinite superposition of Laguerre-Gaussian modes. The conversion from the HG_{00} to the LG_0^1 mode was calculated to be 78% efficient.

A rigorous analysis of the mechanical interaction between the beam and the spiral phaseplate requires the solution of Maxwell's equations at the boundaries of the phaseplate. However, as with many optical and quasi-optical systems, a ray optics analysis gives useful insight into the nature of the interaction. In this paper, we use a ray optics model (Fig. 1) to gain insight into how the orbital angular momentum content of the beam arises from the mode converter. Although the orbital angular momentum is a property of the beam as a whole, it is useful to consider this in terms of the equivalent angular momentum per photon.

Consider a ring of radius r , projected on the spiral surface. The angle, θ , of the local azimuthal slope of the spiral surface is given by

$$\tan \theta = \frac{s}{2\pi r}. \quad (4)$$

A ray parallel to, but a distance r from, the optical axis will be refracted as it emerges from the spiral surface. The deflection angle, α , may be found using Snell's Law:

$$n_2 \sin(\theta + \alpha) = n_1 \sin(\theta). \quad (5)$$

Before refraction, the beam has a linear momentum of $n_2 h/\lambda$ per photon. After refraction, there is a

component of linear momentum in the azimuthal direction, p_ϕ , given by

$$p_\phi = n_2 \frac{h}{\lambda} \sin \alpha. \quad (6)$$

To achieve this there is a transfer of angular momentum, L , between the spiral phaseplate and the beam of light of

$$L = r p_\phi = n_2 \frac{h}{\lambda} r \sin \alpha \quad (7)$$

per photon in the beam.

Let us consider the small-angle case where (4), (5) and (7) reduce to

$$\theta \approx \frac{2}{2\pi r}, \quad (8)$$

$$n_2(\theta + \alpha) \approx n_1 \theta, \quad (9)$$

$$L \approx n_2 \frac{h}{\lambda} r \alpha. \quad (10)$$

Combining Eqs. (8), (9) and (10) with s set by Eq. (3) (the condition for a Laguerre-Gaussian mode), the angular momentum exchanged, L , between the light beam and the phaseplate is

$$L \approx \hbar \frac{s(n_1 - n_2)}{\lambda} \approx \ell \hbar \quad (11)$$

per photon in the beam.

This agrees with the result for Laguerre-Gaussian beams derived from the analysis of Maxwell's equations [3].

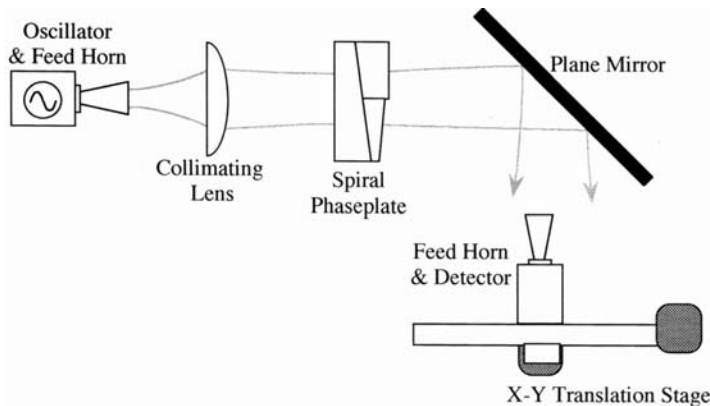


Fig. 2. The experimental configuration for obtaining Laguerre-Gaussian modes at millimetre-wave frequencies.

As r decreases it is unreasonable to approximate $\tan \theta \approx \theta$ and the small-angle calculation for the angular momentum diverges from the exact result. When $r = \ell\lambda$ the difference is approximately 10%. Therefore, we require most of the energy in the incident beam to fall in the region $r > \ell\lambda$. This is easily satisfied in the optical regime, where $\ell\lambda$ is small compared with the typical beam radius. The criterion becomes more important when the wavelength is increased to the millimetre-wave range. Although it is impractical to scale the beam diameter by the same amount as wavelength, it is not difficult to ensure that most of the beam power satisfies the small-angle approximation; this occurs when $w > \ell\lambda$.

The intensity distribution of our experimental results is in good agreement with theory and indicates that most of the energy is being transformed into a single Laguerre-Gaussian mode. It therefore seems reasonable to assume that the prediction of the ray optics model for the orbital angular momentum content of the beam of $\ell\hbar$ per photon is itself valid providing $w > \ell\lambda$.

It follows from (7) that for a fixed step height and constant power, the angular momentum transferred to the beam is independent of the wavelength. Consequently, a spiral phaseplate converter with a step height of a few millimetres may be used to produce an optical beam with a large angular momentum. However, the efficiency of coupling into a single Laguerre-Gaussian mode falls significantly as ℓ increases [6], hence, the generated beam is not a pure mode.

3. Experimental configuration

Fig. 2 shows the experimental configuration to produce millimetre-wave, free-space, Laguerre-Gaussian modes. The source was an InP Gunn diode oscillator with a peak output power of 10–20 mW. Adjusting the dimensions of the resonant cavity tuned the linearly polarised output from 72 to 95 GHz [11]. The circular-aperture, corrugated feed-horn produced a $\sim 98\%$ pure HG_{00} beam with Rayleigh range of 50 mm [12]. A polyethylene lens of focal length 120 mm collimated the beam with $w \approx 25$ mm. We see that $w \gg \ell\lambda$ and therefore the small-angle approxi-

mation is valid in this case. The phaseplate was also made of polyethylene, which has a refractive index of 1.52 at millimetre-wave frequencies [13]. Two different phaseplates were used, one to generate the LG_0^1 mode and the other to generate the LG_0^2 mode. The step heights were 6.7 mm and 13.4 mm respectively to give a single and a double wavelength step at 86 GHz. The planar surface of the phaseplate and both surfaces of the collimating lens were cut with an anti-reflection texture of quarter-wavelength deep concentric grooves.

An aluminium mirror reflected the output from the phaseplate onto a detector mounted on an x -y scanning stage placed in the far field of the converter. The detector used was an Anritsu MP81B/ML83A with an identical feed horn to that on the oscillator. The antenna pattern of the horn is Gaussian in form, and so the measured intensity profile is the convolution of the true far field diffraction pattern and a Gaussian point spread function. The x -y scanning stage and detector were computer controlled to measure a 50×50 grid over a square area with a side of 100 mm. The readings were transferred to *Mathematica* [14], in which they were interpolated and displayed as density plots.

4. Results

Fig. 3(a) shows the result of the conversion from HG_{00} to LG_0^1 . The central minimum, a characteristic of the Laguerre-Gaussian mode, is well defined. Fig. 3(b) shows the corresponding result for the LG_0^2 mode. As expected, the radius of maximum intensity [15] of the LG_0^2 is $\sqrt{2}$ times that of the LG_0^1 . The linear polarisation state of the Laguerre-Gaussian beams was demonstrated using a wire-grid polariser, with which the beam could be completely attenuated. This confirms that, unlike previous waveguide-based work in the microwave regime [9] there is no combination of the spin polarisation and the orbital angular momentum in our case. For both of the generated Laguerre-Gaussian modes, ‘‘hot-spots’’ were observed in the ring with an increased intensity of about 10%. Two possible sources of these have been modelled. One is an imperfection at the centre of the phaseplate, caused by the finite size of the machining

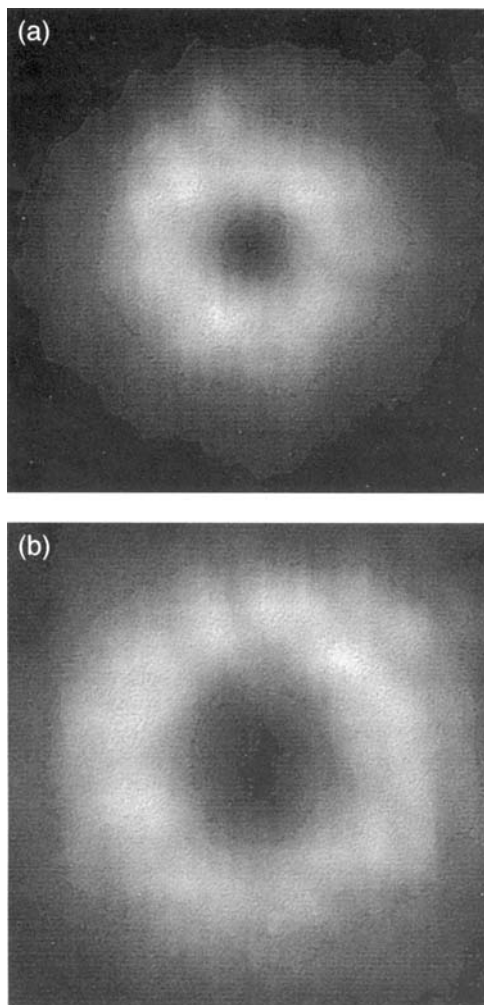


Fig. 3. Far-field intensity distribution for the observed Laguerre-Gaussian modes (a) LG_0^1 and (b) LG_0^2 .

tool; the other is slight misalignment of the axis of the HG_{00} beam and the axis of the phaseplate. The magnitude of the observed ‘‘hot-spots’’ are consistent with the precision of our experimental configuration.

5. Conclusions

In this paper we have used a spiral phaseplate at millimetre-wave frequencies, to transform a free-

space, fundamental Hermite-Gaussian mode into Laguerre-Gaussian modes with an azimuthal phase component. As with earlier work, in the optical regime [6], most of the energy of the incident mode can be converted into a single Laguerre-Gaussian mode, the order of which is set by the step height of the spiral phaseplate. A ray optics analysis of the spiral phaseplate confirms that, in the small-angle approximation, the Laguerre-Gaussian mode produced has a total angular momentum equivalent to $l\hbar$ per photon in the beam.

We believe that the comparatively large angular momentum associated with free-space Laguerre-Gaussian modes in the millimetre-wave range, opens the possibility of observing the transfer of orbital angular momentum from an electro-magnetic wave to a macroscopic object. Previously, angular momentum has been transferred, within a waveguide, from a 10 GHz beam to a suitable antenna. However, complications associated with near-field diffraction and the apparent interchange of spin and orbital angular momentum prevented quantitative agreement between theory and observations [9].

For the free space modes generated in this work we calculate that for a 20 mW Laguerre-Gaussian ($l = 2$) beam, at 86 GHz, the total angular momentum flow per second is $7.4 \times 10^{-14} \text{ kg m}^2 \text{ s}^{-2}$. Total absorption of such a beam and its associated angular momentum by a 4 mm diameter homogeneous sphere of density 1000 kg m^{-3} would result in an angular acceleration of the particle of $0.03^\circ \text{ s}^{-2}$. A suitable material for such a sphere is ferrite loaded epoxy resin, which has a high absorption and can be easily set into any required shape.

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